MA'ADEN معادن

Saudi Arabian Mining Company شركة التعدين العربية السعودية

Project **Mansourah Massarah Gold Project**

Customer: **MA'ADEN GOLD AND BASE METALS (MA'ADEN)**

Title: **DETAIL DRAWING**

GOLD PROCESSING PLANT
DETAIL DIMENSION DRAWING,SLD, CIRCUIT DIAGRAM FOR 49PR POWER GENERATION LOCAL CONTROL PANEL FOR OILY WATER AND SLUDGE TREATMENT

Document Number: **MD-216-49PR-EG-ME-DED-0003**

Revision: **D01**

Revision Date: **18.02.2024**

Status: **AS-BUILT**

L&T Document Number:	O19045	001	X	K	00506	499	R1

REVISION HISTORY						
Rev	**Date**	**Revision Text**	**Prepared By**	**Reviewed By**	**Approved By**	**Employer**
D01	18.02.2024	AS-BUILT	NAEW	SJK	NBR	

Sensitivity: LNT Construction Internal Use

PANEL CODE	DOCUMENT NAME	DOCUMENT DESCRIPTION	SHEET N:o	DOCUMENT N:o	REVISION
BJU906	LIST OF DOCUMENTS	DRAWING LIST	1	DECA00006012	b
BJU906	DIMENSIONAL DRAWING	LAYOUT 1:10	2	DECA00006012	b
BJU906	OVERVIEW DIAGRAM	SINGLE LINE	3	DECA00006012	b
BJU906	CIRCUIT DIAGRAM	LOW VOLTAGE FEEDER	4	DECA00006012	b
BJU906	CIRCUIT DIAGRAM	COMMON INDICATOR/CONTROL	5	DECA00006012	b
BJU906	CIRCUIT DIAGRAM	PUMP CONTROL	6	DECA00006012	b
BJU906	CIRCUIT DIAGRAM	EXTERNAL ALARM	7	DECA00006012	b
BJU906	CONNECTION TABLE	CABLE LIST	8	DECA00006012	b
BJU906	CONNECTION TABLE	INTERNAL CONNECTION LIST	9	DECA00006012	b
BJU906	PARTS LIST	PARTS LIST	10	DECA00006012	b

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Rev.	Date	Created by	Appr. by	Revision text
a	20.7.2021	RSW004	KSU004	Location Box Updated

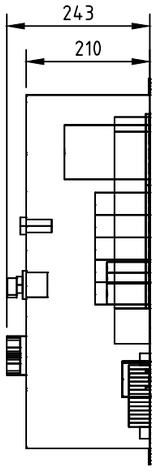
Project name: Mansourah - Massarah Gold Mine
Project id: P/20002
Product id: 6X20V32TS
Based on document id: DAAB198406
Customer document id: -

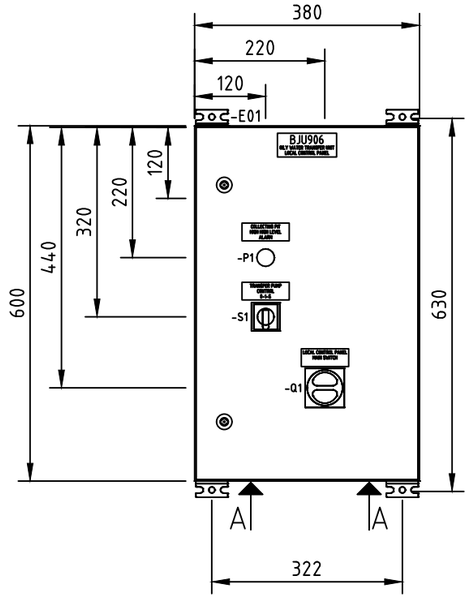
WÄRTSILÄ

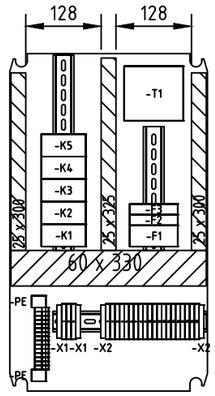
LIST OF DOCUMENTS
OILY WATER TRANSFER UNIT
LOCAL CONTROL PANEL
DRAWING LIST

Created by	Create date	Language	Object designation
MBAX46	8 Jun 2021	EN	BJU906
Reviewed by	Review date	Sheet size	Page
RSW004	8 Jun 2021	A3	1 / 10
Approved by	Approval date	Document Id:	Rev.
KSU004	8 Jun 2021	DECA00006012	C

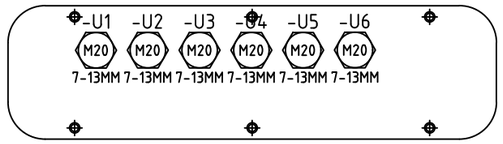
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SECTION A-A SCALE 1:4



PANEL INFORMATION					
WIDTH	380	mm	PROTECTION CLASS		IP65
HEIGHT	600	mm	COLOR		RAL 7035
DEPTH	210	mm	Ik	6	kA/1s

-E01	BJU906	OILY WATER TRANSFER UNIT	LOCAL CONTROL PANEL
-P1	COLLECTING PIT	HIGH HIGH LEVEL	ALARM
-Q1	LOCAL CONTROL PANEL	MAIN SWITCH	
-S1	TRANSFER PUMP	CONTROL	0-1-S

Rev.	Date	Created by	Appr. by	Revision text
a	20.7.2021	RSW004	KSU004	Location Box Updated

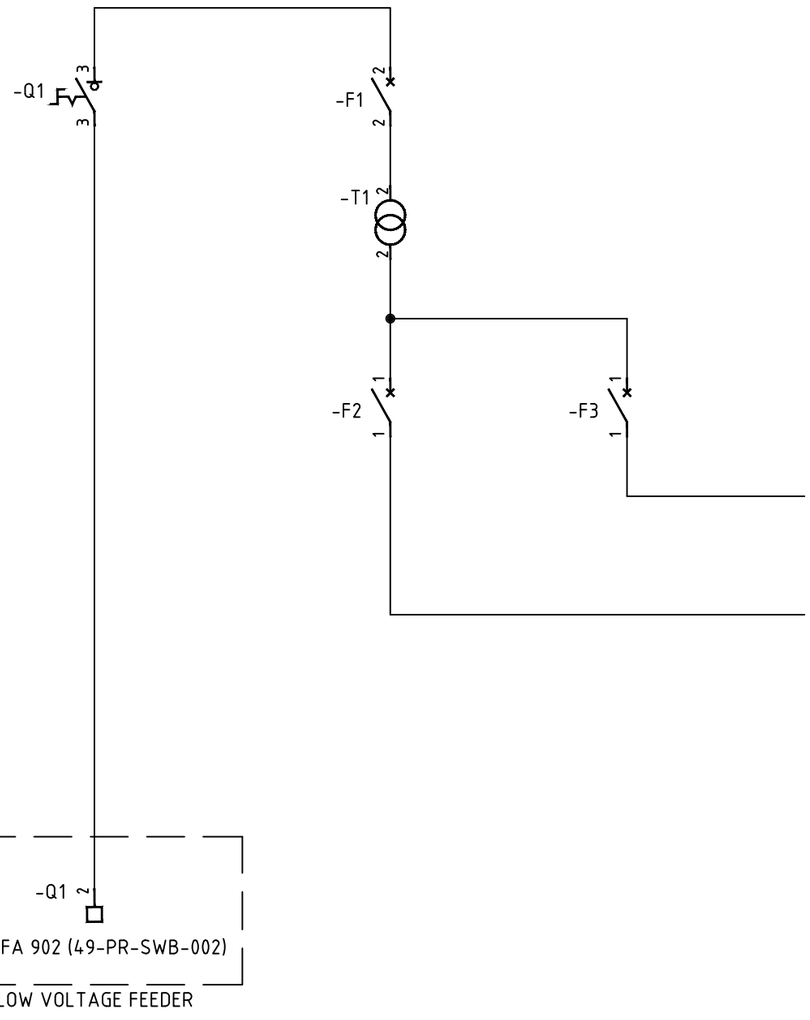
Project name: Mansourah - Massarah Gold Mine
Project id: P/20002
Product id: 6X20V32TS
Based on document id: DAAB198406
Customer document id: -

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DIMENSIONAL DRAWING
OILY WATER TRANSFER UNIT
LOCAL CONTROL PANEL
LAYOUT 1:10

Created by: MBAX46	Create date: 8 Jun 2021	Language: EN	Object designation: BJU906
Reviewed by: RSW004	Review date: 8 Jun 2021	Sheet size: A3	Page: 2 / 10
Approved by: KSU004	Approval date: 8 Jun 2021	Document Id: DECA00006012	Rev.: C

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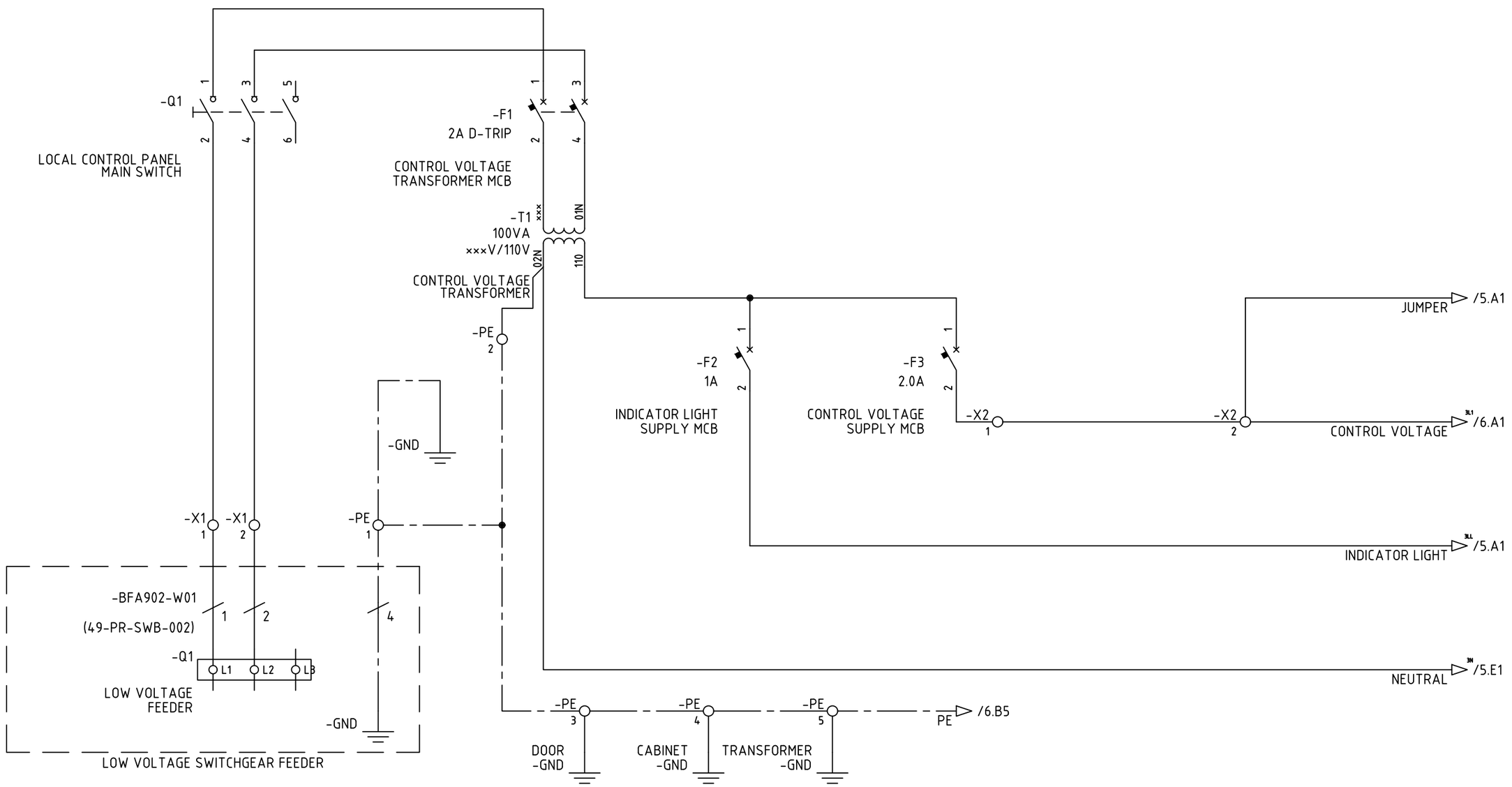


Rev.	Date	Created by	Appr. by	Revision text
a	20.7.2021	RSW004	KSU004	Location Box Updated

Project name: Mansourah - Massarah Gold Mine			OVERVIEW DIAGRAM	Created by: MBAX46	Create date: 8 Jun 2021
Project id: P/20002	Product id: 6X20V32TS	WÄRTSILÄ	OILY WATER TRANSFER UNIT LOCAL CONTROL PANEL SINGLE LINE	Reviewed by: RSW004	Review date: 8 Jun 2021
Based on document id: DAAB198406	Customer document id: -			Approved by: KSU004	Approval date: 8 Jun 2021

Language	Object designation	Sheet size	Page	Document Id:	Rev.
EN	BJU906	A3	3 / 10	DECA00006012	C

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SUPPLY					
Un	400	V	2-phase	60	Hz
In	1	A		10	A

Rev.	Date	Created by	Appr. by	Revision text
a	20.7.2021	RSW004	KSU004	Location Box Updated

Project name	Mansourah - Massarah Gold Mine		
Project id	P/20002	Product id	6X20V32TS
Based on document id	DAAB198406	Customer document id	-

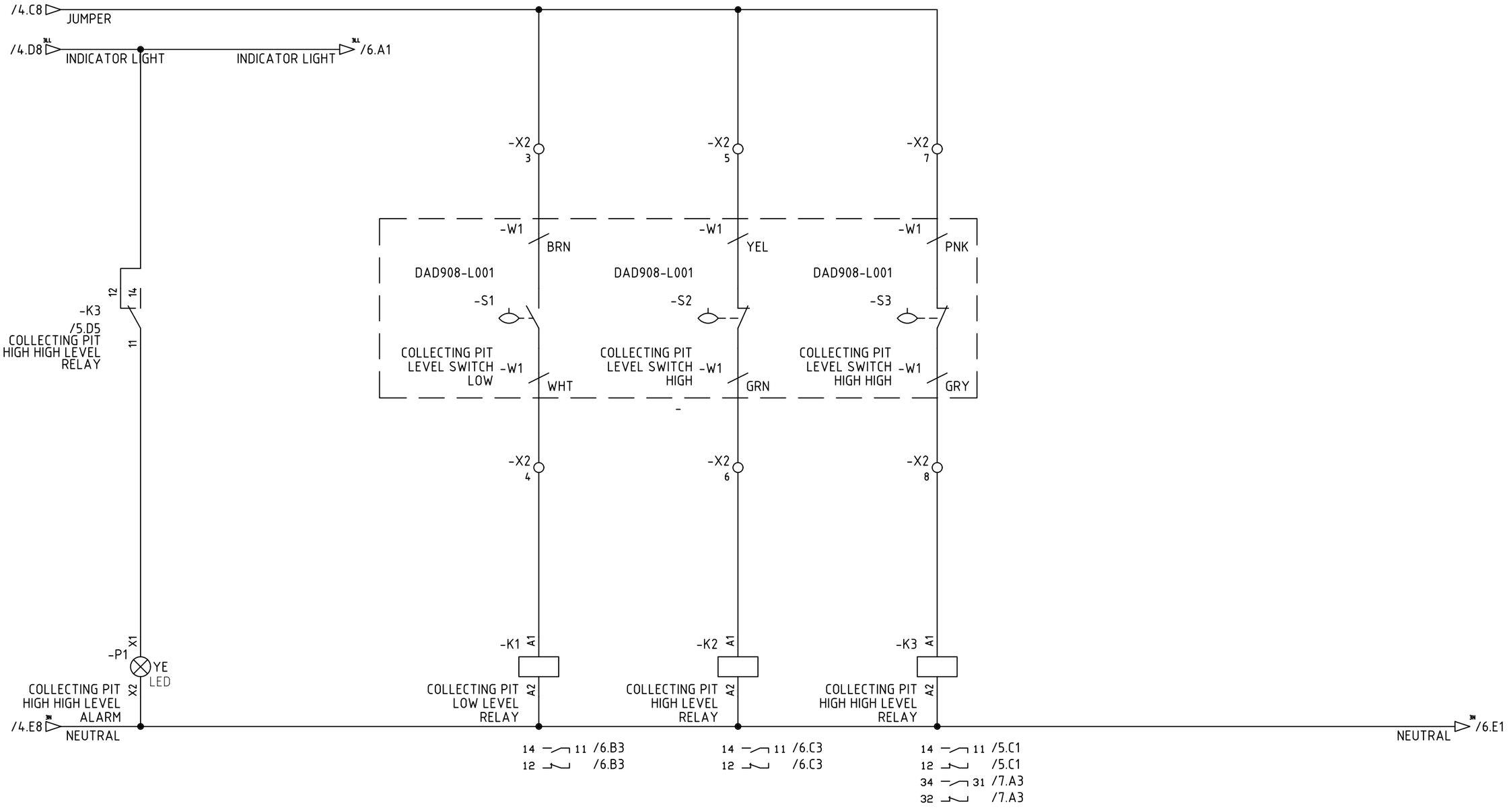
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CIRCUIT DIAGRAM

OILY WATER TRANSFER UNIT
LOCAL CONTROL PANEL
LOW VOLTAGE FEEDER

Created by	MBAX46	Create date	8 Jun 2021
Reviewed by	RSW004	Review date	8 Jun 2021
Approved by	KSU004	Approval date	8 Jun 2021
Language	EN	Object designation	BJU906
Sheet size	A3	Page	4 / 10
Document Id:	DECA00006012	Rev.	b

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Rev.	Date	Created by	Appr. by	Revision text
a	20.7.2021	RSW004	KSU004	Location Box Updated

Project name: Mansourah - Massarah Gold Mine			
Project id: P/20002	Product id: 6X20V32TS		
Based on document id: DAAB198406	Customer document id: -		

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CIRCUIT DIAGRAM	Created by: MBAX46	Create date: 8 Jun 2021	Language: EN	Object designation: BJU906	
OILY WATER TRANSFER UNIT LOCAL CONTROL PANEL COMMON INDICATOR/CONTROL	Reviewed by: RSW004	Review date: 8 Jun 2021	Sheet size: A3	Page: 5 / 10	
	Approved by: KSU004	Approval date: 8 Jun 2021	Document Id: DECA00006012		Rev.: C

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/4.D8 3L1
CONTROL VOLTAGE

/5.A2 3L1
INDICATOR LIGHT

-K4
/6.D3
TRANSFER PUMP
CONTROL
RELAY 1
12 14
11

-K1
/5.D3
COLLECTING PIT
LOW LEVEL
RELAY
12 14
11

-S1
0-1-S
21 22
TRANSFER PUMP
CONTROL
0-1-S

-S1
13 14
/6.B4
TRANSFER PUMP
CONTROL
0-1-S

-S1					
0	1	S	T1	T2	REF
X			21	22	B4
			X1	X2	6E1
		X	13	14	6C4

-K2
/5.D4
COLLECTING PIT
HIGH LEVEL
RELAY
12 14
11

-K4
/6.D3
TRANSFER PUMP
CONTROL
RELAY 1
22 24
21

-K5
/6.D4
TRANSFER PUMP
CONTROL
RELAY2
22 24
21

/4.E6
PE

-PE
6

-X2
9

-W01
3

-W01
1

-GND

DAD908-CV001

-Y1
X1
X2
TRANSFER PUMP
CONTROL VALVE

-W01
2

-X2
10

-S1
X1
/6.B4
WH
LED
X2
TRANSFER PUMP

/5.E8 3W
0-1-S
NEUTRAL

-K4
A1
A2
TRANSFER PUMP
CONTROL
RELAY 1

-K5
A1
A2
TRANSFER PUMP
CONTROL
RELAY2

14 — 11 /6.C1
12 — /6.C1
24 — 21 /6.C3
22 — /6.C3

24 — 21 /6.A6
22 — /6.A6
34 — 31 /7.A4
32 — /7.A4

Rev.	Date	Created by	Appr. by	Revision text
a	20.7.2021	RSW004	KSU004	Location Box Updated

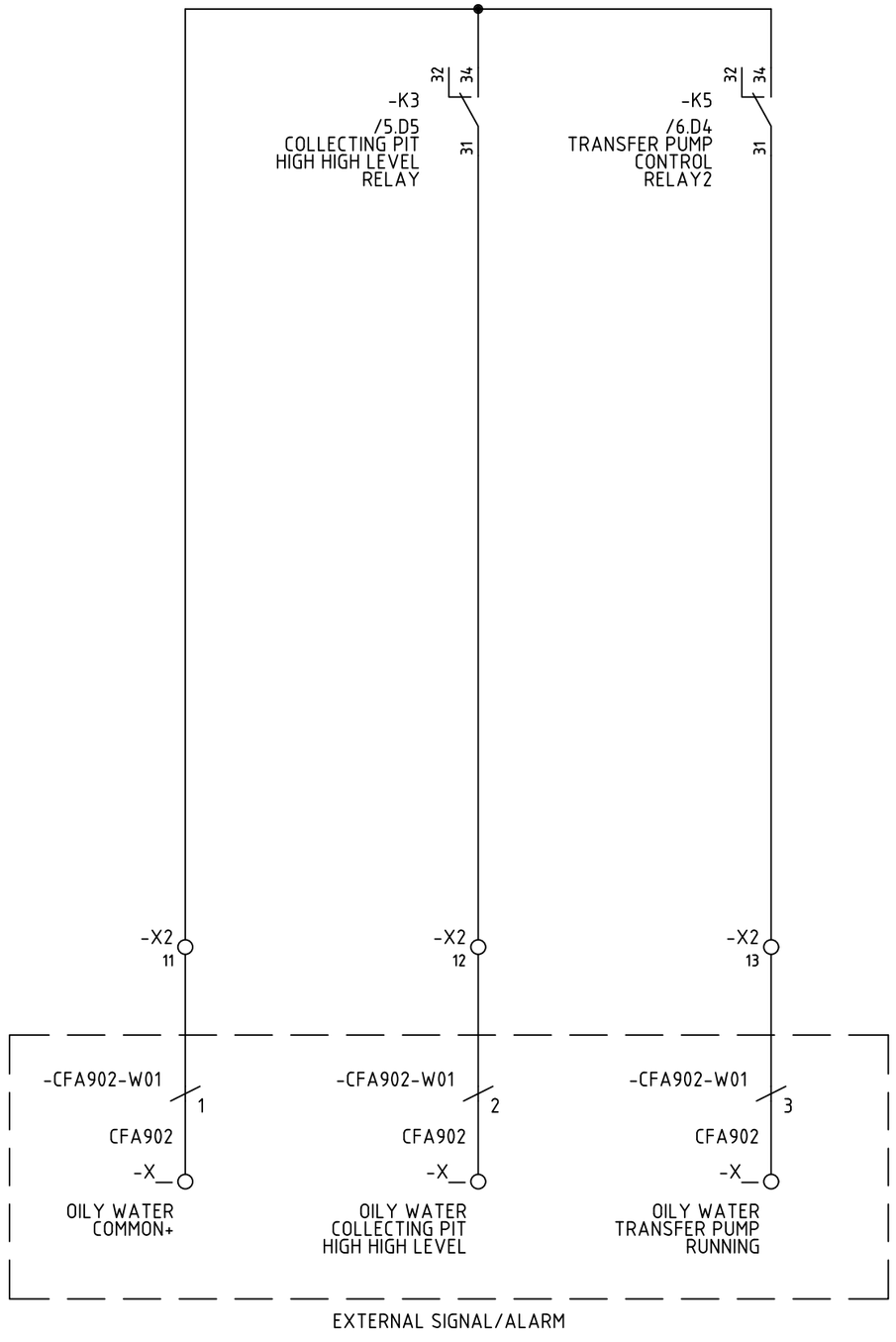
Project name: Mansourah - Massarah Gold Mine
Project id: P/20002
Product id: 6X20V32TS
Based on document id: DAAB198406
Customer document id: -

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CIRCUIT DIAGRAM
OILY WATER TRANSFER UNIT
LOCAL CONTROL PANEL
PUMP CONTROL

Created by: MBAX46	Create date: 8 Jun 2021	Language: EN	Object designation: BJU906
Reviewed by: RSW004	Review date: 8 Jun 2021	Sheet size: A3	Page: 6 / 10
Approved by: KSU004	Approval date: 8 Jun 2021	Document Id: DECA00006012	Rev.: C

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Rev.	Date	Created by	Appr. by	Revision text
a	20.7.2021	RSW004	KSU004	Location Box Updated

Project name	Mansourah - Massarah Gold Mine			
Project id	P/20002	Product id	6X20V32TS	
Based on document id	DAAB198406	Customer document id	-	

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CIRCUIT DIAGRAM	Created by MBAX46	Create date 8 Jun 2021	Language EN	Object designation BJU906			
OILY WATER TRANSFER UNIT LOCAL CONTROL PANEL EXTERNAL ALARM	Reviewed by RSW004	Review date 8 Jun 2021	Sheet size A3	Page 7 / 10			
	Approved by KSU004	Approval date 8 Jun 2021	Document Id: DECA00006012	Rev. C			

FACTORY INSTALLED CABLES TO BE SUPPLIED BY HELKAMA IF NOT OTHERWISE STATED

CABLE SPECIFICATION	CABLE TYPE	CABLE INSTALLATION	CABLE NAME	WIRE	FROM COMPONENT	PIN1	TO COMPONENT	PIN2	SH1	REF1	CABLE REMARK
1kV YY	3G1,5rm		-W01	1	-X2	9	=DAD908-CV001-Y1	X1	6	B6	INSTALLED AT FSITE
			-W01	2	-X2	10	=DAD908-CV001-Y1	X2	6	C6	INSTALLED AT FSITE
			-W01	3	-PE	6	=DAD908-CV001-GND		6	B6	INSTALLED AT FSITE
1kV YCY	3x__+__	BFA902	-BFA902-W01	1	=BFA902-Q1	L1	-X1	1	4	E2	INSTALLED AT SITE
		BFA902	-BFA902-W01	2	=BFA902-Q1	L2	-X1	2	4	E2	INSTALLED AT SITE
		BFA902	-BFA902-W01	4	=BFA902-GND		-PE	1	4	E3	INSTALLED AT SITE
1kV YY	4x1,5rm	CFA902	-CFA902-W01	1	=CFA902-X__		-X2	11	7	E2	INSTALLED AT SITE
		CFA902	-CFA902-W01	2	=CFA902-X__		-X2	12	7	E3	INSTALLED AT SITE
		CFA902	-CFA902-W01	3	=CFA902-X__		-X2	13	7	E4	INSTALLED AT SITE
		DAD908-L001	-W1	BRN	=DAD908-L001-S1		-X2	3	5	C3	INSTALLED AT SITE (INTEGRATED SILICON CABLE)
		DAD908-L001	-W1	GRN	=DAD908-L001-S2		-X2	6	5	C4	INSTALLED AT SITE (INTEGRATED SILICON CABLE)
		DAD908-L001	-W1	GRY	=DAD908-L001-S3		-X2	8	5	C5	INSTALLED AT SITE (INTEGRATED SILICON CABLE)
		DAD908-L001	-W1	PNK	=DAD908-L001-S3		-X2	7	5	C5	INSTALLED AT SITE (INTEGRATED SILICON CABLE)
		DAD908-L001	-W1	WHT	=DAD908-L001-S1		-X2	4	5	C3	INSTALLED AT SITE (INTEGRATED SILICON CABLE)
		DAD908-L001	-W1	YEL	=DAD908-L001-S2		-X2	5	5	C4	INSTALLED AT SITE (INTEGRATED SILICON CABLE)

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Rev.	Date	Created by	Appr. by	Revision text
a	20.7.2021	RSW004	KSU004	Location Box Updated

Project name: Mansourah - Massarah Gold Mine
Project id: P/20002
Product id: 6X20V32TS
Based on document id: DAAB198406
Customer document id: -

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CONNECTION TABLE
OILY WATER TRANSFER UNIT
LOCAL CONTROL PANEL
CABLE LIST

Created by	Create date	Language	Object designation
MBAX46	8 Jun 2021	EN	BJU906
Reviewed by	**Review date**	**Sheet size**	**Page**
RSW004	8 Jun 2021	A3	8 / 10
Approved by	**Approval date**	**Document Id:**	**Rev.**
KSU004	8 Jun 2021	DECA00006012	C

SH1	REF1	CMP1	PIN1	WLAY1	CMP2	PIN2	SH2	REF2
4	C4	-T1	110	AC CONTROL	-F2	1	4	C5
4	C4	-T1	02N	AC CONTROL	-K1	A2	5	E3
4	C5	-F2	1	AC CONTROL	-F3	1	4	C6
4	C5	-F2	2	AC CONTROL	-K3	12	5	C1
4	C6	-F3	2	AC CONTROL	-X2	1	4	D6
4	D6	-X2	1	AC CONTROL	-X2	2	4	D7
4	D7	-X2	2	AC CONTROL	-K1	14	6	B3
5	C1	-K3	11	AC CONTROL	-P1	X1	5	D1
5	C1	-K3	12	AC CONTROL	-K4	14	6	C1
5	E1	-P1	X2	AC CONTROL	-S1	X2	6	E1
5	C3	-X2	4	AC CONTROL	-K1	A1	5	D3
5	E3	-K1	A2	AC CONTROL	-K2	A2	5	E4
5	C4	-X2	6	AC CONTROL	-K2	A1	5	D4
5	E4	-K2	A2	AC CONTROL	-K3	A2	5	E5
5	C5	-X2	8	AC CONTROL	-K3	A1	5	D5
5	E5	-K3	A2	AC CONTROL	-K4	A2	6	E3
6	C1	-K4	11	AC CONTROL	-S1	X1	6	D1
6	B3	-K1	11	AC CONTROL	-S1	21	6	C3
6	B3	-K1	14	AC CONTROL	-K5	24	6	A6
6	C3	-K2	12	AC CONTROL	-K4	24	6	C3
6	C3	-K4	24	AC CONTROL	-S1	22	6	C3
6	D3	-K2	11	AC CONTROL	-K4	21	6	D3
6	D3	-K4	21	AC CONTROL	-K4	A1	6	D3
6	D3	-K4	A1	AC CONTROL	-K5	A1	6	D4
6	E3	-K4	A2	AC CONTROL	-K5	A2	6	E4
6	D4	-K5	A1	AC CONTROL	-S1	14	6	C4
6	E4	-K5	A2	AC CONTROL	-X2	10	6	C6
6	A6	-K5	21	AC CONTROL	-X2	9	6	B6
6	A6	-K5	24	AC CONTROL	-S1	13	6	C4
6	C6	-X2	10	AC CONTROL	-P1	X2	5	E1
7	D2	-X2	11	AC CONTROL	-K3	34	7	A3
7	A3	-K3	34	AC CONTROL	-K5	34	7	A4
7	A3	-K3	31	AC CONTROL	-X2	12	7	D3
7	A4	-K5	31	AC CONTROL	-X2	13	7	D4
4	D7	-X2	2	AC CONTROL JMP	-X2	5	5	B4
5	B3	-X2	3	AC CONTROL JMP	-X2	2	4	D7
5	B4	-X2	5	AC CONTROL JMP	-X2	7	5	B5
4	B2	-Q1	2	AC POWER	-X1	1	4	D2
4	B2	-Q1	1	AC POWER	-F1	1	4	B4
4	B2	-Q1	4	AC POWER	-X1	2	4	D2
4	B2	-Q1	3	AC POWER	-F1	3	4	B4
4	B4	-F1	2	AC POWER	-T1	xxx	4	C4
4	B4	-F1	4	AC POWER	-T1	01N	4	C4
5	C3	-S1		EXTERNAL CONTROL (INTERLOCK)	-X2	3	5	B3
5	C3	-S1		EXTERNAL CONTROL (INTERLOCK)	-X2	4	5	C3
5	C4	-S2		EXTERNAL CONTROL (INTERLOCK)	-X2	5	5	B4
5	C4	-S2		EXTERNAL CONTROL (INTERLOCK)	-X2	6	5	C4
5	C5	-S3		EXTERNAL CONTROL (INTERLOCK)	-X2	7	5	B5
5	C5	-S3		EXTERNAL CONTROL (INTERLOCK)	-X2	8	5	C5
6	B6	-X2	9	EXTERNAL CONTROL (INTERLOCK)	-Y1	X1	6	C6
6	C6	-X2	10	EXTERNAL CONTROL (INTERLOCK)	-Y1	X2	6	C6
7	E2	-X_		EXTERNAL CONTROL (INTERLOCK)	-X2	11	7	D2
7	E3	-X_		EXTERNAL CONTROL (INTERLOCK)	-X2	12	7	D3
7	E4	-X_		EXTERNAL CONTROL (INTERLOCK)	-X2	13	7	D4
4	E2	-Q1	L1	EXTERNAL POWER	-X1	1	4	D2
4	E2	-Q1	L2	EXTERNAL POWER	-X1	2	4	D2
4	E3	-GND		EXTERNAL PROTECTIVE	-PE	1	4	D3
6	B6	-PE	6	EXTERNAL PROTECTIVE	-GND		6	B6
4	D3	-PE	1	PROTECTIVE	-GND		4	D3
4	C4	-T1	02N	PROTECTIVE	-PE	2	4	C3

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Rev.	Date	Created by	Appr. by	Revision text
a	20.7.2021	RSW004	KSU004	Location Box Updated

Project name: Mansourah - Massarah Gold Mine
Project id: P/20002
Product id: 6X20V32TS
Based on document id: DAAB198406
Customer document id: -

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CONNECTION TABLE
OILY WATER TRANSFER UNIT
LOCAL CONTROL PANEL
INTERNAL CONNECTION LIST

Created by: MBAX46	Create date: 8 Jun 2021	Language: EN	Object designation: BJU906
Reviewed by: RSW004	Review date: 8 Jun 2021	Sheet size: A3	Page: 9 / 10
Approved by: KSU004	Approval date: 8 Jun 2021	Document Id: DECA00006012	Rev.: C

COMPONENT NAME	QTY	MANUFACTURER	PART CODE	RATING	DESC
-E01	1	RITTAL	1038000		ENCLOSURE, 380x600x210
-E01	4	RITTAL	2508020		WALL MOUNTING BRACKET
-F1	1	SCHNEIDER ELECTRIC	A9F05202	2A D-TRIP	MINIATURE CIRCUIT BREAKER
-F2	1	SCHNEIDER ELECTRIC	A9F04101	1A	MINIATURE CIRCUIT BREAKER
-F3	1	SCHNEIDER ELECTRIC	A9F04102	2.0A	MINIATURE CIRCUIT BREAKER
-K1	1	CARLO GAVAZZI	RCP11003115120VAC		UNIVERSAL PLUG-IN RELAY, 3 CO
-K1	1	CARLO GAVAZZI	ZPD12A		SOCKET
-K2	1	CARLO GAVAZZI	RCP11003115120VAC		UNIVERSAL PLUG-IN RELAY, 3 CO
-K2	1	CARLO GAVAZZI	ZPD12A		SOCKET
-K3	1	CARLO GAVAZZI	RCP11003115120VAC		UNIVERSAL PLUG-IN RELAY, 3 CO
-K3	1	CARLO GAVAZZI	ZPD12A		SOCKET
-K4	1	CARLO GAVAZZI	RCP11003115120VAC		UNIVERSAL PLUG-IN RELAY, 3 CO
-K4	1	CARLO GAVAZZI	ZPD12A		SOCKET
-K5	1	CARLO GAVAZZI	RCP11003115120VAC		UNIVERSAL PLUG-IN RELAY, 3 CO
-K5	1	CARLO GAVAZZI	ZPD12A		SOCKET
-P1	1	SCHNEIDER ELECTRIC	XB7EV05GP		MONOLITHIC INDICATOR LIGHT, YELLOW, INTEGRAL LED
-PE	2	PHOENIX CONTACT	0404428	AB/SS	END PIECE (FOR PE TERMINAL BLOCK)
-PE	15	PHOENIX CONTACT	0421029		PE TERMINAL BLOCK
-Q1	1	ABB	OT16FT3+OHYS2PJ		MAIN BREAKER WITH IP65 HANDLE
-S1	1	SCHNEIDER ELECTRIC	ZB5AK1813+ZB5AW0G15		CONTROL SWITCH, 3 POSITION
-T1	1	TRAFOMIC	0423192	100VA	CONTROL VOLTAGE TRANSFORMER
-U1	1	AGRO	60010031	M20	CABLE GLAND, M20
-U2	1	AGRO	60010031	M20	CABLE GLAND, M20
-U3	1	AGRO	60010031	M20	CABLE GLAND, M20
-U4	1	AGRO	60010031	M20	CABLE GLAND, M20
-U5	1	AGRO	60010031	M20	CABLE GLAND, M20
-U6	1	AGRO	60010031	M20	CABLE GLAND, M20
-X1	2	WEIDMULLER	1608830000	ZAP/TW5	END PIECE (FOR TERMINAL BLOCK)
-X1	3	WEIDMULLER	1608620000	ZDU 6	TERMINAL BLOCK, FEED-THROUGH, 6MM2
-X2	2	WEIDMULLER	1632090000	ZAP/TW4	END PIECE (FOR TERMINAL BLOCK)
-X2	20	WEIDMULLER	1632050000	ZDU 4	TERMINAL BLOCK, FEED-THROUGH, 4MM2

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Rev.	Date	Created by	Appr. by	Revision text
b	06.04.2022	MBA123	RSW004	Part list updated
a	20.7.2021	RSW004	KSU004	Location Box Updated

Project name: Mansourah - Massarah Gold Mine
Project id: P/20002
Product id: 6X20V32TS
Based on document id: DAAB198406
Customer document id: -

WÄRTSILÄ

PARTS LIST

OILY WATER TRANSFER UNIT
LOCAL CONTROL PANEL
PARTS LIST

Created by: MBAX46	Create date: 8 Jun 2021	Language: EN	Object designation: BJU906
Reviewed by: RSW004	Review date: 8 Jun 2021	Sheet size: A3	Page: 10 / 10
Approved by: KSU004	Approval date: 8 Jun 2021	Document Id: DECA00006012	Rev.: C

PANEL CODE	DOCUMENT NAME	DOCUMENT DESCRIPTION	SHEET N:o	DOCUMENT N:o	REVISION
BJU907	LIST OF DOCUMENTS	DRAWING LIST	1	DECA00006021	b
BJU907	DIMENSIONAL DRAWING	LAYOUT 1:10	2	DECA00006021	b
BJU907	OVERVIEW DIAGRAM	SINGLE LINE	3	DECA00006021	b
BJU907	CIRCUIT DIAGRAM	LOW VOLTAGE FEEDER	4	DECA00006021	b
BJU907	CIRCUIT DIAGRAM	COMMON INDICATOR/CONTROL	5	DECA00006021	b
BJU907	CIRCUIT DIAGRAM	PUMP CONTROL	6	DECA00006021	b
BJU907	CIRCUIT DIAGRAM	EXTERNAL ALARM	7	DECA00006021	b
BJU907	CONNECTION TABLE	CABLE LIST	8	DECA00006021	b
BJU907	CONNECTION TABLE	INTERNAL CONNECTION LIST	9	DECA00006021	b
BJU907	PARTS LIST	PARTS LIST	10	DECA00006021	b

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Rev.	Date	Created by	Appr. by	Revision text
a	20.7.2021	RSW004	KSU004	Location Boxes updated

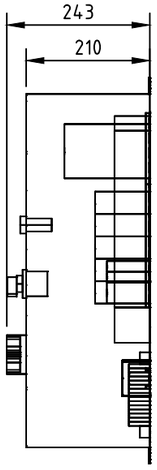
Project name: Mansourah - Massarah Gold Mine
Project id: P/20002
Product id: 6X20V32TS
Based on document id: DAAB198406
Customer document id: -

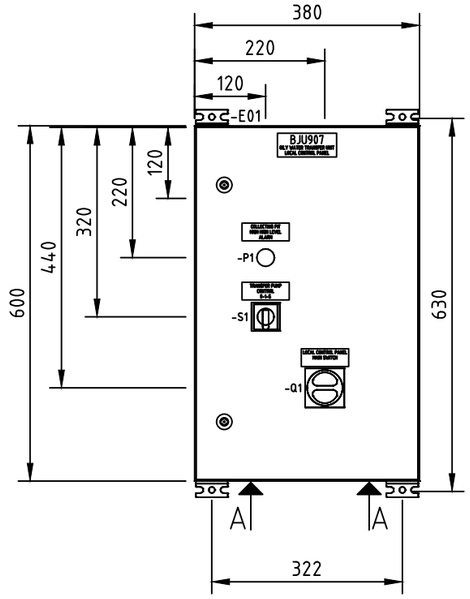
WÄRTSILÄ

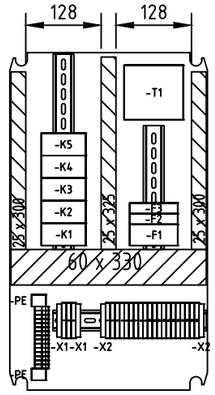
LIST OF DOCUMENTS
OILY WATER TRANSFER UNIT
LOCAL CONTROL PANEL
DRAWING LIST

Created by	Create date	Language	Object designation
MBAX46	8 Jun 2021	EN	BJU907
Reviewed by	**Review date**	**Sheet size**	**Page**
RSW004	8 Jun 2021	A3	1 / 10
Approved by	**Approval date**	**Document Id:**	**Rev.**
KSU004	8 Jun 2021	DECA00006021	C

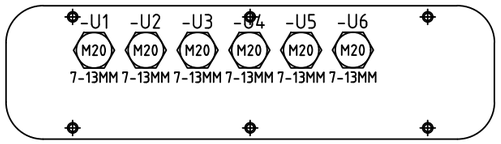
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SECTION A-A SCALE 1:4



PANEL INFORMATION					
WIDTH	380	mm	PROTECTION CLASS		IP65
HEIGHT	600	mm	COLOR		RAL 7035
DEPTH	210	mm	Ik	6	kA/1s

-E01	BJU907	OILY WATER TRANSFER UNIT	LOCAL CONTROL PANEL
-P1	COLLECTING PIT	HIGH HIGH LEVEL	ALARM
-Q1	LOCAL CONTROL PANEL	MAIN SWITCH	
-S1	TRANSFER PUMP	CONTROL	0-1-S

Rev.	Date	Created by	Appr. by	Revision text
a	20.7.2021	RSW004	KSU004	Location Boxes updated

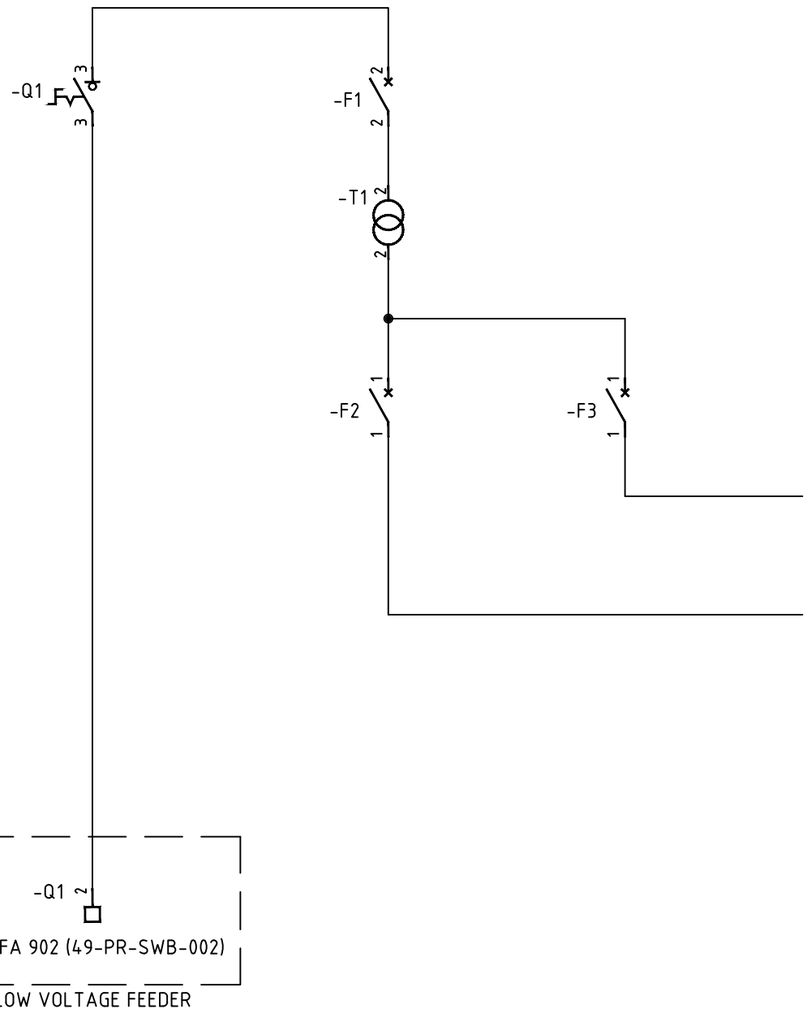
Project name: Mansourah - Massarah Gold Mine
Project id: P/20002
Product id: 6X20V32TS
Based on document id: DAAB198406
Customer document id: -

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DIMENSIONAL DRAWING
OILY WATER TRANSFER UNIT
LOCAL CONTROL PANEL
LAYOUT 1:10

Created by: MBAX46	Create date: 8 Jun 2021	Language: EN	Object designation: BJU907
Reviewed by: RSW004	Review date: 8 Jun 2021	Sheet size: A3	Page: 2 / 10
Approved by: KSU004	Approval date: 8 Jun 2021	Document Id: DECA00006021	Rev.: C

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Rev.	Date	Created by	Appr. by	Revision text
a	20.7.2021	RSW004	KSU004	Location Boxes updated

Project name: Mansourah - Massarah Gold Mine			
Project id: P/20002	Product id: 6X20V32TS		
Based on document id: DAAB198406	Customer document id: -		

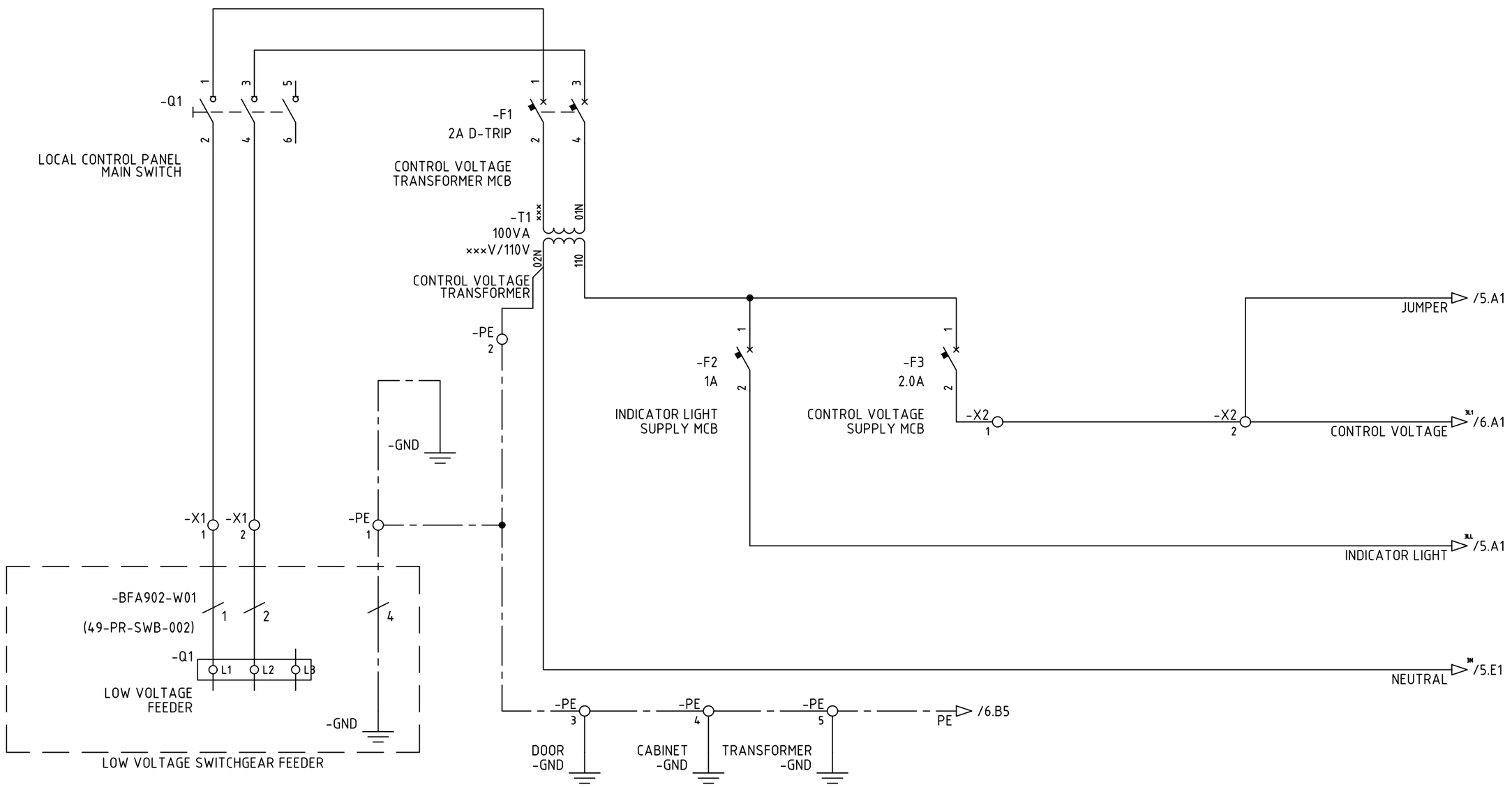
WÄRTSILÄ

OVERVIEW DIAGRAM

OILY WATER TRANSFER UNIT
LOCAL CONTROL PANEL
SINGLE LINE

Created by: MBAX46	Create date: 8 Jun 2021	Language: EN	Object designation: BJU907
Reviewed by: RSW004	Review date: 8 Jun 2021	Sheet size: A3	Page: 3 / 10
Approved by: KSU004	Approval date: 8 Jun 2021	Document Id: DECA0006021	Rev.: C

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SUPPLY					
Un	400	V	2-phase	60	Hz
In	1	A		10	A

Rev.	Date	Created by	Appr. by	Revision text
a	20.7.2021	RSW004	KSU004	Location Boxes updated

Project name	Mansourah - Massarah Gold Mine		
Project id	P/20002	Product id	6X20V32TS
Based on document id	DAAB198406	Customer document id	-

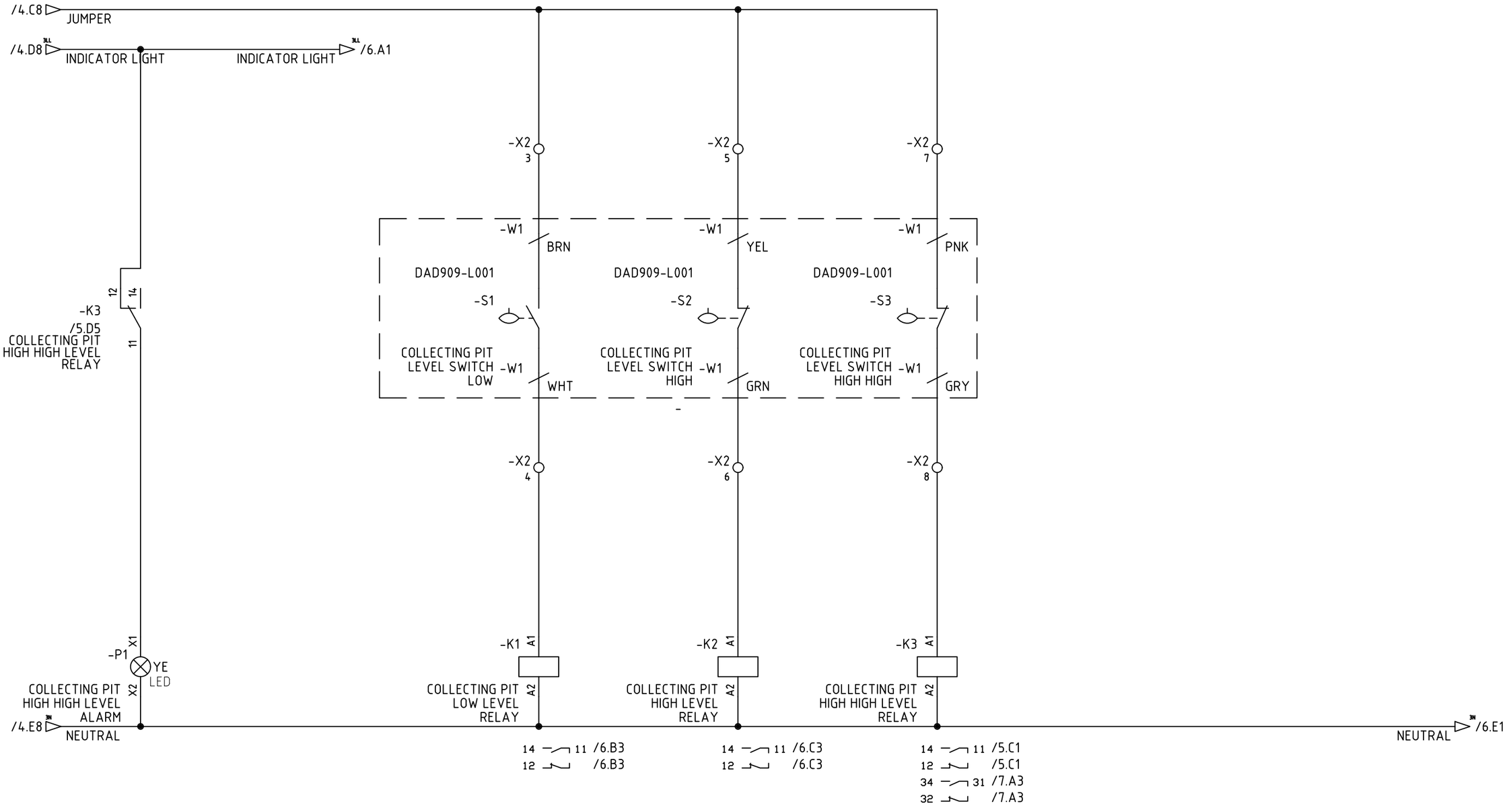
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CIRCUIT DIAGRAM

OILY WATER TRANSFER UNIT
LOCAL CONTROL PANEL
LOW VOLTAGE FEEDER

Created by	MBAX46	Create date	8 Jun 2021
Reviewed by	RSW004	Review date	8 Jun 2021
Approved by	KSU004	Approval date	8 Jun 2021
Language	EN	Object designation	BJU907
Sheet size	A3	Page	4 / 10
Document Id:	DECA00006021	Rev.	C

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Rev.	Date	Created by	Appr. by	Revision text
a	20.7.2021	RSW004	KSU004	Location Boxes updated

Project name	Mansourah - Massarah Gold Mine		
Project id	P/20002	Product id	6X20V32TS
Based on document id	DAAB198406	Customer document id	-

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CIRCUIT DIAGRAM

OILY WATER TRANSFER UNIT
LOCAL CONTROL PANEL
COMMON INDICATOR/CONTROL

Created by	MBAX46	Create date	8 Jun 2021
Reviewed by	RSW004	Review date	8 Jun 2021
Approved by	KSU004	Approval date	8 Jun 2021
Language	EN	Object designation	BJU907
Sheet size	A3	Page	5 / 10
Document Id:	DECA00006021	Rev.	C

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/4.D8 3L1 CONTROL VOLTAGE

/5.A2 3L1 INDICATOR LIGHT

-K4 /6.D3 TRANSFER PUMP CONTROL RELAY 1 — 12 14 / 11

-K1 /5.D3 COLLECTING PIT LOW LEVEL RELAY — 12 14 / 11

-S1 0-1-S — 21 / 22 — TRANSFER PUMP CONTROL 0-1-S

-S1 — 13 / 14 — /6.B4 TRANSFER PUMP CONTROL 0-1-S

0	1	S	-S1		
			T1	T2	REF
X			21	22	B4
			X1	X2	6E1
		X	13	14	6C4

-K2 /5.D4 COLLECTING PIT HIGH LEVEL RELAY — 12 14 / 11

-K4 /6.D3 TRANSFER PUMP CONTROL RELAY 1 — 22 24 / 21

-K5 /6.D4 TRANSFER PUMP CONTROL RELAY2 — 22 24 / 21

/4.E6 PE

-PE 6 — -X2 9

-W01 3 — -W01 1

-GND

DAD909-CV001

-Y1 X1 / X2 — TRANSFER PUMP CONTROL VALVE

-W01 2

-X2 10

-S1 X1 — /6.B4 WH LED — X2 TRANSFER PUMP

/5.E8 3N 0-1-S NEUTRAL

-K4 A1 — TRANSFER PUMP CONTROL RELAY 1 A2

-K5 A1 — TRANSFER PUMP CONTROL RELAY2 A2

14 — 11 /6.C1
12 — /6.C1
24 — 21 /6.C3
22 — /6.C3

24 — 21 /6.A6
22 — /6.A6
34 — 31 /7.A4
32 — /7.A4

Rev.	Date	Created by	Appr. by	Revision text
a	20.7.2021	RSW004	KSU004	Location Boxes updated

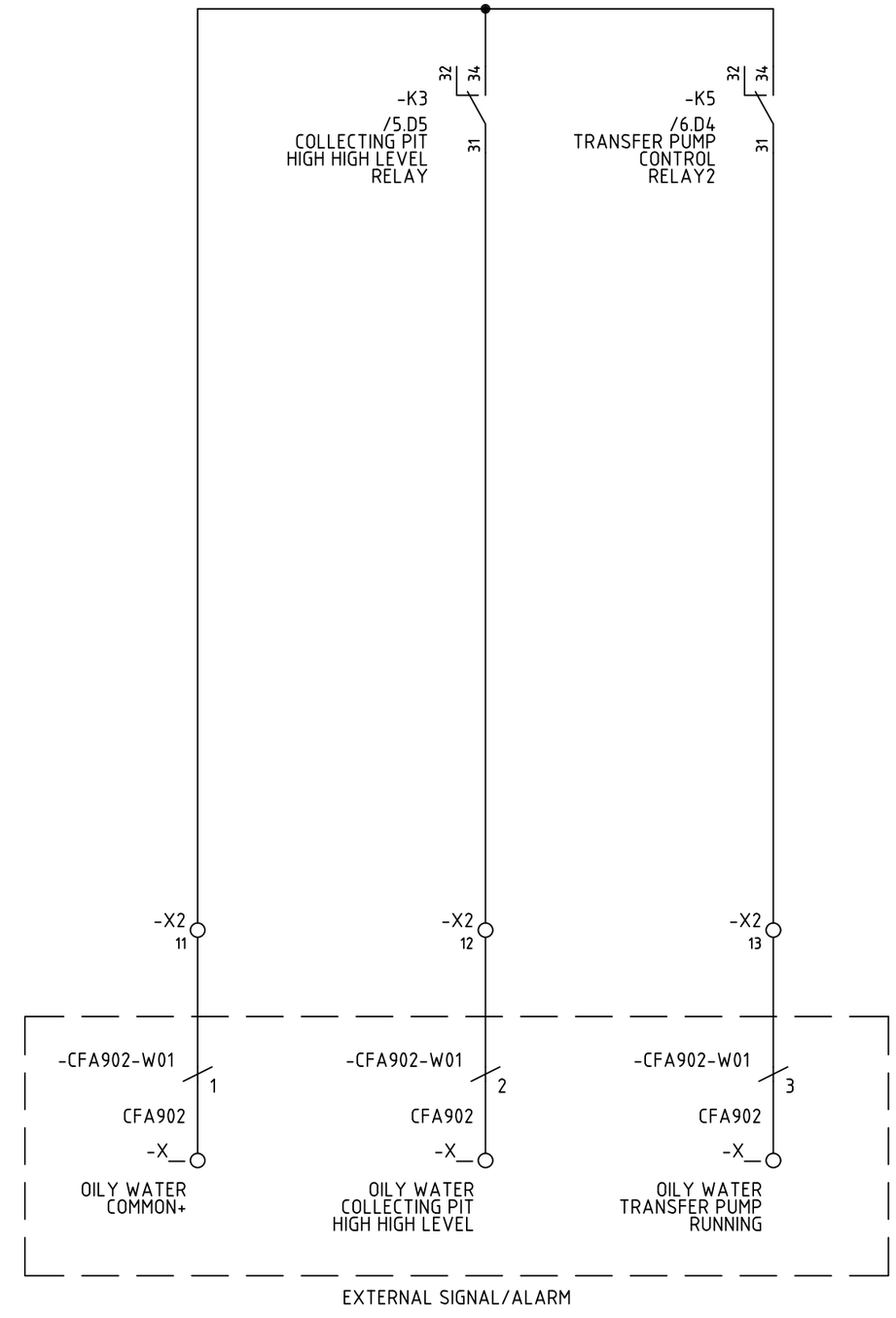
Project name: Mansourah - Massarah Gold Mine
Project id: P/20002
Product id: 6X20V32TS
Based on document id: DAAB198406
Customer document id: -

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CIRCUIT DIAGRAM
OILY WATER TRANSFER UNIT
LOCAL CONTROL PANEL
PUMP CONTROL

Created by: MBAX46	Create date: 8 Jun 2021	Language: EN	Object designation: BJU907
Reviewed by: RSW004	Review date: 8 Jun 2021	Sheet size: A3	Page: 6 / 10
Approved by: KSU004	Approval date: 8 Jun 2021	Document Id: DECA00006021	Rev.: C

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Rev.	Date	Created by	Appr. by	Revision text
a	20.7.2021	RSW004	KSU004	Location Boxes updated

Project name: Mansourah - Massarah Gold Mine			
Project id: P/20002	Product id: 6X20V32TS		
Based on document id: DAAB198406	Customer document id: -		

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CIRCUIT DIAGRAM	Created by: MBAX46	Create date: 8 Jun 2021	Language: EN	Object designation: BJU907			
OILY WATER TRANSFER UNIT LOCAL CONTROL PANEL EXTERNAL ALARM	Reviewed by: RSW004	Review date: 8 Jun 2021	Sheet size: A3	Page: 7 / 10			
	Approved by: KSU004	Approval date: 8 Jun 2021	Document Id: DECA0006021	Rev.: C			

FACTORY INSTALLED CABLES TO BE SUPPLIED BY HELKAMA IF NOT OTHERWISE STATED

CABLE SPECIFICATION	CABLE TYPE	CABLE INSTALLATION	CABLE NAME	WIRE	FROM COMPONENT	PIN1	TO COMPONENT	PIN2	SH1	REF1	CABLE REMARK
1kV YY	3G1,5rm		-W01	1	-X2	9	=DAD909-CV001-Y1	X1	6	B6	INSTALLED AT FSITE
			-W01	2	-X2	10	=DAD909-CV001-Y1	X2	6	C6	INSTALLED AT FSITE
			-W01	3	-PE	6	=DAD909-CV001-GND		6	B6	INSTALLED AT FSITE
1kV YCY	3x__+__	BFA902	-BFA902-W01	1	=BFA902-Q1	L1	-X1	1	4	E2	INSTALLED AT SITE
		BFA902	-BFA902-W01	2	=BFA902-Q1	L2	-X1	2	4	E2	INSTALLED AT SITE
		BFA902	-BFA902-W01	4	=BFA902-GND		-PE	1	4	E3	INSTALLED AT SITE
1kV YY	4x1,5rm	CFA902	-CFA902-W01	1	=CFA902-X__		-X2	11	7	E2	INSTALLED AT SITE
		CFA902	-CFA902-W01	2	=CFA902-X__		-X2	12	7	E3	INSTALLED AT SITE
		CFA902	-CFA902-W01	3	=CFA902-X__		-X2	13	7	E4	INSTALLED AT SITE
		DAD909-L001	-W1	BRN	=DAD909-L001-S1		-X2	3	5	C3	INSTALLED AT SITE (INTEGRATED SILICON CABLE)
		DAD909-L001	-W1	GRN	=DAD909-L001-S2		-X2	6	5	C4	INSTALLED AT SITE (INTEGRATED SILICON CABLE)
		DAD909-L001	-W1	GRY	=DAD909-L001-S3		-X2	8	5	C5	INSTALLED AT SITE (INTEGRATED SILICON CABLE)
		DAD909-L001	-W1	PNK	=DAD909-L001-S3		-X2	7	5	C5	INSTALLED AT SITE (INTEGRATED SILICON CABLE)
		DAD909-L001	-W1	WHT	=DAD909-L001-S1		-X2	4	5	C3	INSTALLED AT SITE (INTEGRATED SILICON CABLE)
		DAD909-L001	-W1	YEL	=DAD909-L001-S2		-X2	5	5	C4	INSTALLED AT SITE (INTEGRATED SILICON CABLE)

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Rev.	Date	Created by	Appr. by	Revision text
a	20.7.2021	RSW004	KSU004	Location Boxes updated

Project name: Mansourah - Massarah Gold Mine
Project id: P/20002
Product id: 6X20V32TS
Based on document id: DAAB198406
Customer document id: -

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CONNECTION TABLE
OILY WATER TRANSFER UNIT
LOCAL CONTROL PANEL
CABLE LIST

Created by	Create date	Language	Object designation
MBAX46	8 Jun 2021	EN	BJU907
Reviewed by	**Review date**	**Sheet size**	**Page**
RSW004	8 Jun 2021	A3	8 / 10
Approved by	**Approval date**	**Document Id:**	**Rev.**
KSU004	8 Jun 2021	DECA0006021	C

SH1	REF1	CMP1	PIN1	WLAY1	CMP2	PIN2	SH2	REF2
4	C4	-T1	110	AC CONTROL	-F2	1	4	C5
4	C4	-T1	02N	AC CONTROL	-K1	A2	5	E3
4	C5	-F2	1	AC CONTROL	-F3	1	4	C6
4	C5	-F2	2	AC CONTROL	-K3	12	5	C1
4	C6	-F3	2	AC CONTROL	-X2	1	4	D6
4	D6	-X2	1	AC CONTROL	-X2	2	4	D7
4	D7	-X2	2	AC CONTROL	-K1	14	6	B3
5	C1	-K3	11	AC CONTROL	-P1	X1	5	D1
5	C1	-K3	12	AC CONTROL	-K4	14	6	C1
5	E1	-P1	X2	AC CONTROL	-S1	X2	6	E1
5	C3	-X2	4	AC CONTROL	-K1	A1	5	D3
5	E3	-K1	A2	AC CONTROL	-K2	A2	5	E4
5	C4	-X2	6	AC CONTROL	-K2	A1	5	D4
5	E4	-K2	A2	AC CONTROL	-K3	A2	5	E5
5	C5	-X2	8	AC CONTROL	-K3	A1	5	D5
5	E5	-K3	A2	AC CONTROL	-K4	A2	6	E3
6	C1	-K4	11	AC CONTROL	-S1	X1	6	D1
6	B3	-K1	11	AC CONTROL	-S1	21	6	C3
6	B3	-K1	14	AC CONTROL	-K5	24	6	A6
6	C3	-K2	12	AC CONTROL	-K4	24	6	C3
6	C3	-K4	24	AC CONTROL	-S1	22	6	C3
6	D3	-K2	11	AC CONTROL	-K4	21	6	D3
6	D3	-K4	21	AC CONTROL	-K4	A1	6	D3
6	D3	-K4	A1	AC CONTROL	-K5	A1	6	D4
6	E3	-K4	A2	AC CONTROL	-K5	A2	6	E4
6	D4	-K5	A1	AC CONTROL	-S1	14	6	C4
6	E4	-K5	A2	AC CONTROL	-X2	10	6	C6
6	A6	-K5	21	AC CONTROL	-X2	9	6	B6
6	A6	-K5	24	AC CONTROL	-S1	13	6	C4
6	C6	-X2	10	AC CONTROL	-P1	X2	5	E1
7	D2	-X2	11	AC CONTROL	-K3	34	7	A3
7	A3	-K3	34	AC CONTROL	-K5	34	7	A4
7	A3	-K3	31	AC CONTROL	-X2	12	7	D3
7	A4	-K5	31	AC CONTROL	-X2	13	7	D4
4	D7	-X2	2	AC CONTROL JMP	-X2	5	5	B4
5	B3	-X2	3	AC CONTROL JMP	-X2	2	4	D7
5	B4	-X2	5	AC CONTROL JMP	-X2	7	5	B5
4	B2	-Q1	2	AC POWER	-X1	1	4	D2
4	B2	-Q1	1	AC POWER	-F1	1	4	B4
4	B2	-Q1	4	AC POWER	-X1	2	4	D2
4	B2	-Q1	3	AC POWER	-F1	3	4	B4
4	B4	-F1	2	AC POWER	-T1	xxx	4	C4
4	B4	-F1	4	AC POWER	-T1	01N	4	C4
5	C3	-S1		EXTERNAL CONTROL (INTERLOCK)	-X2	3	5	B3
5	C3	-S1		EXTERNAL CONTROL (INTERLOCK)	-X2	4	5	C3
5	C4	-S2		EXTERNAL CONTROL (INTERLOCK)	-X2	5	5	B4
5	C4	-S2		EXTERNAL CONTROL (INTERLOCK)	-X2	6	5	C4
5	C5	-S3		EXTERNAL CONTROL (INTERLOCK)	-X2	7	5	B5
5	C5	-S3		EXTERNAL CONTROL (INTERLOCK)	-X2	8	5	C5
6	B6	-X2	9	EXTERNAL CONTROL (INTERLOCK)	-Y1	X1	6	C6
6	C6	-X2	10	EXTERNAL CONTROL (INTERLOCK)	-Y1	X2	6	C6
7	E2	-X__		EXTERNAL CONTROL (INTERLOCK)	-X2	11	7	D2
7	E3	-X__		EXTERNAL CONTROL (INTERLOCK)	-X2	12	7	D3
7	E4	-X__		EXTERNAL CONTROL (INTERLOCK)	-X2	13	7	D4
4	E2	-Q1	L1	EXTERNAL POWER	-X1	1	4	D2
4	E2	-Q1	L2	EXTERNAL POWER	-X1	2	4	D2
4	E3	-GND		EXTERNAL PROTECTIVE	-PE	1	4	D3
6	B6	-PE	6	EXTERNAL PROTECTIVE	-GND		6	B6
4	D3	-PE	1	PROTECTIVE	-GND		4	D3
4	C4	-T1	02N	PROTECTIVE	-PE	2	4	C3

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Rev.	Date	Created by	Appr. by	Revision text
a	20.7.2021	RSW004	KSU004	Location Boxes updated

Project name: Mansourah - Massarah Gold Mine
Project id: P/20002
Product id: 6X20V32TS
Based on document id: DAAB198406
Customer document id: -

WÄRTSILÄ

CONNECTION TABLE
OILY WATER TRANSFER UNIT
LOCAL CONTROL PANEL
INTERNAL CONNECTION LIST

Created by: MBAX46	Create date: 8 Jun 2021	Language: EN	Object designation: BJU907
Reviewed by: RSW004	Review date: 8 Jun 2021	Sheet size: A3	Page: 9 / 10
Approved by: KSU004	Approval date: 8 Jun 2021	Document Id: DECA00006021	Rev.: C

COMPONENT NAME	QTY	MANUFACTURER	PART CODE	RATING	DESC
-E01	1	RITTAL	1038000		ENCLOSURE, 380x600x210
	4	RITTAL	2508020		WALL MOUNTING BRACKET
-F1	1	SCHNEIDER ELECTRIC	A9F05202	2A D-TRIP	MINIATURE CIRCUIT BREAKER
-F2	1	SCHNEIDER ELECTRIC	A9F04101	1A	MINIATURE CIRCUIT BREAKER
-F3	1	SCHNEIDER ELECTRIC	A9F04102	2.0A	MINIATURE CIRCUIT BREAKER
-K1	1	CARLO GAVAZZI	RCP11003115120VAC		UNIVERSAL PLUG-IN RELAY, 3 CO
	1	CARLO GAVAZZI	ZPD12A		SOCKET
-K2	1	CARLO GAVAZZI	RCP11003115120VAC		UNIVERSAL PLUG-IN RELAY, 3 CO
	1	CARLO GAVAZZI	ZPD12A		SOCKET
-K3	1	CARLO GAVAZZI	RCP11003115120VAC		UNIVERSAL PLUG-IN RELAY, 3 CO
	1	CARLO GAVAZZI	ZPD12A		SOCKET
-K4	1	CARLO GAVAZZI	RCP11003115120VAC		UNIVERSAL PLUG-IN RELAY, 3 CO
	1	CARLO GAVAZZI	ZPD12A		SOCKET
-K5	1	CARLO GAVAZZI	RCP11003115120VAC		UNIVERSAL PLUG-IN RELAY, 3 CO
	1	CARLO GAVAZZI	ZPD12A		SOCKET
-P1	1	SCHNEIDER ELECTRIC	XB7EV05GP		MONOLITHIC INDICATOR LIGHT, YELLOW, INTEGRAL LED
-PE	2	PHOENIX CONTACT	0404428	AB/SS	END PIECE (FOR PE TERMINAL BLOCK)
-PE	15	PHOENIX CONTACT	0421029		PE TERMINAL BLOCK
-Q1	1	ABB	OT16FT3+OHYS2PJ		MAIN BREAKER WITH IP65 HANDLE
-S1	1	SCHNEIDER ELECTRIC	ZB5AK1813+ZB5AW0G15		CONTROL SWITCH, 3 POSITION
-T1	1	TRAFOMIC	0423192	100VA	CONTROL VOLTAGE TRANSFORMER
-U1	1	AGRO	60010031	M20	CABLE GLAND, M20
-U2	1	AGRO	60010031	M20	CABLE GLAND, M20
-U3	1	AGRO	60010031	M20	CABLE GLAND, M20
-U4	1	AGRO	60010031	M20	CABLE GLAND, M20
-U5	1	AGRO	60010031	M20	CABLE GLAND, M20
-U6	1	AGRO	60010031	M20	CABLE GLAND, M20
-X1	2	WEIDMULLER	1608830000	ZAP/TW5	END PIECE (FOR TERMINAL BLOCK)
	3	WEIDMULLER	1608620000	ZDU 6	TERMINAL BLOCK, FEED-THROUGH, 6MM2
-X2	2	WEIDMULLER	1632090000	ZAP/TW4	END PIECE (FOR TERMINAL BLOCK)
	20	WEIDMULLER	1632050000	ZDU 4	TERMINAL BLOCK, FEED-THROUGH, 4MM2

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Rev.	Date	Created by	Appr. by	Revision text
b	07.04.2022	MBA123	RSW004	Part list updated
a	20.7.2021	RSW004	KSU004	Location Boxes updated

Project name: Mansourah - Massarah Gold Mine
Project id: P/20002
Product id: 6X20V32TS
Based on document id: DAAB198406
Customer document id: -

WÄRTSILÄ

PARTS LIST
OILY WATER TRANSFER UNIT
LOCAL CONTROL PANEL
PARTS LIST

Created by: MBAX46	Create date: 8 Jun 2021	Language: EN	Object designation: BJU907
Reviewed by: RSW004	Review date: 8 Jun 2021	Sheet size: A3	Page: 10 / 10
Approved by: KSU004	Approval date: 8 Jun 2021	Document Id: DECA0006021	Rev.: C

PANEL CODE	DOCUMENT NAME	DOCUMENT DESCRIPTION	SHEET N:o	DOCUMENT N:o	REVISION
BJU908	LIST OF DOCUMENTS	DRAWING LIST	1	DECA00006022	b
BJU908	DIMENSIONAL DRAWING	LAYOUT 1:10	2	DECA00006022	b
BJU908	OVERVIEW DIAGRAM	SINGLE LINE	3	DECA00006022	b
BJU908	CIRCUIT DIAGRAM	LOW VOLTAGE FEEDER	4	DECA00006022	b
BJU908	CIRCUIT DIAGRAM	COMMON INDICATOR/CONTROL	5	DECA00006022	b
BJU908	CIRCUIT DIAGRAM	PUMP CONTROL	6	DECA00006022	b
BJU908	CIRCUIT DIAGRAM	EXTERNAL ALARM	7	DECA00006022	b
BJU908	CONNECTION TABLE	CABLE LIST	8	DECA00006022	b
BJU908	CONNECTION TABLE	INTERNAL CONNECTION LIST	9	DECA00006022	b
BJU908	PARTS LIST	PARTS LIST	10	DECA00006022	b

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Rev.	Date	Created by	Appr. by	Revision text
a	21.7.2021	RSW004	KSU004	Location Boxes Updated

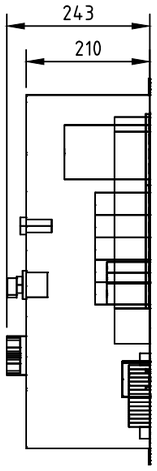
Project name: Mansourah - Massarah Gold Mine
Project id: P/20002
Product id: 6X20V32TS
Based on document id: DAAB198406
Customer document id: -

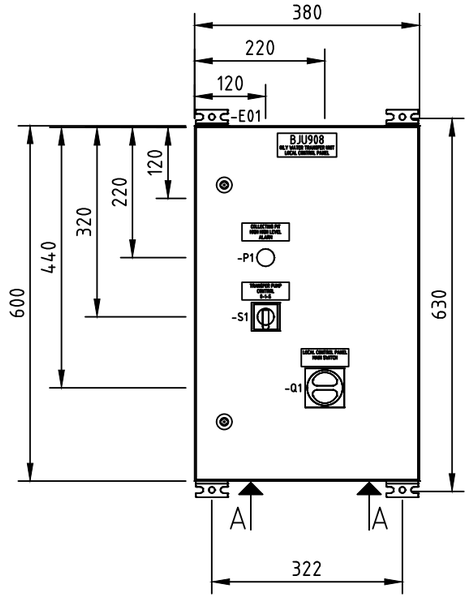
WÄRTSILÄ

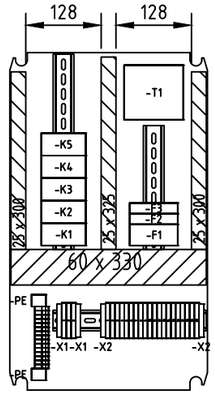
LIST OF DOCUMENTS
OILY WATER TRANSFER UNIT
LOCAL CONTROL PANEL
DRAWING LIST

Created by	Create date	Language	Object designation
MBAX46	8 Jun 2021	EN	BJU908
Reviewed by	Review date	Sheet size	Page
RSW004	8 Jun 2021	A3	1 / 10
Approved by	Approval date	Document Id:	Rev.
KSU004	8 Jun 2021	DECA00006022	C

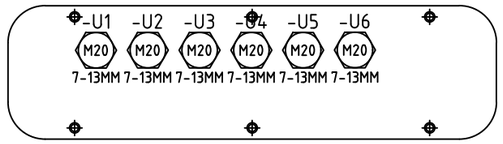
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SECTION A-A SCALE 1:4



PANEL INFORMATION					
WIDTH	380	mm	PROTECTION CLASS		IP65
HEIGHT	600	mm	COLOR		RAL 7035
DEPTH	210	mm	Ik	6	kA/1s

-E01	BJU908	OILY WATER TRANSFER UNIT	LOCAL CONTROL PANEL
-P1	COLLECTING PIT	HIGH HIGH LEVEL	ALARM
-Q1	LOCAL CONTROL PANEL	MAIN SWITCH	
-S1	TRANSFER PUMP	CONTROL	0-1-S

Rev.	Date	Created by	Appr. by	Revision text
a	21.7.2021	RSW004	KSU004	Location Boxes Updated

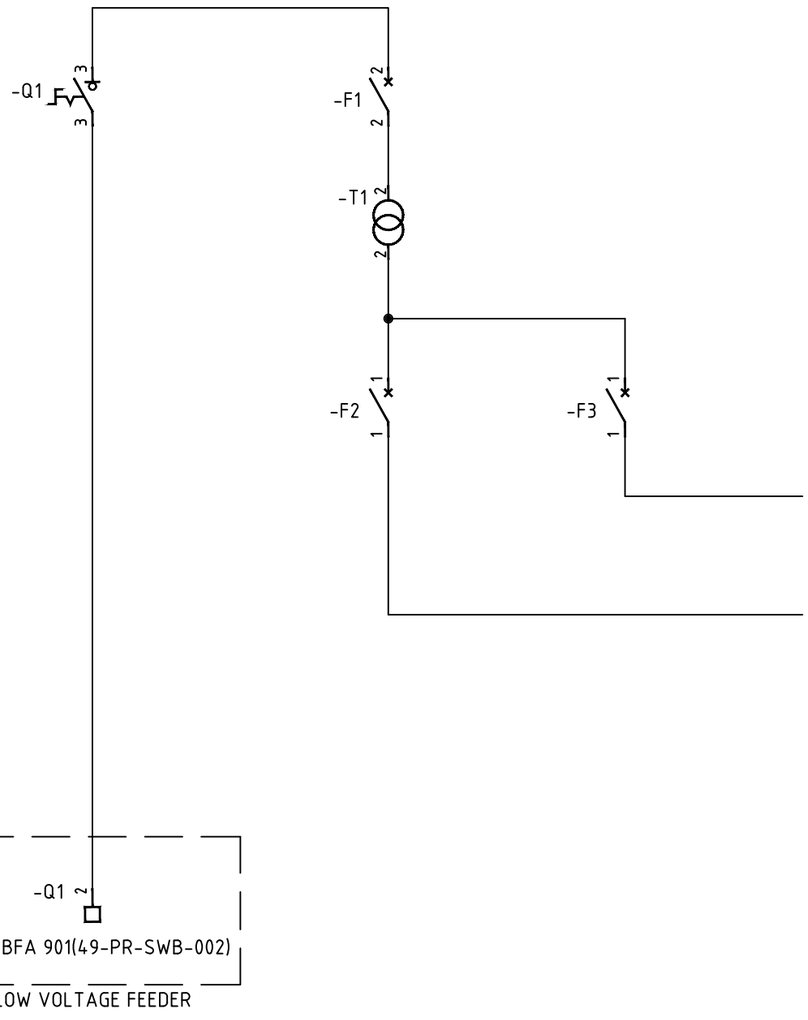
Project name: Mansourah - Massarah Gold Mine
Project id: P/20002
Product id: 6X20V32TS
Based on document id: DAAB198406
Customer document id: -

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DIMENSIONAL DRAWING
OILY WATER TRANSFER UNIT
LOCAL CONTROL PANEL
LAYOUT 1:10

Created by: MBAX46	Create date: 8 Jun 2021	Language: EN	Object designation: BJU908
Reviewed by: RSW004	Review date: 8 Jun 2021	Sheet size: A3	Page: 2 / 10
Approved by: KSU004	Approval date: 8 Jun 2021	Document Id: DECA00006022	Rev.: C

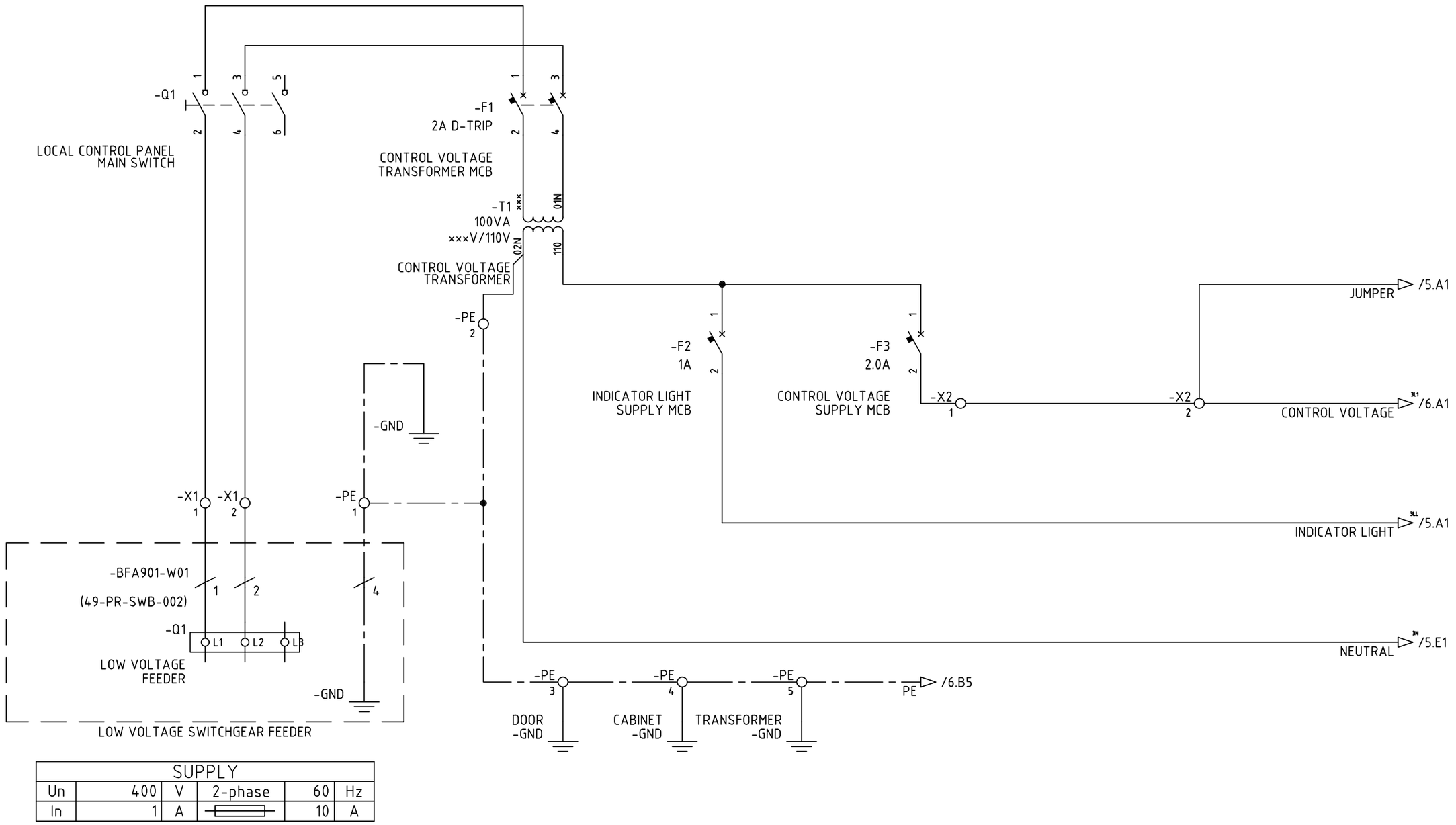
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Rev.	Date	Created by	Appr. by	Revision text
a	21.7.2021	RSW004	KSU004	Location Boxes Updated

Project name: Mansourah - Massarah Gold Mine			
Project id: P/20002	Product id: 6X20V32TS		
Based on document id: DAAB198406	Customer document id: -		
OVERVIEW DIAGRAM	Created by: MBAX46	Create date: 8 Jun 2021	Language: EN
OILY WATER TRANSFER UNIT LOCAL CONTROL PANEL SINGLE LINE	Reviewed by: RSW004	Review date: 8 Jun 2021	Sheet size: A3
	Approved by: KSU004	Approval date: 8 Jun 2021	Object designation: BJU908
			Page: 3 / 10
			Document Id: DECA00006022 Rev. C1

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SUPPLY					
Un	400	V	2-phase	60	Hz
In	1	A		10	A

Rev.	Date	Created by	Appr. by	Revision text
a	21.7.2021	RSW004	KSU004	Location Boxes Updated

Project name	Mansourah - Massarah Gold Mine		
Project id	P/20002	Product id	6X20V32TS
Based on document id	DAAB198406	Customer document id	-

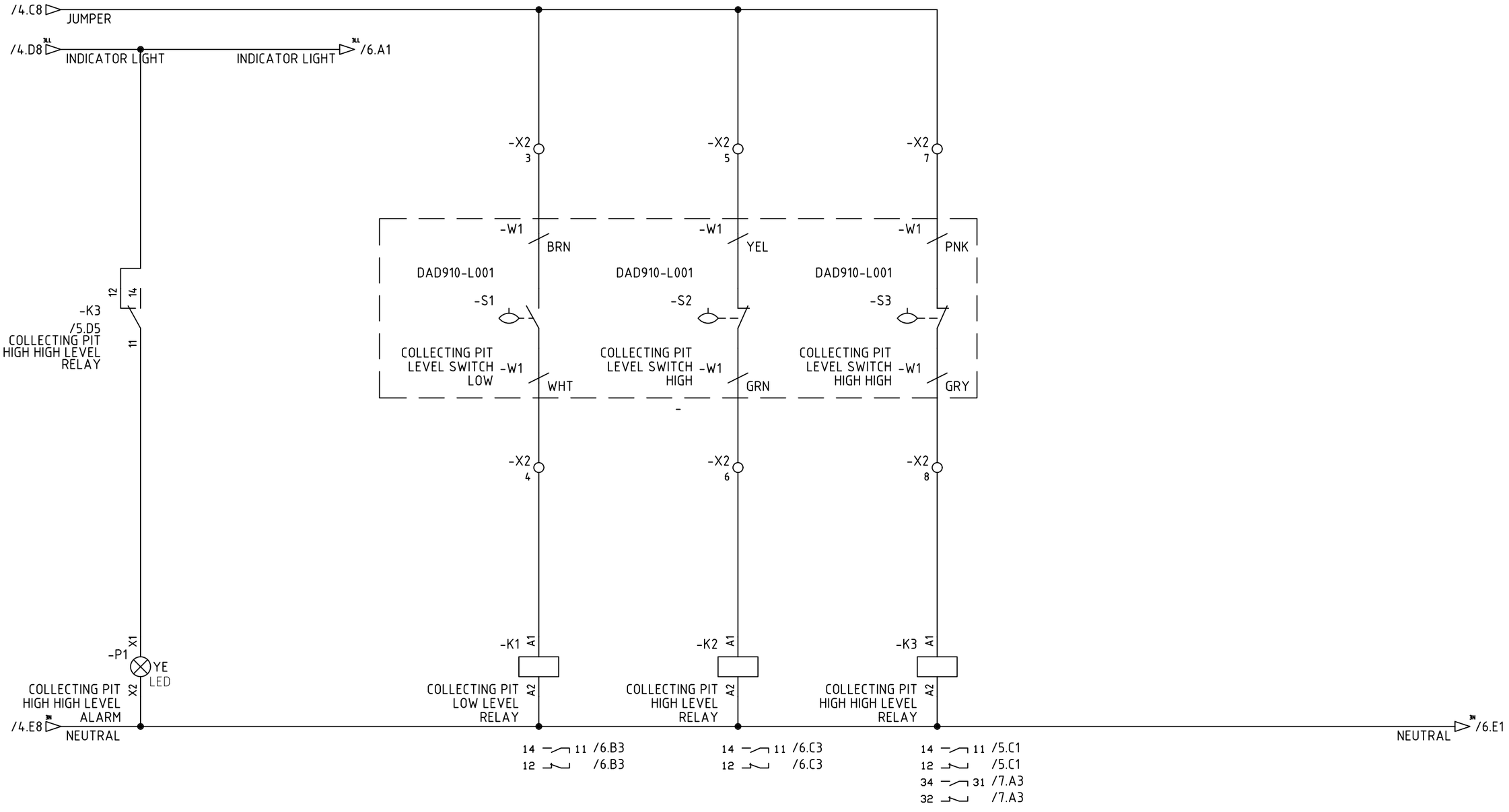
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CIRCUIT DIAGRAM

OILY WATER TRANSFER UNIT
LOCAL CONTROL PANEL
LOW VOLTAGE FEEDER

Created by	MBAX46	Create date	8 Jun 2021
Reviewed by	RSW004	Review date	8 Jun 2021
Approved by	KSU004	Approval date	8 Jun 2021
Language	EN	Object designation	BJU908
Sheet size	A3	Page	4 / 10
Document Id:	DECA00006022	Rev.	C

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Rev.	Date	Created by	Appr. by	Revision text
a	21.7.2021	RSW004	KSU004	Location Boxes Updated

Project name	Mansourah - Massarah Gold Mine		
Project id	P/20002	Product id	6X20V32TS
Based on document id	DAAB198406	Customer document id	-

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CIRCUIT DIAGRAM

OILY WATER TRANSFER UNIT
LOCAL CONTROL PANEL
COMMON INDICATOR/CONTROL

Created by	MBAX46	Create date	8 Jun 2021
Reviewed by	RSW004	Review date	8 Jun 2021
Approved by	KSU004	Approval date	8 Jun 2021
Language	EN	Object designation	BJU908
Sheet size	A3	Page	5 / 10
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/4.D8 3L1 CONTROL VOLTAGE

/5.A2 3L1 INDICATOR LIGHT

-K4 /6.D3 TRANSFER PUMP CONTROL RELAY 1 — 12 14 / 11

-K1 /5.D3 COLLECTING PIT LOW LEVEL RELAY — 12 14 / 11

-S1 0-1-S — 21 22 — TRANSFER PUMP CONTROL 0-1-S

-S1 — 13 14 — /6.B4 TRANSFER PUMP CONTROL 0-1-S

-S1					
0	1	S	T1	T2	REF
X			21	22	B4
			X1	X2	6E1
		X	13	14	6C4

-K2 /5.D4 COLLECTING PIT HIGH LEVEL RELAY — 12 14 / 11

-K4 /6.D3 TRANSFER PUMP CONTROL RELAY 1 — 22 24 / 21

-K5 /6.D4 TRANSFER PUMP CONTROL RELAY2 — 22 24 / 21

/4.E6 PE

-PE 6 — -X2 9

-W01 3 — -W01 1

-GND

DAD910-CV001

-Y1 X1 X2 — TRANSFER PUMP CONTROL VALVE

-W01 2

-X2 10

-S1 X1 /6.B4 WH LED — TRANSFER PUMP X2

/5.E8 0-1-S NEUTRAL

-K4 A1 — TRANSFER PUMP CONTROL RELAY 1 A2

-K5 A1 — TRANSFER PUMP CONTROL RELAY2 A2

14 — 11 /6.C1
12 — /6.C1
24 — 21 /6.C3
22 — /6.C3

24 — 21 /6.A6
22 — /6.A6
34 — 31 /7.A4
32 — /7.A4

Rev.	Date	Created by	Appr. by	Revision text
a	21.7.2021	RSW004	KSU004	Location Boxes Updated

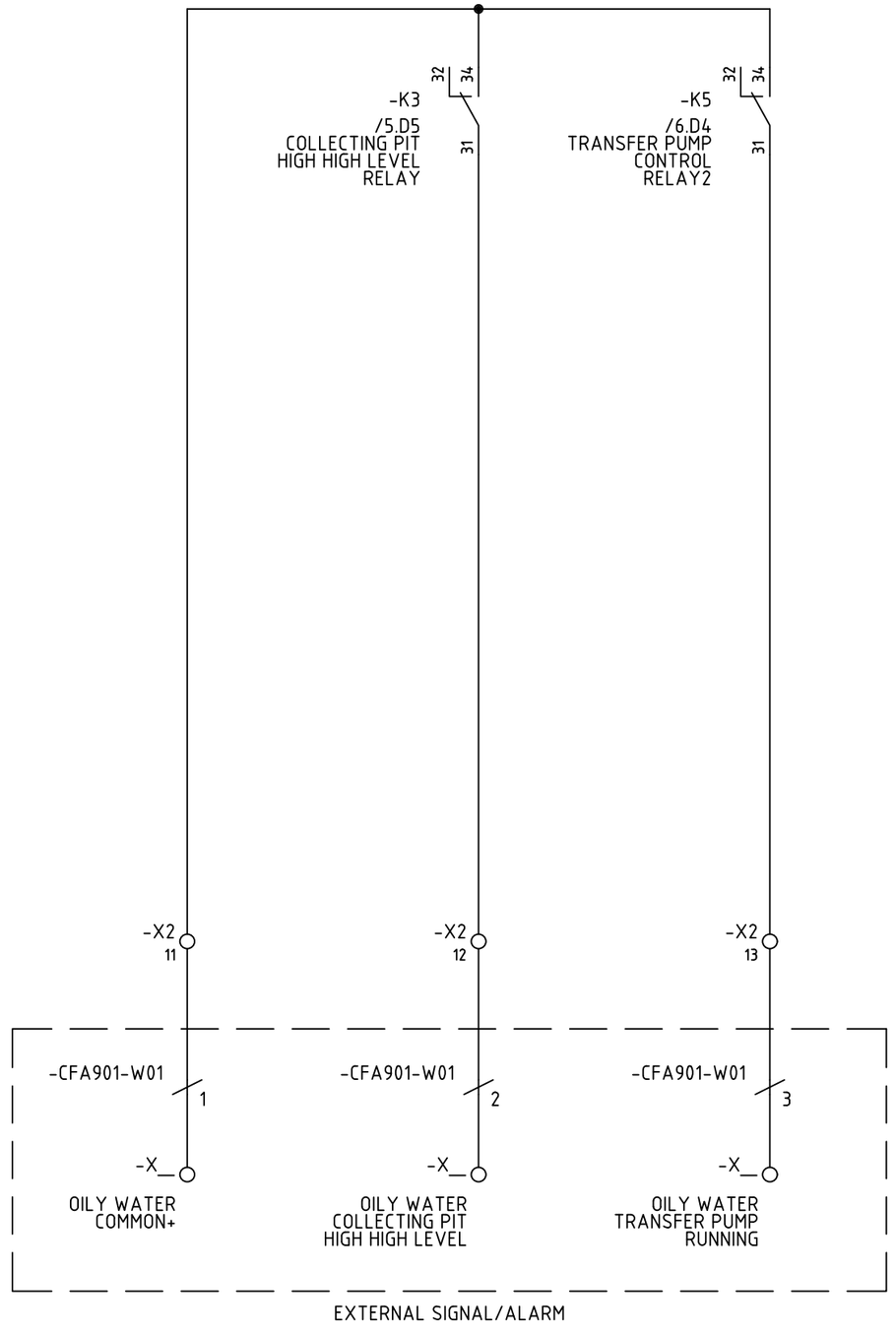
Project name: Mansourah - Massarah Gold Mine
Project id: P/20002
Product id: 6X20V32TS
Based on document id: DAAB198406
Customer document id: -

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CIRCUIT DIAGRAM
OILY WATER TRANSFER UNIT
LOCAL CONTROL PANEL
PUMP CONTROL

Created by: MBAX46	Create date: 8 Jun 2021	Language: EN	Object designation: BJU908
Reviewed by: RSW004	Review date: 8 Jun 2021	Sheet size: A3	Page: 6 / 10
Approved by: KSU004	Approval date: 8 Jun 2021	Document Id: DECA00006022	Rev.: C

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Rev.	Date	Created by	Appr. by	Revision text
a	21.7.2021	RSW004	KSU004	Location Boxes Updated

Project name: Mansourah - Massarah Gold Mine			
Project id: P/20002	Product id: 6X20V32TS		
Based on document id: DAAB198406	Customer document id: -		

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CIRCUIT DIAGRAM

OILY WATER TRANSFER UNIT
LOCAL CONTROL PANEL
EXTERNAL ALARM

Created by: MBAX46	Create date: 8 Jun 2021	Language: EN	Object designation: BJU908
Reviewed by: RSW004	Review date: 8 Jun 2021	Sheet size: A3	Page: 7 / 10
Approved by: KSU004	Approval date: 8 Jun 2021	Document Id: DECA00006022	Rev.: C

FACTORY INSTALLED CABLES TO BE SUPPLIED BY HELKAMA IF NOT OTHERWISE STATED

CABLE SPECIFICATION	CABLE TYPE	CABLE INSTALLATION	CABLE NAME	WIRE	FROM COMPONENT	PIN1	TO COMPONENT	PIN2	SH1	REF1	CABLE REMARK
1kV YY	3G1,5rm		-W01	1	-X2	9	=DAD910-CV001-Y1	X1	6	B6	INSTALLED AT FSITE
			-W01	2	-X2	10	=DAD910-CV001-Y1	X2	6	C6	INSTALLED AT FSITE
			-W01	3	-PE	6	=DAD910-CV001-GND		6	B6	INSTALLED AT FSITE
1kV YCY	3x__+__	BFA901	-BFA901-W01	1	=BFA901-Q1	L1	-X1	1	4	E2	INSTALLED AT SITE
		BFA901	-BFA901-W01	2	=BFA901-Q1	L2	-X1	2	4	E2	INSTALLED AT SITE
		BFA901	-BFA901-W01	4	=BFA901-GND		-PE	1	4	E3	INSTALLED AT SITE
1kV YY	4x1,5rm	CFA901	-CFA901-W01	1	=CFA901-X__		-X2	11	7	E2	INSTALLED AT SITE
		CFA901	-CFA901-W01	2	=CFA901-X__		-X2	12	7	E3	INSTALLED AT SITE
		CFA901	-CFA901-W01	3	=CFA901-X__		-X2	13	7	E4	INSTALLED AT SITE
		DAD910-L001	-W1	BRN	=DAD910-L001-S1		-X2	3	5	C3	INSTALLED AT SITE (INTEGRATED SILICON CABLE)
		DAD910-L001	-W1	GRN	=DAD910-L001-S2		-X2	6	5	C4	INSTALLED AT SITE (INTEGRATED SILICON CABLE)
		DAD910-L001	-W1	GRY	=DAD910-L001-S3		-X2	8	5	C5	INSTALLED AT SITE (INTEGRATED SILICON CABLE)
		DAD910-L001	-W1	PNK	=DAD910-L001-S3		-X2	7	5	C5	INSTALLED AT SITE (INTEGRATED SILICON CABLE)
		DAD910-L001	-W1	WHT	=DAD910-L001-S1		-X2	4	5	C3	INSTALLED AT SITE (INTEGRATED SILICON CABLE)
		DAD910-L001	-W1	YEL	=DAD910-L001-S2		-X2	5	5	C4	INSTALLED AT SITE (INTEGRATED SILICON CABLE)

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Rev.	Date	Created by	Appr. by	Revision text
a	21.7.2021	RSW004	KSU004	Location Boxes Updated

Project name: Mansourah - Massarah Gold Mine
Project id: P/20002
Product id: 6X20V32TS
Based on document id: DAAB198406
Customer document id: -

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CONNECTION TABLE
OILY WATER TRANSFER UNIT
LOCAL CONTROL PANEL
CABLE LIST

Created by	Create date	Language	Object designation
MBAX46	8 Jun 2021	EN	BJU908
Reviewed by	**Review date**	**Sheet size**	**Page**
RSW004	8 Jun 2021	A3	8 / 10
Approved by	**Approval date**	**Document Id:**	**Rev.**
KSU004	8 Jun 2021	DECA0006022	C

SH1	REF1	CMP1	PIN1	WLAY1	CMP2	PIN2	SH2	REF2
4	C4	-T1	110	AC CONTROL	-F2	1	4	C5
4	C4	-T1	02N	AC CONTROL	-K1	A2	5	E3
4	C5	-F2	1	AC CONTROL	-F3	1	4	C6
4	C5	-F2	2	AC CONTROL	-K3	12	5	C1
4	C6	-F3	2	AC CONTROL	-X2	1	4	D6
4	D6	-X2	1	AC CONTROL	-X2	2	4	D7
4	D7	-X2	2	AC CONTROL	-K1	14	6	B3
5	C1	-K3	11	AC CONTROL	-P1	X1	5	D1
5	C1	-K3	12	AC CONTROL	-K4	14	6	C1
5	E1	-P1	X2	AC CONTROL	-S1	X2	6	E1
5	C3	-X2	4	AC CONTROL	-K1	A1	5	D3
5	E3	-K1	A2	AC CONTROL	-K2	A2	5	E4
5	C4	-X2	6	AC CONTROL	-K2	A1	5	D4
5	E4	-K2	A2	AC CONTROL	-K3	A2	5	E5
5	C5	-X2	8	AC CONTROL	-K3	A1	5	D5
5	E5	-K3	A2	AC CONTROL	-K4	A2	6	E3
6	C1	-K4	11	AC CONTROL	-S1	X1	6	D1
6	B3	-K1	11	AC CONTROL	-S1	21	6	C3
6	B3	-K1	14	AC CONTROL	-K5	24	6	A6
6	C3	-K2	12	AC CONTROL	-K4	24	6	C3
6	C3	-K4	24	AC CONTROL	-S1	22	6	C3
6	D3	-K2	11	AC CONTROL	-K4	21	6	D3
6	D3	-K4	21	AC CONTROL	-K4	A1	6	D3
6	D3	-K4	A1	AC CONTROL	-K5	A1	6	D4
6	E3	-K4	A2	AC CONTROL	-K5	A2	6	E4
6	D4	-K5	A1	AC CONTROL	-S1	14	6	C4
6	E4	-K5	A2	AC CONTROL	-X2	10	6	C6
6	A6	-K5	21	AC CONTROL	-X2	9	6	B6
6	A6	-K5	24	AC CONTROL	-S1	13	6	C4
6	C6	-X2	10	AC CONTROL	-P1	X2	5	E1
7	D2	-X2	11	AC CONTROL	-K3	34	7	A3
7	A3	-K3	34	AC CONTROL	-K5	34	7	A4
7	A3	-K3	31	AC CONTROL	-X2	12	7	D3
7	A4	-K5	31	AC CONTROL	-X2	13	7	D4
4	D7	-X2	2	AC CONTROL JMP	-X2	5	5	B4
5	B3	-X2	3	AC CONTROL JMP	-X2	2	4	D7
5	B4	-X2	5	AC CONTROL JMP	-X2	7	5	B5
4	B2	-Q1	2	AC POWER	-X1	1	4	D2
4	B2	-Q1	1	AC POWER	-F1	1	4	B4
4	B2	-Q1	4	AC POWER	-X1	2	4	D2
4	B2	-Q1	3	AC POWER	-F1	3	4	B4
4	B4	-F1	2	AC POWER	-T1	xxx	4	C4
4	B4	-F1	4	AC POWER	-T1	01N	4	C4
5	C3	-S1		EXTERNAL CONTROL (INTERLOCK)	-X2	3	5	B3
5	C3	-S1		EXTERNAL CONTROL (INTERLOCK)	-X2	4	5	C3
5	C4	-S2		EXTERNAL CONTROL (INTERLOCK)	-X2	5	5	B4
5	C4	-S2		EXTERNAL CONTROL (INTERLOCK)	-X2	6	5	C4
5	C5	-S3		EXTERNAL CONTROL (INTERLOCK)	-X2	7	5	B5
5	C5	-S3		EXTERNAL CONTROL (INTERLOCK)	-X2	8	5	C5
6	B6	-X2	9	EXTERNAL CONTROL (INTERLOCK)	-Y1	X1	6	C6
6	C6	-X2	10	EXTERNAL CONTROL (INTERLOCK)	-Y1	X2	6	C6
7	E2	-X__		EXTERNAL CONTROL (INTERLOCK)	-X2	11	7	D2
7	E3	-X__		EXTERNAL CONTROL (INTERLOCK)	-X2	12	7	D3
7	E4	-X__		EXTERNAL CONTROL (INTERLOCK)	-X2	13	7	D4
4	E2	-Q1	L1	EXTERNAL POWER	-X1	1	4	D2
4	E2	-Q1	L2	EXTERNAL POWER	-X1	2	4	D2
4	E3	-GND		EXTERNAL PROTECTIVE	-PE	1	4	D3
6	B6	-PE	6	EXTERNAL PROTECTIVE	-GND		6	B6
4	D3	-PE	1	PROTECTIVE	-GND		4	D3
4	C4	-T1	02N	PROTECTIVE	-PE	2	4	C3

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Rev.	Date	Created by	Appr. by	Revision text
a	21.7.2021	RSW004	KSU004	Location Boxes Updated

Project name: Mansourah - Massarah Gold Mine
Project id: P/20002
Product id: 6X20V32TS
Based on document id: DAAB198406
Customer document id: -

WÄRTSILÄ

CONNECTION TABLE
OILY WATER TRANSFER UNIT
LOCAL CONTROL PANEL
INTERNAL CONNECTION LIST

Created by: MBAX46	Create date: 8 Jun 2021	Language: EN	Object designation: BJU908
Reviewed by: RSW004	Review date: 8 Jun 2021	Sheet size: A3	Page: 9 / 10
Approved by: KSU004	Approval date: 8 Jun 2021	Document Id: DECA00006022	Rev.: C

COMPONENT NAME	QTY	MANUFACTURER	PART CODE	RATING	DESC
-E01	1	RITTAL	1038000		ENCLOSURE, 380x600x210
-E01	4	RITTAL	2508020		WALL MOUNTING BRACKET
-F1	1	SCHNEIDER ELECTRIC	A9F05202	2A D-TRIP	MINIATURE CIRCUIT BREAKER
-F2	1	SCHNEIDER ELECTRIC	A9F04101	1A	MINIATURE CIRCUIT BREAKER
-F3	1	SCHNEIDER ELECTRIC	A9F04102	2.0A	MINIATURE CIRCUIT BREAKER
-K1	1	CARLO GAVAZZI	RCP11003115120VAC		UNIVERSAL PLUG-IN RELAY, 3 CO
-K1	1	CARLO GAVAZZI	ZPD12A		SOCKET
-K2	1	CARLO GAVAZZI	RCP11003115120VAC		UNIVERSAL PLUG-IN RELAY, 3 CO
-K2	1	CARLO GAVAZZI	ZPD12A		SOCKET
-K3	1	CARLO GAVAZZI	RCP11003115120VAC		UNIVERSAL PLUG-IN RELAY, 3 CO
-K3	1	CARLO GAVAZZI	ZPD12A		SOCKET
-K4	1	CARLO GAVAZZI	RCP11003115120VAC		UNIVERSAL PLUG-IN RELAY, 3 CO
-K4	1	CARLO GAVAZZI	ZPD12A		SOCKET
-K5	1	CARLO GAVAZZI	RCP11003115120VAC		UNIVERSAL PLUG-IN RELAY, 3 CO
-K5	1	CARLO GAVAZZI	ZPD12A		SOCKET
-P1	1	SCHNEIDER ELECTRIC	XB7EV05GP		MONOLITHIC INDICATOR LIGHT, YELLOW, INTEGRAL LED
-PE	2	PHOENIX CONTACT	0404428	AB/SS	END PIECE (FOR PE TERMINAL BLOCK)
-PE	15	PHOENIX CONTACT	0421029		PE TERMINAL BLOCK
-Q1	1	ABB	OT16FT3+OHYS2PJ		MAIN BREAKER WITH IP65 HANDLE
-S1	1	SCHNEIDER ELECTRIC	ZB5AK1813+ZB5AW0G15		CONTROL SWITCH, 3 POSITION
-T1	1	TRAFOMIC	0423192	100VA	CONTROL VOLTAGE TRANSFORMER
-U1	1	AGRO	60010031	M20	CABLE GLAND, M20
-U2	1	AGRO	60010031	M20	CABLE GLAND, M20
-U3	1	AGRO	60010031	M20	CABLE GLAND, M20
-U4	1	AGRO	60010031	M20	CABLE GLAND, M20
-U5	1	AGRO	60010031	M20	CABLE GLAND, M20
-U6	1	AGRO	60010031	M20	CABLE GLAND, M20
-X1	2	WEIDMULLER	1608830000	ZAP/TW5	END PIECE (FOR TERMINAL BLOCK)
-X1	3	WEIDMULLER	1608620000	ZDU 6	TERMINAL BLOCK, FEED-THROUGH, 6MM2
-X2	2	WEIDMULLER	1632090000	ZAP/TW4	END PIECE (FOR TERMINAL BLOCK)
-X2	20	WEIDMULLER	1632050000	ZDU 4	TERMINAL BLOCK, FEED-THROUGH, 4MM2

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Rev.	Date	Created by	Appr. by	Revision text
b	07.04.2022	MBA123	RSW004	Part list updated
a	21.7.2021	RSW004	KSU004	Location Boxes Updated

Project name: Mansourah - Massarah Gold Mine
Project id: P/20002
Product id: 6X20V32TS
Based on document id: DAAB198406
Customer document id: -

WÄRTSILÄ

PARTS LIST
OILY WATER TRANSFER UNIT
LOCAL CONTROL PANEL
PARTS LIST

Created by	Create date	Language	Object designation
MBAX46	8 Jun 2021	EN	BJU908
Reviewed by	**Review date**	**Sheet size**	**Page**
RSW004	8 Jun 2021	A3	10 / 10
Approved by	**Approval date**	**Document Id:**	**Rev.**
KSU004	8 Jun 2021	DECA0006022	C

PANEL CODE	DOCUMENT NAME	DOCUMENT DESCRIPTION	SHEET N:o	DOCUMENT N:o	REVISION
BJU909	LIST OF DOCUMENTS	DRAWING LIST	1	DECA00006026	b
BJU909	DIMENSIONAL DRAWING	LAYOUT 1:10	2	DECA00006026	b
BJU909	OVERVIEW DIAGRAM	SINGLE LINE	3	DECA00006026	b
BJU909	CIRCUIT DIAGRAM	LOW VOLTAGE FEEDER	4	DECA00006026	b
BJU909	CIRCUIT DIAGRAM	COMMON INDICATOR/CONTROL	5	DECA00006026	b
BJU909	CIRCUIT DIAGRAM	PUMP CONTROL	6	DECA00006026	b
BJU909	CIRCUIT DIAGRAM	EXTERNAL ALARM	7	DECA00006026	b
BJU909	CONNECTION TABLE	CABLE LIST	8	DECA00006026	b
BJU909	CONNECTION TABLE	INTERNAL CONNECTION LIST	9	DECA00006026	b
BJU909	PARTS LIST	PARTS LIST	10	DECA00006026	b

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Rev.	Date	Created by	Appr. by	Revision text
a	27.7.2021	RSW004	KSU004	Location Box Updated

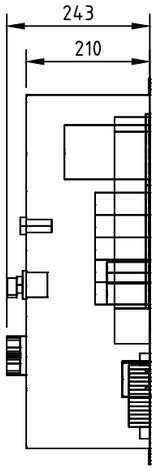
Project name: Mansourah - Massarah Gold Mine
Project id: P/20002
Product id: 6X20V32TS
Based on document id: DAAB198406
Customer document id: -

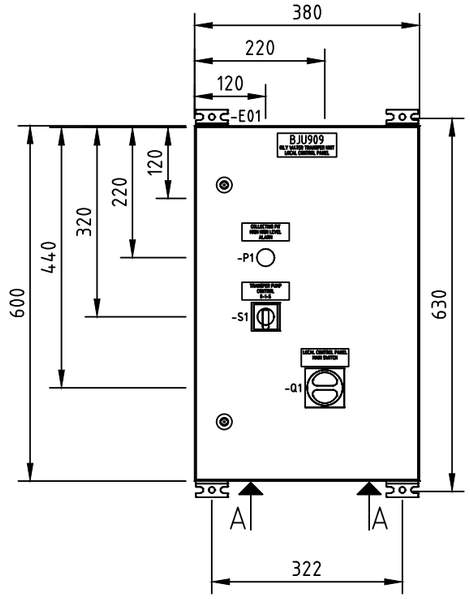
WÄRTSILÄ

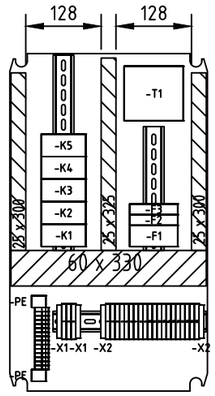
LIST OF DOCUMENTS
OILY WATER TRANSFER UNIT
LOCAL CONTROL PANEL
DRAWING LIST

Created by	Create date	Language	Object designation
MBAX46	8 Jun 2021	EN	BJU909
Reviewed by	**Review date**	**Sheet size**	**Page**
RSW004	8 Jun 2021	A3	1 / 10
Approved by	**Approval date**	**Document Id:**	**Rev.**
KSU004	8 Jun 2021	DECA00006026	C

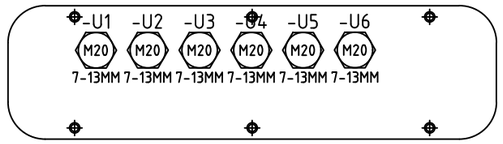
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SECTION A-A SCALE 1:4



PANEL INFORMATION					
WIDTH	380	mm	PROTECTION CLASS		IP65
HEIGHT	600	mm	COLOR		RAL 7035
DEPTH	210	mm	Ik	6	kA/1s

-E01	BJU909	OILY WATER TRANSFER UNIT	LOCAL CONTROL PANEL
-P1	COLLECTING PIT	HIGH HIGH LEVEL	ALARM
-Q1	LOCAL CONTROL PANEL	MAIN SWITCH	
-S1	TRANSFER PUMP	CONTROL	0-1-S

Rev.	Date	Created by	Appr. by	Revision text
a	27.7.2021	RSW004	KSU004	Location Box Updated

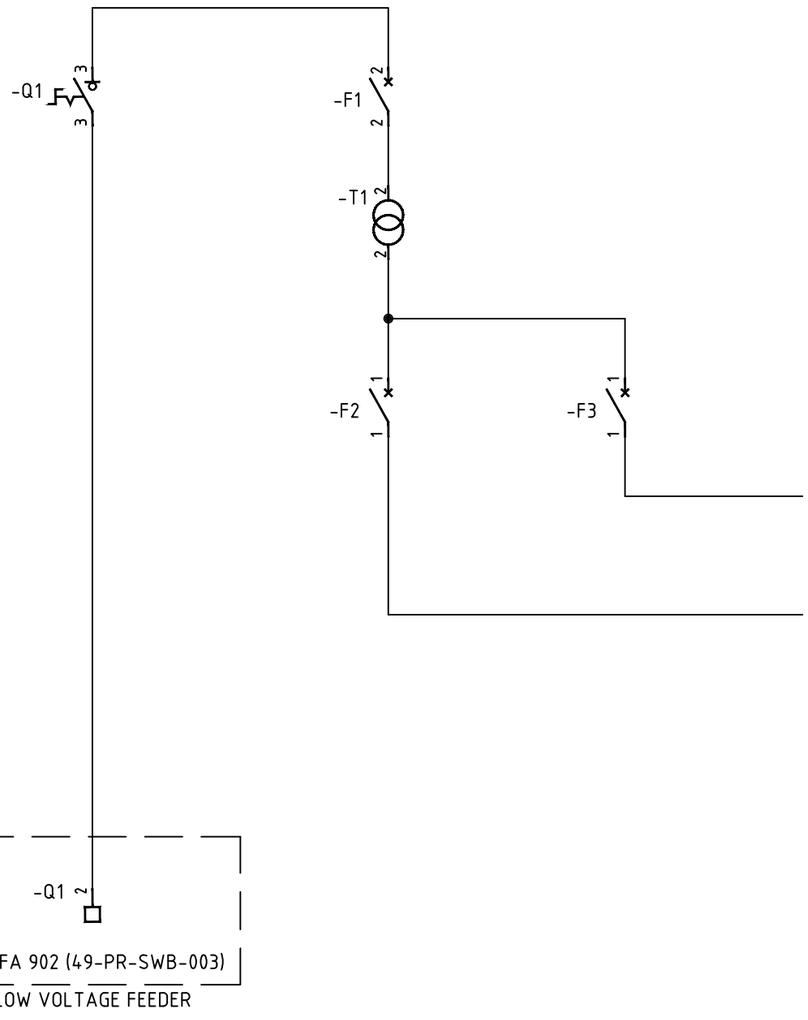
Project name: Mansourah - Massarah Gold Mine
Project id: P/20002
Product id: 6X20V32TS
Based on document id: DAAB198406
Customer document id: -

WÄRTSILÄ

DIMENSIONAL DRAWING
OILY WATER TRANSFER UNIT
LOCAL CONTROL PANEL
LAYOUT 1:10

Created by: MBAX46	Create date: 8 Jun 2021	Language: EN	Object designation: BJU909
Reviewed by: RSW004	Review date: 8 Jun 2021	Sheet size: A3	Page: 2 / 10
Approved by: KSU004	Approval date: 8 Jun 2021	Document Id: DECA00006026	Rev.: C

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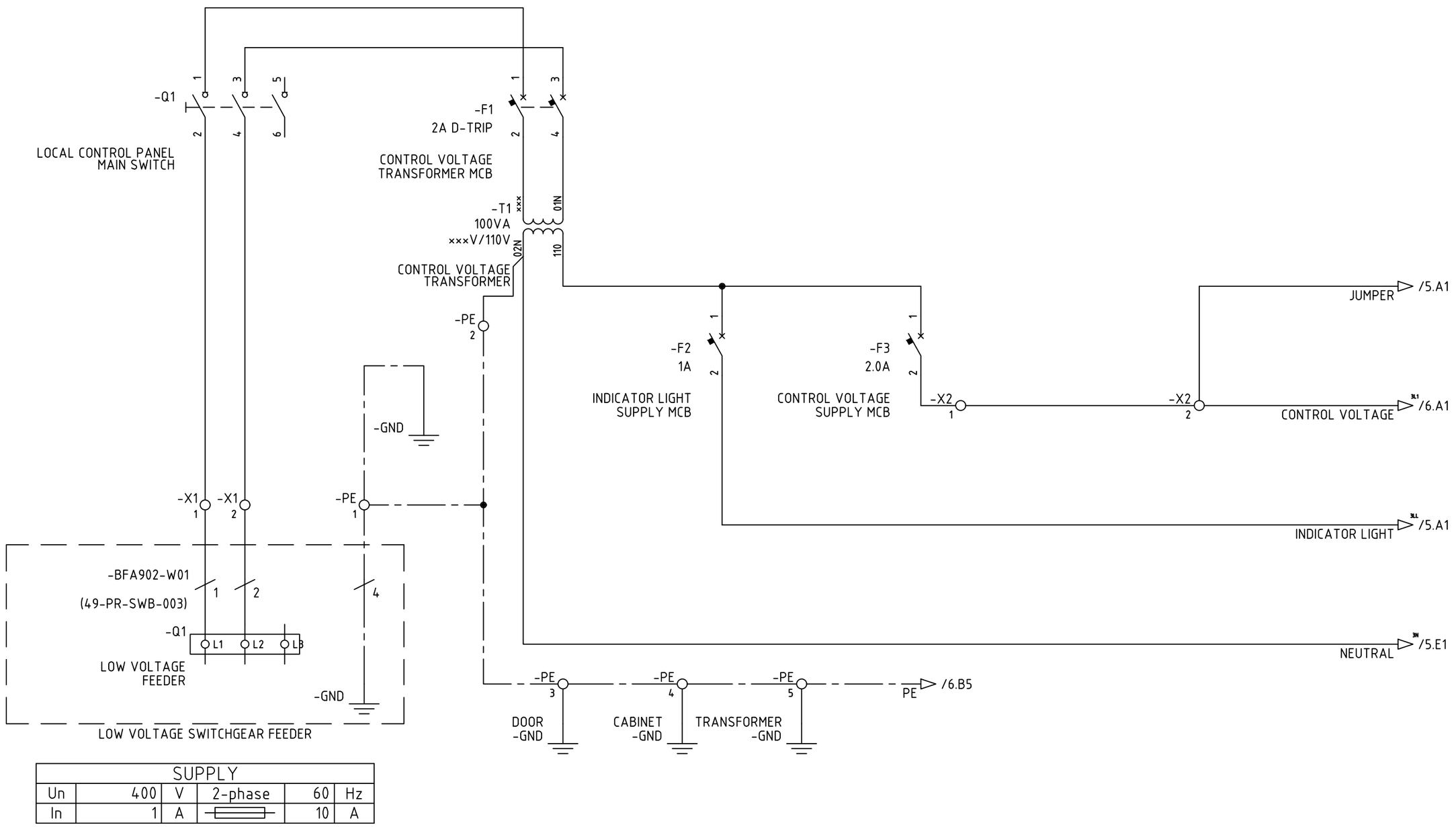


Rev.	Date	Created by	Appr. by	Revision text
a	27.7.2021	RSW004	KSU004	Location Box Updated

Project name: Mansourah - Massarah Gold Mine		WÄRTSILÄ	OVERVIEW DIAGRAM	Created by: MBAX46	Create date: 8 Jun 2021
Project id: P/20002	Product id: 6X20V32TS		OILY WATER TRANSFER UNIT LOCAL CONTROL PANEL SINGLE LINE	Reviewed by: RSW004	Review date: 8 Jun 2021
Based on document id: DAAB198406	Customer document id: -			Approved by: KSU004	Approval date: 8 Jun 2021

Language	Object designation	Sheet size	Page	Document Id:	Rev.
EN	BJU909	A3	3 / 10	DECA00006026	C

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SUPPLY					
Un	400	V	2-phase	60	Hz
In	1	A		10	A

Rev.	Date	Created by	Appr. by	Revision text
a	27.7.2021	RSW004	KSU004	Location Box Updated

Project name	Mansourah - Massarah Gold Mine		
Project id	P/20002	Product id	6X20V32TS
Based on document id	DAAB198406	Customer document id	-

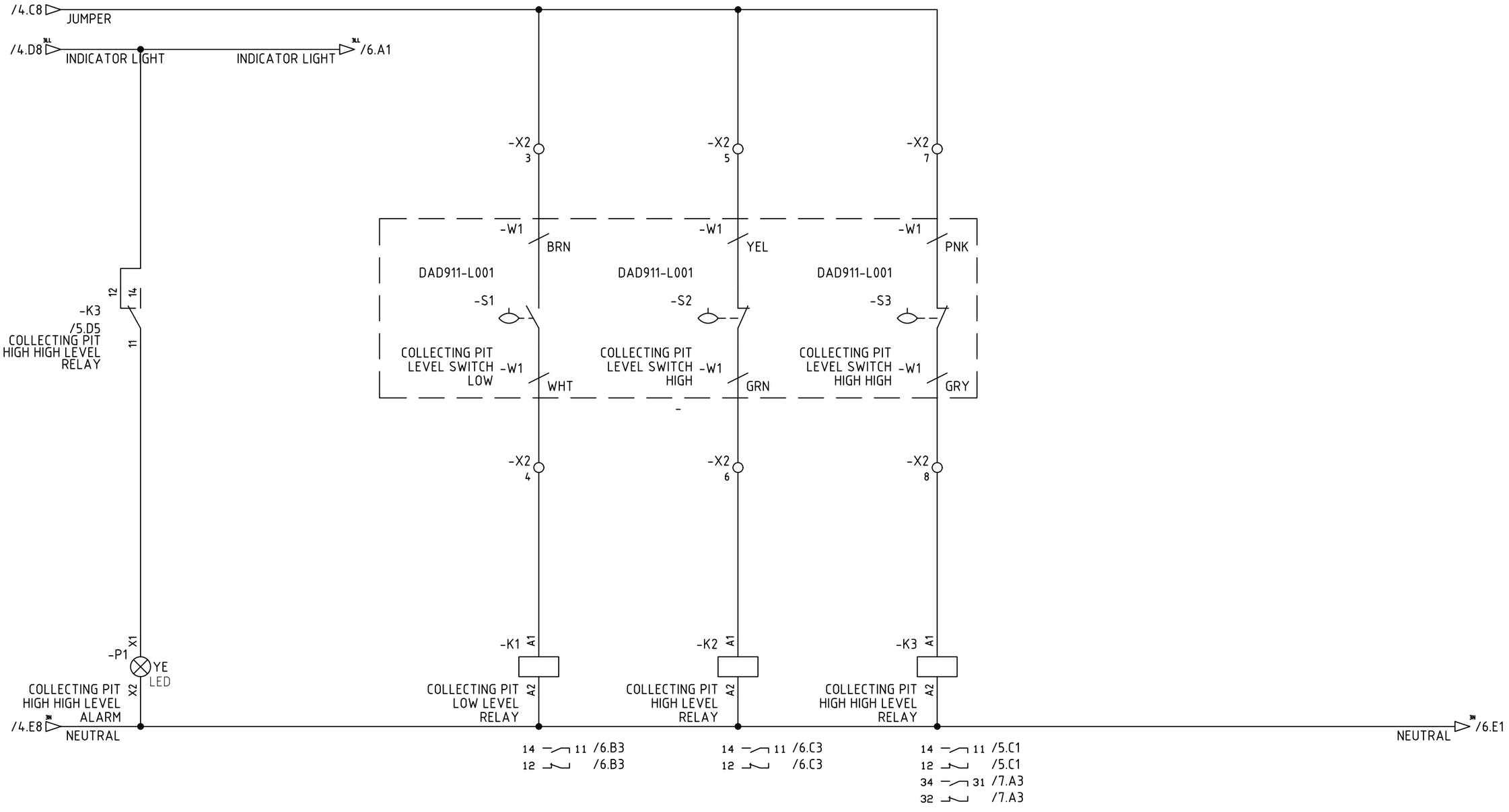
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CIRCUIT DIAGRAM

OILY WATER TRANSFER UNIT
LOCAL CONTROL PANEL
LOW VOLTAGE FEEDER

Created by	MBAX46	Create date	8 Jun 2021
Reviewed by	RSW004	Review date	8 Jun 2021
Approved by	KSU004	Approval date	8 Jun 2021
Language	EN	Object designation	BJU909
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Rev.	Date	Created by	Appr. by	Revision text
a	27.7.2021	RSW004	KSU004	Location Box Updated

Project name	Mansourah - Massarah Gold Mine		
Project id	P/20002	Product id	6X20V32TS
Based on document id	DAAB198406	Customer document id	-

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CIRCUIT DIAGRAM

OILY WATER TRANSFER UNIT
LOCAL CONTROL PANEL
COMMON INDICATOR/CONTROL

Created by	MBAX46	Create date	8 Jun 2021
Reviewed by	RSW004	Review date	8 Jun 2021
Approved by	KSU004	Approval date	8 Jun 2021
Language	EN	Object designation	BJU909
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/4.D8 3L1 CONTROL VOLTAGE

/5.A2 3L1 INDICATOR LIGHT

-K4 /6.D3 TRANSFER PUMP CONTROL RELAY 1 — 12 14 / 11

-K1 /5.D3 COLLECTING PIT LOW LEVEL RELAY — 12 14 / 11

-S1 0-1-S — 21 / 22 — TRANSFER PUMP CONTROL 0-1-S

-S1 — 13 / 14 — /6.B4 TRANSFER PUMP CONTROL 0-1-S

0	1	S	-S1		
			T1	T2	REF
X			21	22	B4
			X1	X2	6E1
		X	13	14	6C4

-K2 /5.D4 COLLECTING PIT HIGH LEVEL RELAY — 12 14 / 11

-K4 /6.D3 TRANSFER PUMP CONTROL RELAY 1 — 22 24 / 21

-K5 /6.D4 TRANSFER PUMP CONTROL RELAY2 — 22 24 / 21

/4.E6 PE

-PE 6 — -X2 9

-W01 3 — -W01 1

-GND — DAD911-CV001

-Y1 X1 / X2 — TRANSFER PUMP CONTROL VALVE

-W01 2

-X2 10

-S1 X1 — /6.B4 WH LED — X2 TRANSFER PUMP

/5.E8 3N 0-1-S NEUTRAL

-K4 A1 — TRANSFER PUMP CONTROL RELAY 1 A2

-K5 A1 — TRANSFER PUMP CONTROL RELAY2 A2

14 — 11 /6.C1
12 — /6.C1
24 — 21 /6.C3
22 — /6.C3

24 — 21 /6.A6
22 — /6.A6
34 — 31 /7.A4
32 — /7.A4

Rev.	Date	Created by	Appr. by	Revision text
a	27.7.2021	RSW004	KSU004	Location Box Updated

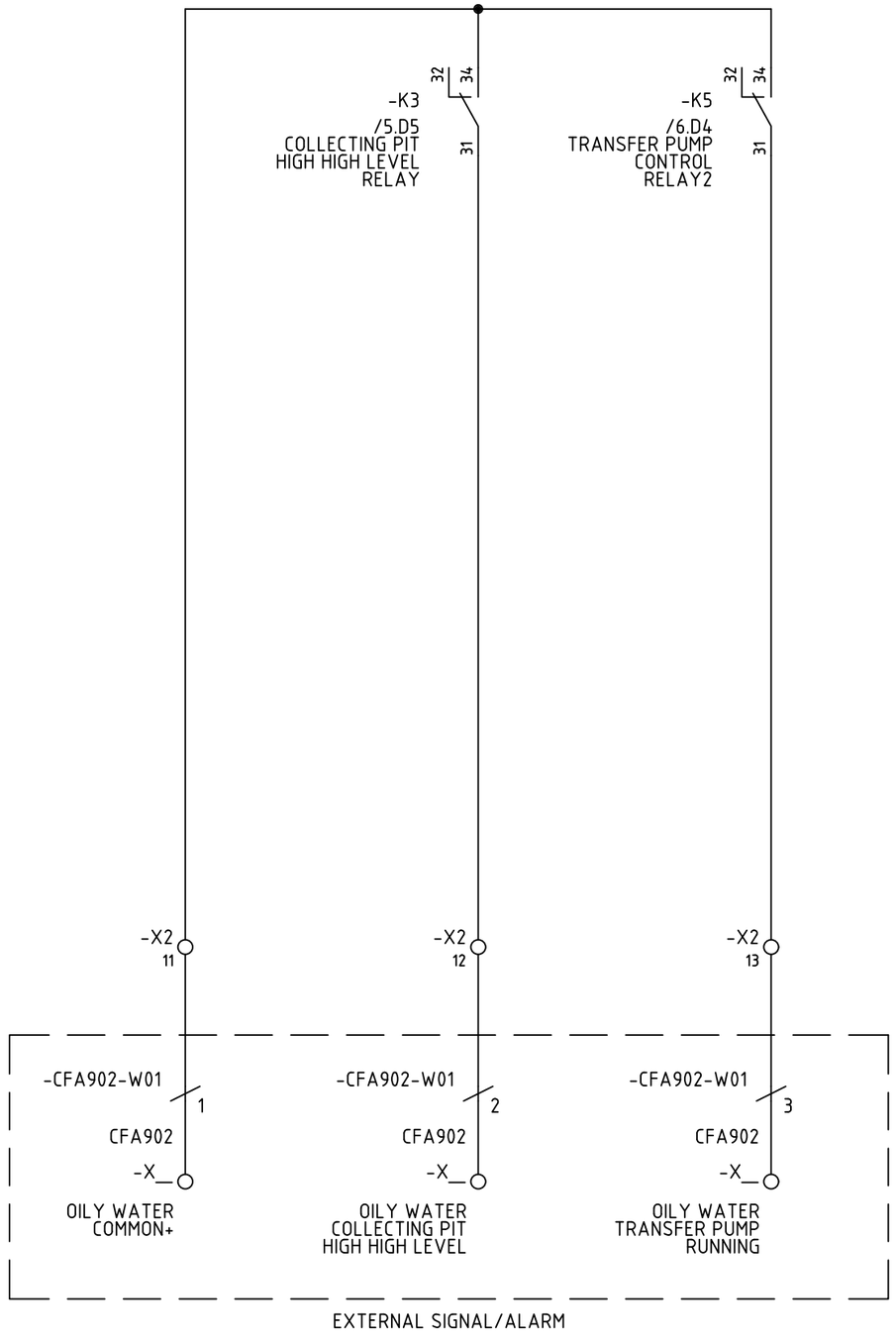
Project name: Mansourah - Massarah Gold Mine
Project id: P/20002 | Product id: 6X20V32TS
Based on document id: DAAB198406 | Customer document id: -

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CIRCUIT DIAGRAM
OILY WATER TRANSFER UNIT
LOCAL CONTROL PANEL
PUMP CONTROL

Created by: MBAX46	Create date: 8 Jun 2021	Language: EN	Object designation: BJU909
Reviewed by: RSW004	Review date: 8 Jun 2021	Sheet size: A3	Page: 6 / 10
Approved by: KSU004	Approval date: 8 Jun 2021	Document Id: DECA00006026	Rev.: C

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Rev.	Date	Created by	Appr. by	Revision text
a	27.7.2021	RSW004	KSU004	Location Box Updated

Project name	Mansourah - Massarah Gold Mine		
Project id	P/20002	Product id	6X20V32TS
Based on document id	DAAB198406	Customer document id	-

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CIRCUIT DIAGRAM

OILY WATER TRANSFER UNIT
LOCAL CONTROL PANEL
EXTERNAL ALARM

Created by	MBAX46	Create date	8 Jun 2021
Reviewed by	RSW004	Review date	8 Jun 2021
Approved by	KSU004	Approval date	8 Jun 2021
Language	EN	Object designation	BJU909
Sheet size	A3	Page	7 / 10
Document Id:	DECA00006026	Rev.	C

FACTORY INSTALLED CABLES TO BE SUPPLIED BY HELKAMA IF NOT OTHERWISE STATED

CABLE SPECIFICATION	CABLE TYPE	CABLE INSTALLATION	CABLE NAME	WIRE	FROM COMPONENT	PIN1	TO COMPONENT	PIN2	SH1	REF1	CABLE REMARK
1kV YY	3G1,5rm		-W01	1	-X2	9	=DAD911-CV001-Y1	X1	6	B6	INSTALLED AT FSITE
			-W01	2	-X2	10	=DAD911-CV001-Y1	X2	6	C6	INSTALLED AT FSITE
			-W01	3	-PE	6	=DAD911-CV001-GND		6	B6	INSTALLED AT FSITE
1kV YCY	3x__+__	BFA902	-BFA902-W01	1	=BFA902-Q1	L1	-X1	1	4	E2	INSTALLED AT SITE
		BFA902	-BFA902-W01	2	=BFA902-Q1	L2	-X1	2	4	E2	INSTALLED AT SITE
		BFA902	-BFA902-W01	4	=BFA902-GND		-PE	1	4	E3	INSTALLED AT SITE
1kV YY	4x1,5rm	CFA902	-CFA902-W01	1	=CFA902-X__		-X2	11	7	E2	INSTALLED AT SITE
		CFA902	-CFA902-W01	2	=CFA902-X__		-X2	12	7	E3	INSTALLED AT SITE
		CFA902	-CFA902-W01	3	=CFA902-X__		-X2	13	7	E4	INSTALLED AT SITE
		DAD911-L001	-W1	BRN	=DAD911-L001-S1		-X2	3	5	C3	INSTALLED AT SITE (INTEGRATED SILICON CABLE)
		DAD911-L001	-W1	GRN	=DAD911-L001-S2		-X2	6	5	C4	INSTALLED AT SITE (INTEGRATED SILICON CABLE)
		DAD911-L001	-W1	GRY	=DAD911-L001-S3		-X2	8	5	C5	INSTALLED AT SITE (INTEGRATED SILICON CABLE)
		DAD911-L001	-W1	PNK	=DAD911-L001-S3		-X2	7	5	C5	INSTALLED AT SITE (INTEGRATED SILICON CABLE)
		DAD911-L001	-W1	WHT	=DAD911-L001-S1		-X2	4	5	C3	INSTALLED AT SITE (INTEGRATED SILICON CABLE)
		DAD911-L001	-W1	YEL	=DAD911-L001-S2		-X2	5	5	C4	INSTALLED AT SITE (INTEGRATED SILICON CABLE)

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Rev.	Date	Created by	Appr. by	Revision text
a	27.7.2021	RSW004	KSU004	Location Box Updated

Project name: Mansourah - Massarah Gold Mine
Project id: P/20002
Product id: 6X20V32TS
Based on document id: DAAB198406
Customer document id: -

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CONNECTION TABLE
OILY WATER TRANSFER UNIT
LOCAL CONTROL PANEL
CABLE LIST

Created by	Create date	Language	Object designation
MBAX46	8 Jun 2021	EN	BJU909
Reviewed by	**Review date**	**Sheet size**	**Page**
RSW004	8 Jun 2021	A3	8 / 10
Approved by	**Approval date**	**Document Id:**	**Rev.**
KSU004	8 Jun 2021	DECA0006026	C

SH1	REF1	CMP1	PIN1	WLAY1	CMP2	PIN2	SH2	REF2
4	C4	-T1	110	AC CONTROL	-F2	1	4	C5
4	C4	-T1	02N	AC CONTROL	-K1	A2	5	E3
4	C5	-F2	1	AC CONTROL	-F3	1	4	C6
4	C5	-F2	2	AC CONTROL	-K3	12	5	C1
4	C6	-F3	2	AC CONTROL	-X2	1	4	D6
4	D6	-X2	1	AC CONTROL	-X2	2	4	D7
4	D7	-X2	2	AC CONTROL	-K1	14	6	B3
5	C1	-K3	11	AC CONTROL	-P1	X1	5	D1
5	C1	-K3	12	AC CONTROL	-K4	14	6	C1
5	E1	-P1	X2	AC CONTROL	-S1	X2	6	E1
5	C3	-X2	4	AC CONTROL	-K1	A1	5	D3
5	E3	-K1	A2	AC CONTROL	-K2	A2	5	E4
5	C4	-X2	6	AC CONTROL	-K2	A1	5	D4
5	E4	-K2	A2	AC CONTROL	-K3	A2	5	E5
5	C5	-X2	8	AC CONTROL	-K3	A1	5	D5
5	E5	-K3	A2	AC CONTROL	-K4	A2	6	E3
6	C1	-K4	11	AC CONTROL	-S1	X1	6	D1
6	B3	-K1	11	AC CONTROL	-S1	21	6	C3
6	B3	-K1	14	AC CONTROL	-K5	24	6	A6
6	C3	-K2	12	AC CONTROL	-K4	24	6	C3
6	C3	-K4	24	AC CONTROL	-S1	22	6	C3
6	D3	-K2	11	AC CONTROL	-K4	21	6	D3
6	D3	-K4	21	AC CONTROL	-K4	A1	6	D3
6	D3	-K4	A1	AC CONTROL	-K5	A1	6	D4
6	E3	-K4	A2	AC CONTROL	-K5	A2	6	E4
6	D4	-K5	A1	AC CONTROL	-S1	14	6	C4
6	E4	-K5	A2	AC CONTROL	-X2	10	6	C6
6	A6	-K5	21	AC CONTROL	-X2	9	6	B6
6	A6	-K5	24	AC CONTROL	-S1	13	6	C4
6	C6	-X2	10	AC CONTROL	-P1	X2	5	E1
7	D2	-X2	11	AC CONTROL	-K3	34	7	A3
7	A3	-K3	34	AC CONTROL	-K5	34	7	A4
7	A3	-K3	31	AC CONTROL	-X2	12	7	D3
7	A4	-K5	31	AC CONTROL	-X2	13	7	D4
4	D7	-X2	2	AC CONTROL JMP	-X2	5	5	B4
5	B3	-X2	3	AC CONTROL JMP	-X2	2	4	D7
5	B4	-X2	5	AC CONTROL JMP	-X2	7	5	B5
4	B2	-Q1	2	AC POWER	-X1	1	4	D2
4	B2	-Q1	1	AC POWER	-F1	1	4	B4
4	B2	-Q1	4	AC POWER	-X1	2	4	D2
4	B2	-Q1	3	AC POWER	-F1	3	4	B4
4	B4	-F1	2	AC POWER	-T1	xxx	4	C4
4	B4	-F1	4	AC POWER	-T1	01N	4	C4
5	C3	-S1		EXTERNAL CONTROL (INTERLOCK)	-X2	3	5	B3
5	C3	-S1		EXTERNAL CONTROL (INTERLOCK)	-X2	4	5	C3
5	C4	-S2		EXTERNAL CONTROL (INTERLOCK)	-X2	5	5	B4
5	C4	-S2		EXTERNAL CONTROL (INTERLOCK)	-X2	6	5	C4
5	C5	-S3		EXTERNAL CONTROL (INTERLOCK)	-X2	7	5	B5
5	C5	-S3		EXTERNAL CONTROL (INTERLOCK)	-X2	8	5	C5
6	B6	-X2	9	EXTERNAL CONTROL (INTERLOCK)	-Y1	X1	6	C6
6	C6	-X2	10	EXTERNAL CONTROL (INTERLOCK)	-Y1	X2	6	C6
7	E2	-X__		EXTERNAL CONTROL (INTERLOCK)	-X2	11	7	D2
7	E3	-X__		EXTERNAL CONTROL (INTERLOCK)	-X2	12	7	D3
7	E4	-X__		EXTERNAL CONTROL (INTERLOCK)	-X2	13	7	D4
4	E2	-Q1	L1	EXTERNAL POWER	-X1	1	4	D2
4	E2	-Q1	L2	EXTERNAL POWER	-X1	2	4	D2
4	E3	-GND		EXTERNAL PROTECTIVE	-PE	1	4	D3
6	B6	-PE	6	EXTERNAL PROTECTIVE	-GND		6	B6
4	D3	-PE	1	PROTECTIVE	-GND		4	D3
4	C4	-T1	02N	PROTECTIVE	-PE	2	4	C3

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Rev.	Date	Created by	Appr. by	Revision text
a	27.7.2021	RSW004	KSU004	Location Box Updated

Project name: Mansourah - Massarah Gold Mine
Project id: P/20002
Product id: 6X20V32TS
Based on document id: DAAB198406
Customer document id: -

WÄRTSILÄ

CONNECTION TABLE
OILY WATER TRANSFER UNIT
LOCAL CONTROL PANEL
INTERNAL CONNECTION LIST

Created by: MBAX46	Create date: 8 Jun 2021	Language: EN	Object designation: BJU909
Reviewed by: RSW004	Review date: 8 Jun 2021	Sheet size: A3	Page: 9 / 10
Approved by: KSU004	Approval date: 8 Jun 2021	Document Id: DECA00006026	Rev.: C

COMPONENT NAME	QTY	MANUFACTURER	PART CODE	RATING	DESC
-E01	1	RITTAL	1038000		ENCLOSURE, 380x600x210
	4	RITTAL	2508020		WALL MOUNTING BRACKET
-F1	1	SCHNEIDER ELECTRIC	A9F05202	2A D-TRIP	MINIATURE CIRCUIT BREAKER
-F2	1	SCHNEIDER ELECTRIC	A9F04101	1A	MINIATURE CIRCUIT BREAKER
-F3	1	SCHNEIDER ELECTRIC	A9F04102	2.0A	MINIATURE CIRCUIT BREAKER
-K1	1	CARLO GAVAZZI	RCP11003115120VAC		UNIVERSAL PLUG-IN RELAY, 3 CO
	1	CARLO GAVAZZI	ZPD12A		SOCKET
-K2	1	CARLO GAVAZZI	RCP11003115120VAC		UNIVERSAL PLUG-IN RELAY, 3 CO
	1	CARLO GAVAZZI	ZPD12A		SOCKET
-K3	1	CARLO GAVAZZI	RCP11003115120VAC		UNIVERSAL PLUG-IN RELAY, 3 CO
	1	CARLO GAVAZZI	ZPD12A		SOCKET
-K4	1	CARLO GAVAZZI	RCP11003115120VAC		UNIVERSAL PLUG-IN RELAY, 3 CO
	1	CARLO GAVAZZI	ZPD12A		SOCKET
-K5	1	CARLO GAVAZZI	RCP11003115120VAC		UNIVERSAL PLUG-IN RELAY, 3 CO
	1	CARLO GAVAZZI	ZPD12A		SOCKET
-P1	1	SCHNEIDER ELECTRIC	XB7EV05GP		MONOLITHIC INDICATOR LIGHT, YELLOW, INTEGRAL LED
-PE	2	PHOENIX CONTACT	0404428	AB/SS	END PIECE (FOR PE TERMINAL BLOCK)
-PE	15	PHOENIX CONTACT	0421029		PE TERMINAL BLOCK
-Q1	1	ABB	OT16FT3+OHYS2PJ		MAIN BREAKER WITH IP65 HANDLE
-S1	1	SCHNEIDER ELECTRIC	ZB5AK1813+ZB5AW0G15		CONTROL SWITCH, 3 POSITION
-T1	1	TRAFOMIC	0423192	100VA	CONTROL VOLTAGE TRANSFORMER
-U1	1	AGRO	60010031	M20	CABLE GLAND, M20
-U2	1	AGRO	60010031	M20	CABLE GLAND, M20
-U3	1	AGRO	60010031	M20	CABLE GLAND, M20
-U4	1	AGRO	60010031	M20	CABLE GLAND, M20
-U5	1	AGRO	60010031	M20	CABLE GLAND, M20
-U6	1	AGRO	60010031	M20	CABLE GLAND, M20
-X1	2	WEIDMULLER	1608830000	ZAP/TW5	END PIECE (FOR TERMINAL BLOCK)
	3	WEIDMULLER	1608620000	ZDU 6	TERMINAL BLOCK, FEED-THROUGH, 6MM2
-X2	2	WEIDMULLER	1632090000	ZAP/TW4	END PIECE (FOR TERMINAL BLOCK)
	20	WEIDMULLER	1632050000	ZDU 4	TERMINAL BLOCK, FEED-THROUGH, 4MM2

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Rev.	Date	Created by	Appr. by	Revision text
b	07.04.2022	MBA123	RSW004	Part list updated
a	27.7.2021	RSW004	KSU004	Location Box Updated

Project name: Mansourah - Massarah Gold Mine
Project id: P/20002
Product id: 6X20V32TS
Based on document id: DAAB198406
Customer document id: -

WÄRTSILÄ

PARTS LIST
OILY WATER TRANSFER UNIT
LOCAL CONTROL PANEL
PARTS LIST

Created by: MBAX46	Create date: 8 Jun 2021	Language: EN	Object designation: BJU909
Reviewed by: RSW004	Review date: 8 Jun 2021	Sheet size: A3	Page: 10 / 10
Approved by: KSU004	Approval date: 8 Jun 2021	Document Id: DECA00006026	Rev.: C

PANEL CODE	DOCUMENT NAME	DOCUMENT DESCRIPTION	SHEET N:o	DOCUMENT N:o	REVISION
BJU910	LIST OF DOCUMENTS	DRAWING LIST	1	DECA00006027	b
BJU910	DIMENSIONAL DRAWING	LAYOUT 1:10	2	DECA00006027	b
BJU910	OVERVIEW DIAGRAM	SINGLE LINE	3	DECA00006027	b
BJU910	CIRCUIT DIAGRAM	LOW VOLTAGE FEEDER	4	DECA00006027	b
BJU910	CIRCUIT DIAGRAM	COMMON INDICATOR/CONTROL	5	DECA00006027	b
BJU910	CIRCUIT DIAGRAM	PUMP CONTROL	6	DECA00006027	b
BJU910	CIRCUIT DIAGRAM	EXTERNAL ALARM	7	DECA00006027	b
BJU910	CONNECTION TABLE	CABLE LIST	8	DECA00006027	b
BJU910	CONNECTION TABLE	INTERNAL CONNECTION LIST	9	DECA00006027	b
BJU910	PARTS LIST	PARTS LIST	10	DECA00006027	b

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Rev.	Date	Created by	Appr. by	Revision text
a	27.7.2021	RSW004	KSU004	Location Box Update

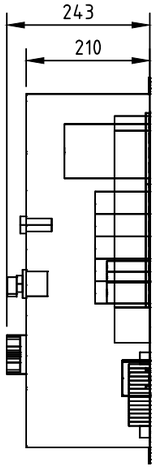
Project name: Mansourah - Massarah Gold Mine
Project id: P/20002
Product id: 6X20V32TS
Based on document id: DAAB198406
Customer document id: -

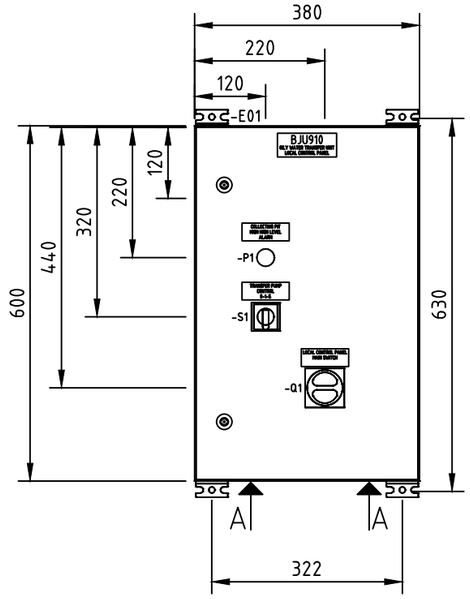
WÄRTSILÄ

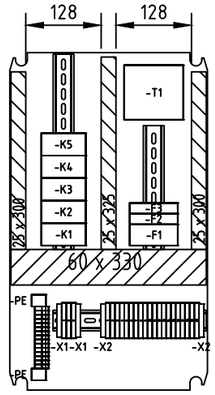
LIST OF DOCUMENTS
OILY WATER TRANSFER UNIT
LOCAL CONTROL PANEL
DRAWING LIST

Created by: MBAX46	Create date: 8 Jun 2021	Language: EN	Object designation: BJU910
Reviewed by: RSW004	Review date: 8 Jun 2021	Sheet size: A3	Page: 1 / 10
Approved by: KSU004	Approval date: 8 Jun 2021	Document Id: DECA00006027	Rev.: C

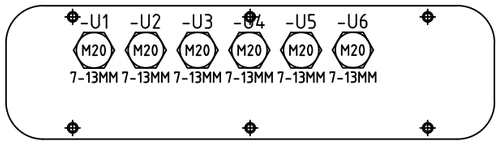
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SECTION A-A SCALE 1:4



PANEL INFORMATION					
WIDTH	380	mm	PROTECTION CLASS		IP65
HEIGHT	600	mm	COLOR		RAL 7035
DEPTH	210	mm	Ik	6	kA/1s

-E01	BJU910	OILY WATER TRANSFER UNIT	LOCAL CONTROL PANEL
-P1	COLLECTING PIT	HIGH HIGH LEVEL	ALARM
-Q1	LOCAL CONTROL PANEL	MAIN SWITCH	
-S1	TRANSFER PUMP	CONTROL	0-1-S

Rev.	Date	Created by	Appr. by	Revision text
a	27.7.2021	RSW004	KSU004	Location Box Update

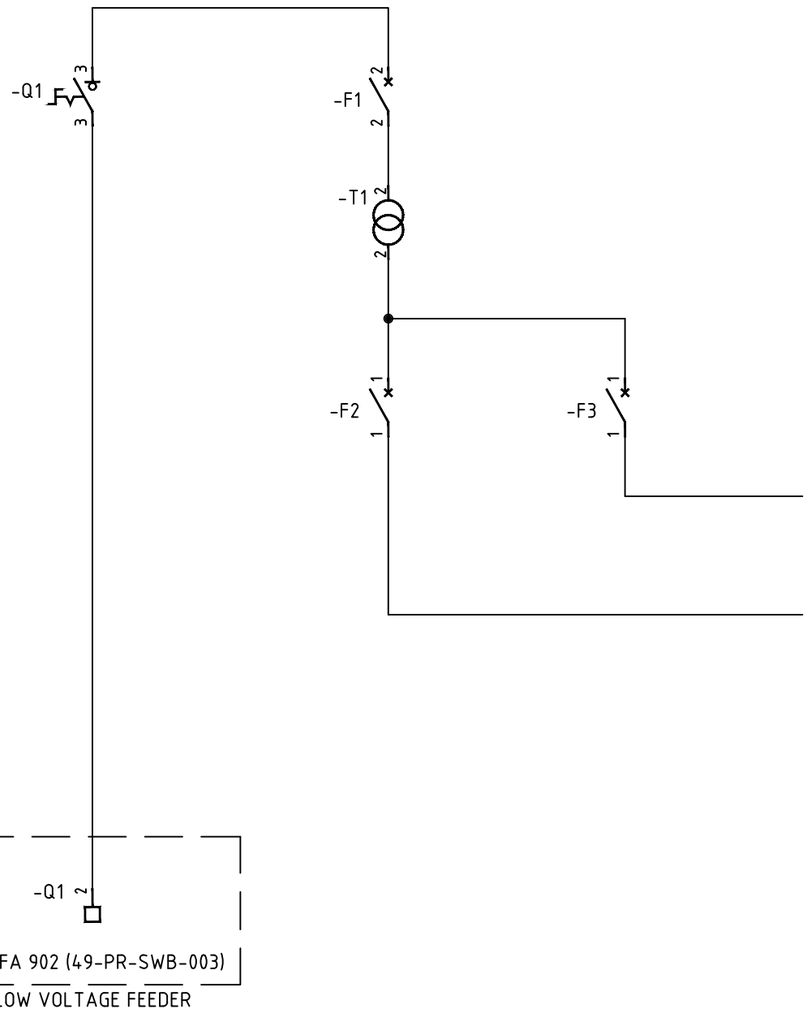
Project name: Mansourah - Massarah Gold Mine
Project id: P/20002
Product id: 6X20V32TS
Based on document id: DAAB198406
Customer document id: -

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DIMENSIONAL DRAWING
OILY WATER TRANSFER UNIT
LOCAL CONTROL PANEL
LAYOUT 1:10

Created by: MBAX46	Create date: 8 Jun 2021	Language: EN	Object designation: BJU910
Reviewed by: RSW004	Review date: 8 Jun 2021	Sheet size: A3	Page: 2 / 10
Approved by: KSU004	Approval date: 8 Jun 2021	Document Id: DECA00006027	Rev.: C

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a	27.7.2021	RSW004	KSU004	Location Box Update

Project name	Mansourah - Massarah Gold Mine		
Project id	P/20002	Product id	6X20V32TS
Based on document id	DAAB198406	Customer document id	-

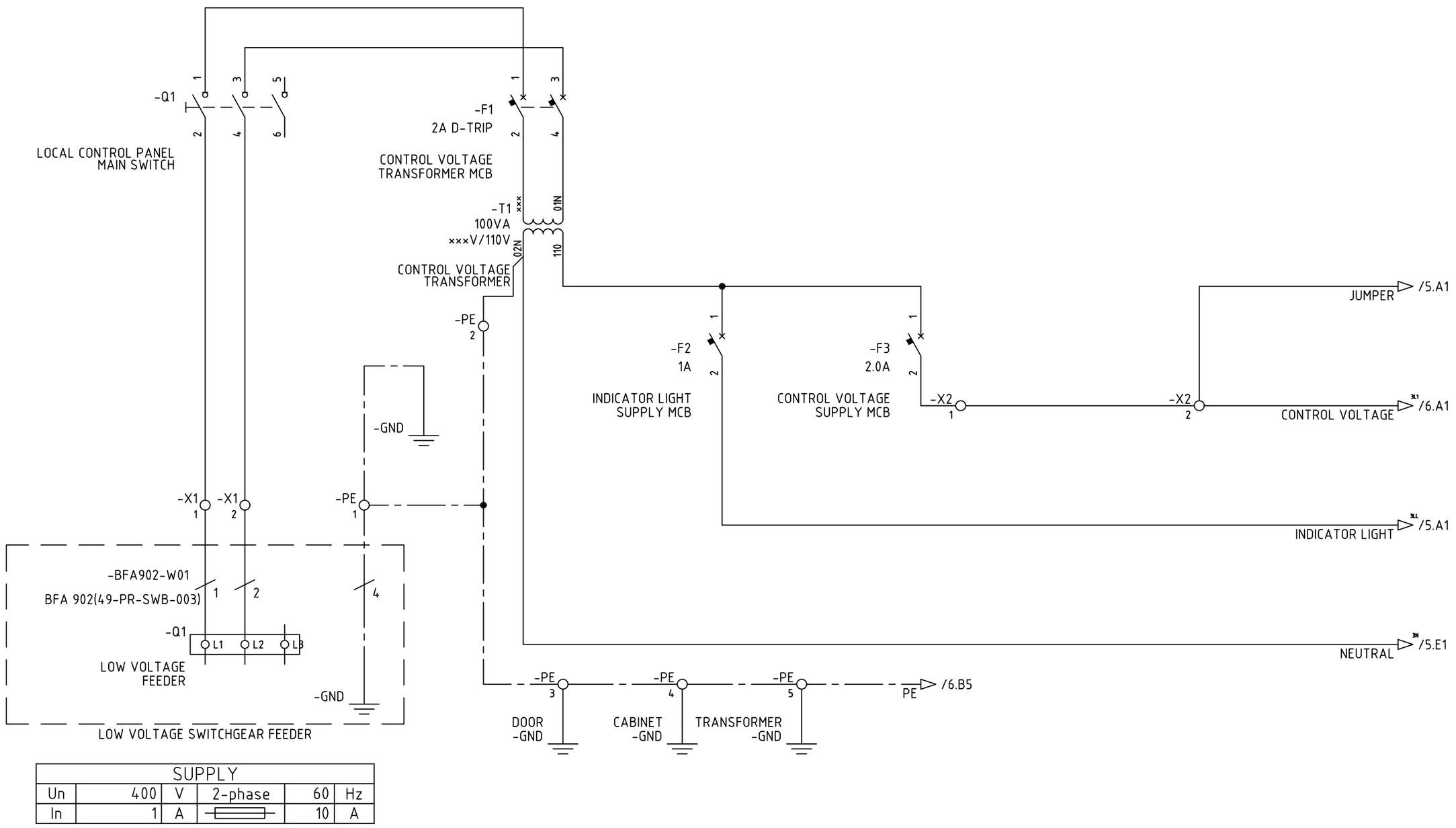
WÄRTSILÄ

OVERVIEW DIAGRAM

OILY WATER TRANSFER UNIT
LOCAL CONTROL PANEL
SINGLE LINE

Created by	MBAX46	Create date	8 Jun 2021
Reviewed by	RSW004	Review date	8 Jun 2021
Approved by	KSU004	Approval date	8 Jun 2021
Language	EN	Object designation	BJU910
Sheet size	A3	Page	3 / 10
Document Id:	DECA0006027	Rev.	C

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SUPPLY					
Un	400	V	2-phase	60	Hz
In	1	A		10	A

Rev.	Date	Created by	Appr. by	Revision text
a	27.7.2021	RSW004	KSU004	Location Box Update

Project name	Mansourah - Massarah Gold Mine		
Project id	P/20002	Product id	6X20V32TS
Based on document id	DAAB198406	Customer document id	-

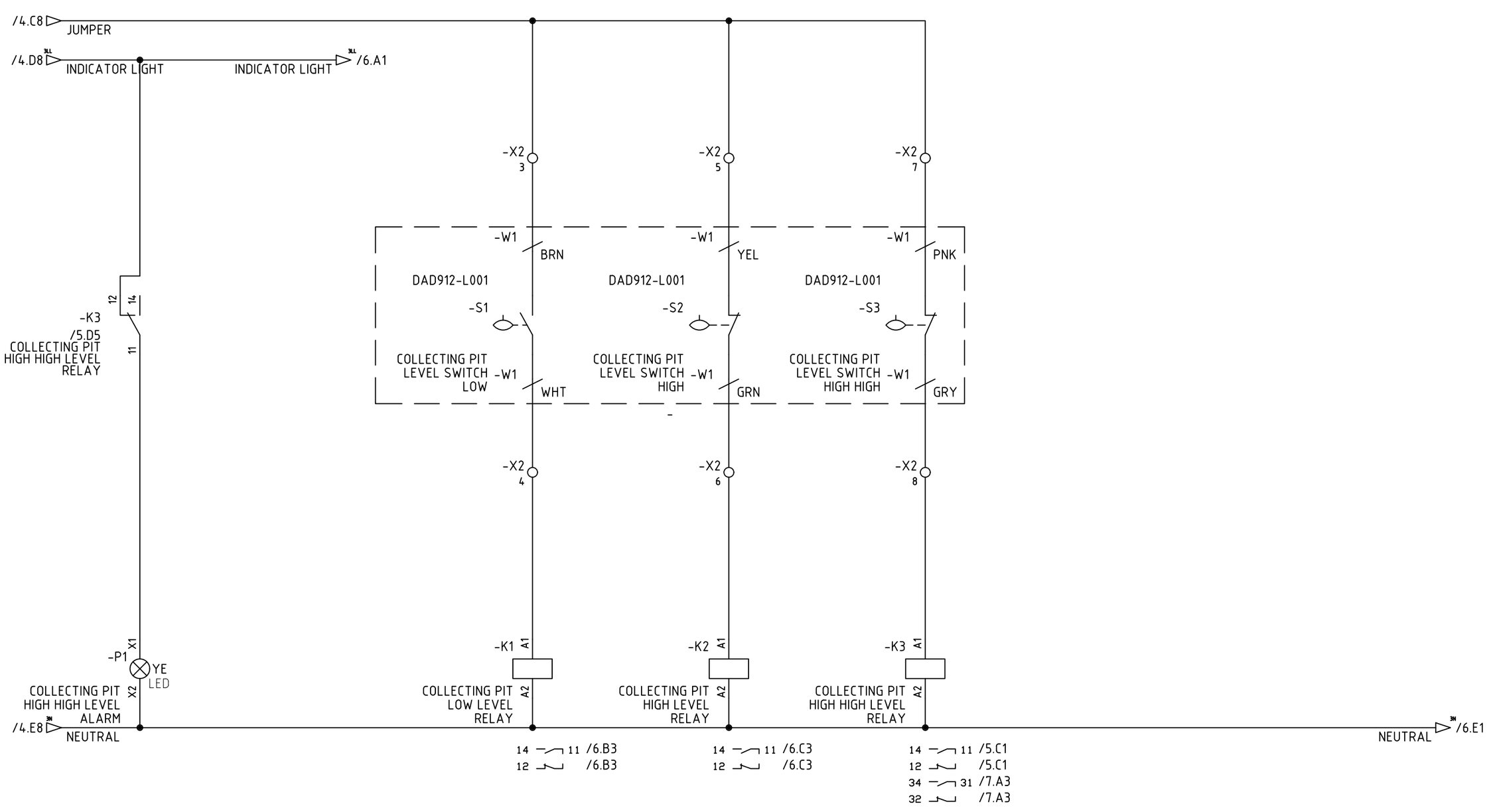
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CIRCUIT DIAGRAM

OILY WATER TRANSFER UNIT
LOCAL CONTROL PANEL
LOW VOLTAGE FEEDER

Created by	MBAX46	Create date	8 Jun 2021
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Approved by	KSU004	Approval date	8 Jun 2021
Language	EN	Object designation	BJU910
Sheet size	A3	Page	4 / 10
Document Id:	DECA00006027	Rev.	C

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Rev.	Date	Created by	Appr. by	Revision text
a	27.7.2021	RSW004	KSU004	Location Box Update

Project name	Mansourah - Massarah Gold Mine		
Project id	P/20002	Product id	6X20V32TS
Based on document id	DAAB198406	Customer document id	-

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CIRCUIT DIAGRAM

OILY WATER TRANSFER UNIT
LOCAL CONTROL PANEL
COMMON INDICATOR/CONTROL

Created by	MBAX46	Create date	8 Jun 2021
Reviewed by	RSW004	Review date	8 Jun 2021
Approved by	KSU004	Approval date	8 Jun 2021
Language	EN	Object designation	BJU910
Sheet size	A3	Page	5 / 10
Document Id:	DECA00006027	Rev.	C

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/4.D8 3L1 CONTROL VOLTAGE

/5.A2 3L1 INDICATOR LIGHT

-K4 /6.D3 TRANSFER PUMP CONTROL RELAY 1 — 12 14 / 11

-K1 /5.D3 COLLECTING PIT LOW LEVEL RELAY — 12 14 / 11

-S1 0-1-S — 21 22 — TRANSFER PUMP CONTROL 0-1-S

-S1 — 13 14 — /6.B4 TRANSFER PUMP CONTROL 0-1-S

-S1					
0	1	S	T1	T2	REF
X			21	22	B4
			X1	X2	6E1
		X	13	14	6C4

-K2 /5.D4 COLLECTING PIT HIGH LEVEL RELAY — 12 14 / 11

-K4 /6.D3 TRANSFER PUMP CONTROL RELAY 1 — 22 24 / 21

-K5 /6.D4 TRANSFER PUMP CONTROL RELAY2 — 22 24 / 21

/4.E6 PE

-PE 6 — -X2 9

-W01 3 — -W01 1

-GND

DAD912-CV001

-Y1 X1 / X2 — TRANSFER PUMP CONTROL VALVE

-W01 2

-X2 10

-S1 X1 — /6.B4 WH LED — X2 TRANSFER PUMP

/5.E8 3N 0-1-S NEUTRAL

-K4 A1 — TRANSFER PUMP CONTROL RELAY 1 A2

-K5 A1 — TRANSFER PUMP CONTROL RELAY2 A2

14 — 11 /6.C1
12 — /6.C1
24 — 21 /6.C3
22 — /6.C3

24 — 21 /6.A6
22 — /6.A6
34 — 31 /7.A4
32 — /7.A4

Rev.	Date	Created by	Appr. by	Revision text
a	27.7.2021	RSW004	KSU004	Location Box Update

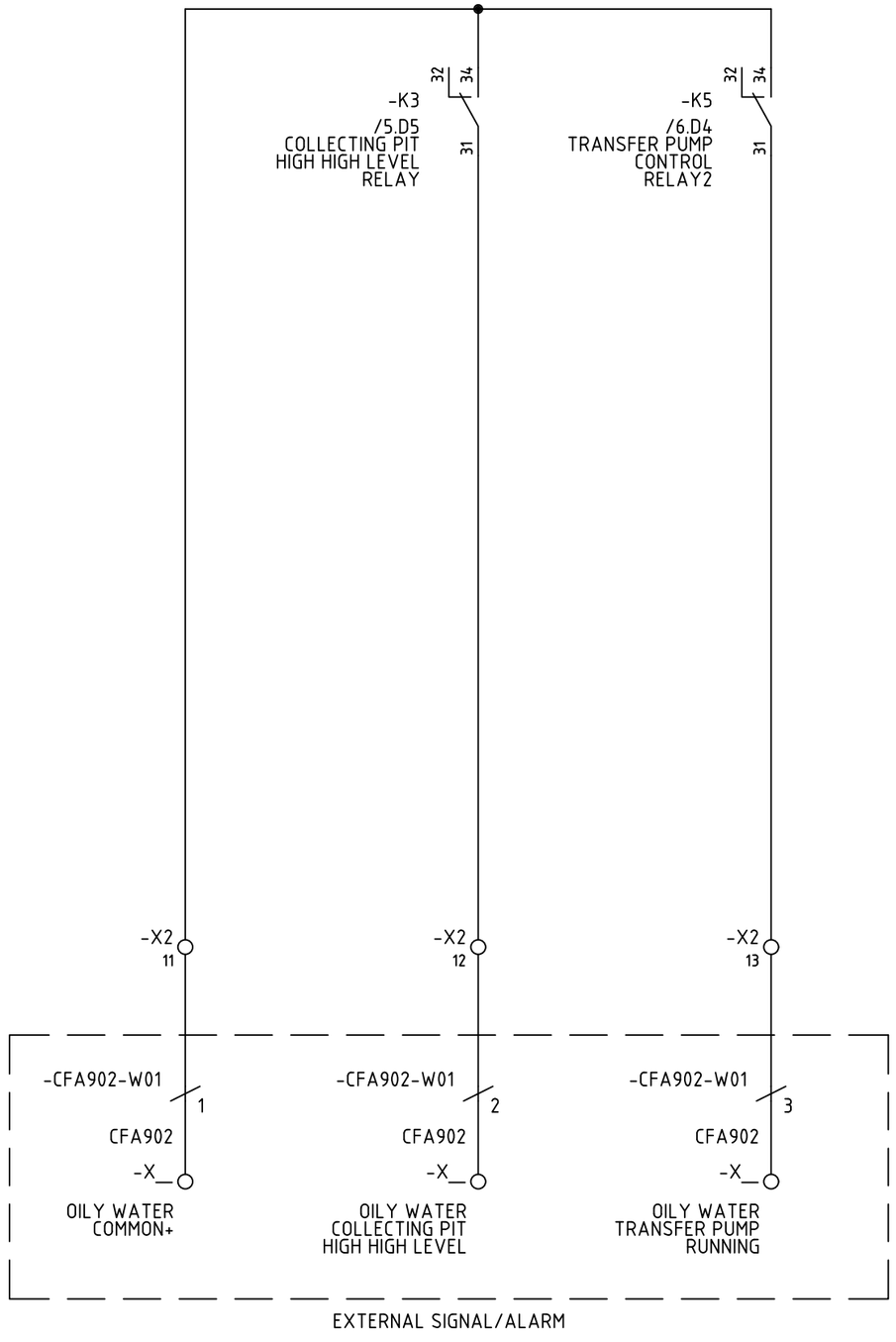
Project name: Mansourah - Massarah Gold Mine
Project id: P/20002 | Product id: 6X20V32TS
Based on document id: DAAB198406 | Customer document id: -

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CIRCUIT DIAGRAM
OILY WATER TRANSFER UNIT
LOCAL CONTROL PANEL
PUMP CONTROL

Created by: MBAX46	Create date: 8 Jun 2021	Language: EN	Object designation: BJU910
Reviewed by: RSW004	Review date: 8 Jun 2021	Sheet size: A3	Page: 6 / 10
Approved by: KSU004	Approval date: 8 Jun 2021	Document Id: DECA00006027	Rev.: C

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Rev.	Date	Created by	Appr. by	Revision text
a	27.7.2021	RSW004	KSU004	Location Box Update

Project name	Mansourah - Massarah Gold Mine			
Project id	P/20002	Product id	6X20V32TS	
Based on document id	DAAB198406	Customer document id	-	

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CIRCUIT DIAGRAM

OILY WATER TRANSFER UNIT
LOCAL CONTROL PANEL
EXTERNAL ALARM

Created by: MBAX46	Create date: 8 Jun 2021	Language: EN	Object designation: BJU910
Reviewed by: RSW004	Review date: 8 Jun 2021	Sheet size: A3	Page: 7 / 10
Approved by: KSU004	Approval date: 8 Jun 2021	Document Id: DECA00006027	Rev.: C

FACTORY INSTALLED CABLES TO BE SUPPLIED BY HELKAMA IF NOT OTHERWISE STATED

CABLE SPECIFICATION	CABLE TYPE	CABLE INSTALLATION	CABLE NAME	WIRE	FROM COMPONENT	PIN1	TO COMPONENT	PIN2	SH1	REF1	CABLE REMARK
1kV YY	3G1,5rm		-W01	1	-X2	9	=DAD912-CV001-Y1	X1	6	B6	INSTALLED AT FSITE
			-W01	2	-X2	10	=DAD912-CV001-Y1	X2	6	C6	INSTALLED AT FSITE
			-W01	3	-PE	6	=DAD912-CV001-GND		6	B6	INSTALLED AT FSITE
1kV YCY	3x__+__	BFA902	-BFA902-W01	1	=BFA902-Q1	L1	-X1	1	4	E2	INSTALLED AT SITE
		BFA902	-BFA902-W01	2	=BFA902-Q1	L2	-X1	2	4	E2	INSTALLED AT SITE
		BFA902	-BFA902-W01	4	=BFA902-GND		-PE	1	4	E3	INSTALLED AT SITE
1kV YY	4x1,5rm	CFA902	-CFA902-W01	1	=CFA902-X__		-X2	11	7	E2	INSTALLED AT SITE
		CFA902	-CFA902-W01	2	=CFA902-X__		-X2	12	7	E3	INSTALLED AT SITE
		CFA902	-CFA902-W01	3	=CFA902-X__		-X2	13	7	E4	INSTALLED AT SITE
		DAD912-L001	-W1	BRN	=DAD912-L001-S1		-X2	3	5	C3	INSTALLED AT SITE (INTEGRATED SILICON CABLE)
		DAD912-L001	-W1	GRN	=DAD912-L001-S2		-X2	6	5	C4	INSTALLED AT SITE (INTEGRATED SILICON CABLE)
		DAD912-L001	-W1	GRY	=DAD912-L001-S3		-X2	8	5	C5	INSTALLED AT SITE (INTEGRATED SILICON CABLE)
		DAD912-L001	-W1	PNK	=DAD912-L001-S3		-X2	7	5	C5	INSTALLED AT SITE (INTEGRATED SILICON CABLE)
		DAD912-L001	-W1	WHT	=DAD912-L001-S1		-X2	4	5	C3	INSTALLED AT SITE (INTEGRATED SILICON CABLE)
		DAD912-L001	-W1	YEL	=DAD912-L001-S2		-X2	5	5	C4	INSTALLED AT SITE (INTEGRATED SILICON CABLE)

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Rev.	Date	Created by	Appr. by	Revision text
a	27.7.2021	RSW004	KSU004	Location Box Update

Project name: Mansourah - Massarah Gold Mine
Project id: P/20002
Product id: 6X20V32TS
Based on document id: DAAB198406
Customer document id: -

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CONNECTION TABLE
OILY WATER TRANSFER UNIT
LOCAL CONTROL PANEL
CABLE LIST

Created by	Create date	Language	Object designation
MBAX46	8 Jun 2021	EN	BJU910
Reviewed by	**Review date**	**Sheet size**	**Page**
RSW004	8 Jun 2021	A3	8 / 10
Approved by	**Approval date**	**Document Id:**	**Rev.**
KSU004	8 Jun 2021	DECA0006027	C

SH1	REF1	CMP1	PIN1	WLAY1	CMP2	PIN2	SH2	REF2
4	C4	-T1	110	AC CONTROL	-F2	1	4	C5
4	C4	-T1	02N	AC CONTROL	-K1	A2	5	E3
4	C5	-F2	1	AC CONTROL	-F3	1	4	C6
4	C5	-F2	2	AC CONTROL	-K3	12	5	C1
4	C6	-F3	2	AC CONTROL	-X2	1	4	D6
4	D6	-X2	1	AC CONTROL	-X2	2	4	D7
4	D7	-X2	2	AC CONTROL	-K1	14	6	B3
5	C1	-K3	11	AC CONTROL	-P1	X1	5	D1
5	C1	-K3	12	AC CONTROL	-K4	14	6	C1
5	E1	-P1	X2	AC CONTROL	-S1	X2	6	E1
5	C3	-X2	4	AC CONTROL	-K1	A1	5	D3
5	E3	-K1	A2	AC CONTROL	-K2	A2	5	E4
5	C4	-X2	6	AC CONTROL	-K2	A1	5	D4
5	E4	-K2	A2	AC CONTROL	-K3	A2	5	E5
5	C5	-X2	8	AC CONTROL	-K3	A1	5	D5
5	E5	-K3	A2	AC CONTROL	-K4	A2	6	E3
6	C1	-K4	11	AC CONTROL	-S1	X1	6	D1
6	B3	-K1	11	AC CONTROL	-S1	21	6	C3
6	B3	-K1	14	AC CONTROL	-K5	24	6	A6
6	C3	-K2	12	AC CONTROL	-K4	24	6	C3
6	C3	-K4	24	AC CONTROL	-S1	22	6	C3
6	D3	-K2	11	AC CONTROL	-K4	21	6	D3
6	D3	-K4	21	AC CONTROL	-K4	A1	6	D3
6	D3	-K4	A1	AC CONTROL	-K5	A1	6	D4
6	E3	-K4	A2	AC CONTROL	-K5	A2	6	E4
6	D4	-K5	A1	AC CONTROL	-S1	14	6	C4
6	E4	-K5	A2	AC CONTROL	-X2	10	6	C6
6	A6	-K5	21	AC CONTROL	-X2	9	6	B6
6	A6	-K5	24	AC CONTROL	-S1	13	6	C4
6	C6	-X2	10	AC CONTROL	-P1	X2	5	E1
7	D2	-X2	11	AC CONTROL	-K3	34	7	A3
7	A3	-K3	34	AC CONTROL	-K5	34	7	A4
7	A3	-K3	31	AC CONTROL	-X2	12	7	D3
7	A4	-K5	31	AC CONTROL	-X2	13	7	D4
4	D7	-X2	2	AC CONTROL JMP	-X2	5	5	B4
5	B3	-X2	3	AC CONTROL JMP	-X2	2	4	D7
5	B4	-X2	5	AC CONTROL JMP	-X2	7	5	B5
4	B2	-Q1	2	AC POWER	-X1	1	4	D2
4	B2	-Q1	1	AC POWER	-F1	1	4	B4
4	B2	-Q1	4	AC POWER	-X1	2	4	D2
4	B2	-Q1	3	AC POWER	-F1	3	4	B4
4	B4	-F1	2	AC POWER	-T1	xxx	4	C4
4	B4	-F1	4	AC POWER	-T1	01N	4	C4
5	C3	-S1		EXTERNAL CONTROL (INTERLOCK)	-X2	3	5	B3
5	C3	-S1		EXTERNAL CONTROL (INTERLOCK)	-X2	4	5	C3
5	C4	-S2		EXTERNAL CONTROL (INTERLOCK)	-X2	5	5	B4
5	C4	-S2		EXTERNAL CONTROL (INTERLOCK)	-X2	6	5	C4
5	C5	-S3		EXTERNAL CONTROL (INTERLOCK)	-X2	7	5	B5
5	C5	-S3		EXTERNAL CONTROL (INTERLOCK)	-X2	8	5	C5
6	B6	-X2	9	EXTERNAL CONTROL (INTERLOCK)	-Y1	X1	6	C6
6	C6	-X2	10	EXTERNAL CONTROL (INTERLOCK)	-Y1	X2	6	C6
7	E2	-X_		EXTERNAL CONTROL (INTERLOCK)	-X2	11	7	D2
7	E3	-X_		EXTERNAL CONTROL (INTERLOCK)	-X2	12	7	D3
7	E4	-X_		EXTERNAL CONTROL (INTERLOCK)	-X2	13	7	D4
4	E2	-Q1	L1	EXTERNAL POWER	-X1	1	4	D2
4	E2	-Q1	L2	EXTERNAL POWER	-X1	2	4	D2
4	E3	-GND		EXTERNAL PROTECTIVE	-PE	1	4	D3
6	B6	-PE	6	EXTERNAL PROTECTIVE	-GND		6	B6
4	D3	-PE	1	PROTECTIVE	-GND		4	D3
4	C4	-T1	02N	PROTECTIVE	-PE	2	4	C3

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Rev.	Date	Created by	Appr. by	Revision text
a	27.7.2021	RSW004	KSU004	Location Box Update

Project name: Mansourah - Massarah Gold Mine
Project id: P/20002
Product id: 6X20V32TS
Based on document id: DAAB198406
Customer document id: -

WÄRTSILÄ

CONNECTION TABLE
OILY WATER TRANSFER UNIT
LOCAL CONTROL PANEL
INTERNAL CONNECTION LIST

Created by: MBAX46	Create date: 8 Jun 2021	Language: EN	Object designation: BJU910
Reviewed by: RSW004	Review date: 8 Jun 2021	Sheet size: A3	Page: 9 / 10
Approved by: KSU004	Approval date: 8 Jun 2021	Document Id: DECA00006027	Rev.: C

COMPONENT NAME	QTY	MANUFACTURER	PART CODE	RATING	DESC
-E01	1	RITTAL	1038000		ENCLOSURE, 380x600x210
	4	RITTAL	2508020		WALL MOUNTING BRACKET
-F1	1	SCHNEIDER ELECTRIC	A9F05202	2A D-TRIP	MINIATURE CIRCUIT BREAKER
-F2	1	SCHNEIDER ELECTRIC	A9F04101	1A	MINIATURE CIRCUIT BREAKER
-F3	1	SCHNEIDER ELECTRIC	A9F04102	2.0A	MINIATURE CIRCUIT BREAKER
-K1	1	CARLO GAVAZZI	RCP11003115120VAC		UNIVERSAL PLUG-IN RELAY, 3 CO
	1	CARLO GAVAZZI	ZPD12A		SOCKET
-K2	1	CARLO GAVAZZI	RCP11003115120VAC		UNIVERSAL PLUG-IN RELAY, 3 CO
	1	CARLO GAVAZZI	ZPD12A		SOCKET
-K3	1	CARLO GAVAZZI	RCP11003115120VAC		UNIVERSAL PLUG-IN RELAY, 3 CO
	1	CARLO GAVAZZI	ZPD12A		SOCKET
-K4	1	CARLO GAVAZZI	RCP11003115120VAC		UNIVERSAL PLUG-IN RELAY, 3 CO
	1	CARLO GAVAZZI	ZPD12A		SOCKET
-K5	1	CARLO GAVAZZI	RCP11003115120VAC		UNIVERSAL PLUG-IN RELAY, 3 CO
	1	CARLO GAVAZZI	ZPD12A		SOCKET
-P1	1	SCHNEIDER ELECTRIC	XB7EV05GP		MONOLITHIC INDICATOR LIGHT, YELLOW, INTEGRAL LED
-PE	2	PHOENIX CONTACT	0404428	AB/SS	END PIECE (FOR PE TERMINAL BLOCK)
-PE	15	PHOENIX CONTACT	0421029		PE TERMINAL BLOCK
-Q1	1	ABB	OT16FT3+OHYS2PJ		MAIN BREAKER WITH IP65 HANDLE
-S1	1	SCHNEIDER ELECTRIC	ZB5AK1813+ZB5AW0G15		CONTROL SWITCH, 3 POSITION
-T1	1	TRAFOMIC	0423192	100VA	CONTROL VOLTAGE TRANSFORMER
-U1	1	AGRO	60010031	M20	CABLE GLAND, M20
-U2	1	AGRO	60010031	M20	CABLE GLAND, M20
-U3	1	AGRO	60010031	M20	CABLE GLAND, M20
-U4	1	AGRO	60010031	M20	CABLE GLAND, M20
-U5	1	AGRO	60010031	M20	CABLE GLAND, M20
-U6	1	AGRO	60010031	M20	CABLE GLAND, M20
-X1	2	WEIDMULLER	1608830000	ZAP/TW5	END PIECE (FOR TERMINAL BLOCK)
	3	WEIDMULLER	1608620000	ZDU 6	TERMINAL BLOCK, FEED-THROUGH, 6MM2
-X2	2	WEIDMULLER	1632090000	ZAP/TW4	END PIECE (FOR TERMINAL BLOCK)
	20	WEIDMULLER	1632050000	ZDU 4	TERMINAL BLOCK, FEED-THROUGH, 4MM2

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Rev.	Date	Created by	Appr. by	Revision text
b	07.04.2022	MBA123	RSW004	Part list updated
a	27.7.2021	RSW004	KSU004	Location Box Update

Project name: Mansourah - Massarah Gold Mine
Project id: P/20002
Product id: 6X20V32TS
Based on document id: DAAB198406
Customer document id: -

WÄRTSILÄ

PARTS LIST
OILY WATER TRANSFER UNIT
LOCAL CONTROL PANEL
PARTS LIST

Created by: MBAX46	Create date: 8 Jun 2021	Language: EN	Object designation: BJU910
Reviewed by: RSW004	Review date: 8 Jun 2021	Sheet size: A3	Page: 10 / 10
Approved by: KSU004	Approval date: 8 Jun 2021	Document Id: DECA0006027	Rev.: C

PANEL CODE	DOCUMENT NAME	DOCUMENT DESCRIPTION	SHEET N:o	DOCUMENT N:o	REVISION
BJU911	LIST OF DOCUMENTS	DRAWING LIST	1	DECA00006028	b
BJU911	DIMENSIONAL DRAWING	LAYOUT 1:10	2	DECA00006028	b
BJU911	OVERVIEW DIAGRAM	SINGLE LINE	3	DECA00006028	b
BJU911	CIRCUIT DIAGRAM	LOW VOLTAGE FEEDER	4	DECA00006028	b
BJU911	CIRCUIT DIAGRAM	COMMON INDICATOR/CONTROL	5	DECA00006028	b
BJU911	CIRCUIT DIAGRAM	PUMP CONTROL	6	DECA00006028	b
BJU911	CIRCUIT DIAGRAM	EXTERNAL ALARM	7	DECA00006028	b
BJU911	CONNECTION TABLE	CABLE LIST	8	DECA00006028	b
BJU911	CONNECTION TABLE	INTERNAL CONNECTION LIST	9	DECA00006028	b
BJU911	PARTS LIST	PARTS LIST	10	DECA00006028	b

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Rev.	Date	Created by	Appr. by	Revision text
a	27.7.2021	RSW004	KSU004	Location Box Updated

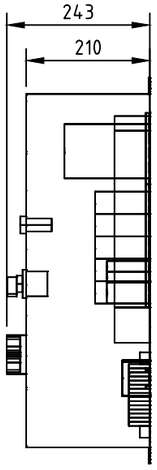
Project name: Mansourah - Massarah Gold Mine
Project id: P/20002
Product id: 6X20V32TS
Based on document id: DAAB198406
Customer document id: -

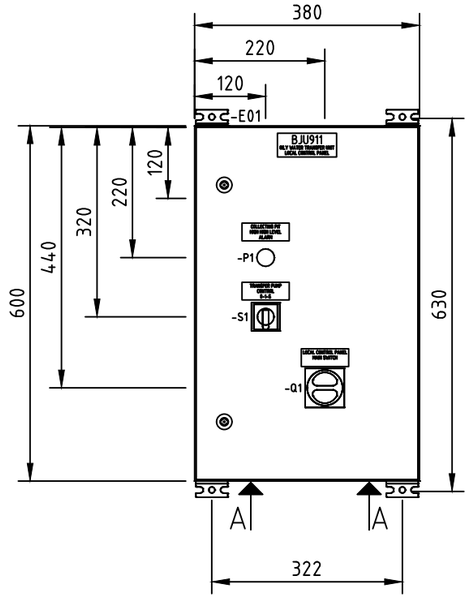
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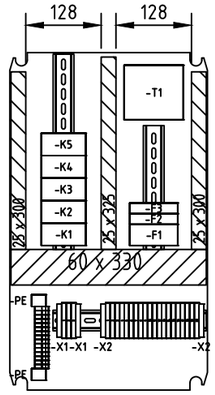
LIST OF DOCUMENTS
OILY WATER TRANSFER UNIT
LOCAL CONTROL PANEL
DRAWING LIST

Created by	Create date	Language	Object designation
MBAX46	8 Jun 2021	EN	BJU911
Reviewed by	**Review date**	**Sheet size**	**Page**
RSW004	8 Jun 2021	A3	1 / 10
Approved by	**Approval date**	**Document Id:**	**Rev.**
KSU004	8 Jun 2021	DECA00006028	C

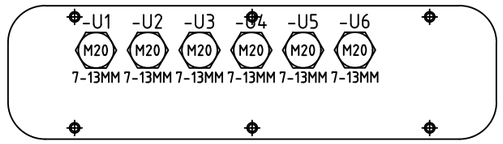
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SECTION A-A SCALE 1:4



PANEL INFORMATION					
WIDTH	380	mm	PROTECTION CLASS		IP65
HEIGHT	600	mm	COLOR		RAL 7035
DEPTH	210	mm	Ik	6	kA/1s

-E01	BJU911	OILY WATER TRANSFER UNIT	LOCAL CONTROL PANEL
-P1	COLLECTING PIT	HIGH HIGH LEVEL	ALARM
-Q1	LOCAL CONTROL PANEL	MAIN SWITCH	
-S1	TRANSFER PUMP	CONTROL	0-1-S

Rev.	Date	Created by	Appr. by	Revision text
a	27.7.2021	RSW004	KSU004	Location Box Updated

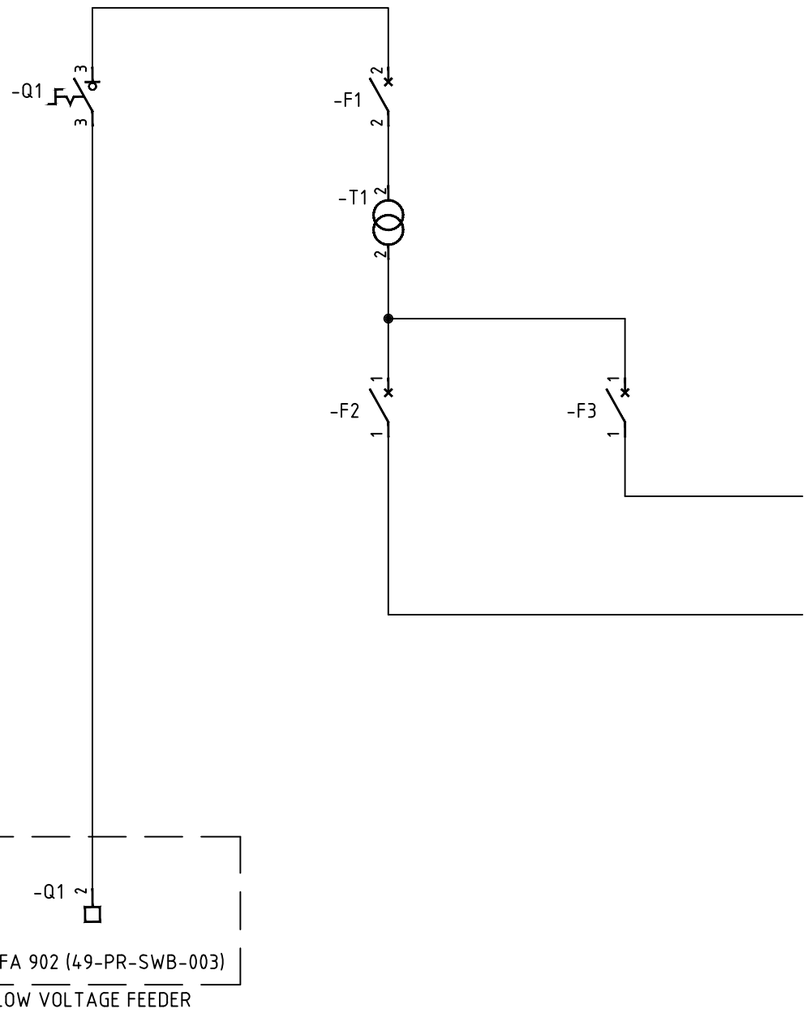
Project name: Mansourah - Massarah Gold Mine
Project id: P/20002
Product id: 6X20V32TS
Based on document id: DAAB198406
Customer document id: -

WÄRTSILÄ

DIMENSIONAL DRAWING
OILY WATER TRANSFER UNIT
LOCAL CONTROL PANEL
LAYOUT 1:10

Created by: MBAX46	Create date: 8 Jun 2021	Language: EN	Object designation: BJU911
Reviewed by: RSW004	Review date: 8 Jun 2021	Sheet size: A3	Page: 2 / 10
Approved by: KSU004	Approval date: 8 Jun 2021	Document Id: DECA00006028	Rev.: C

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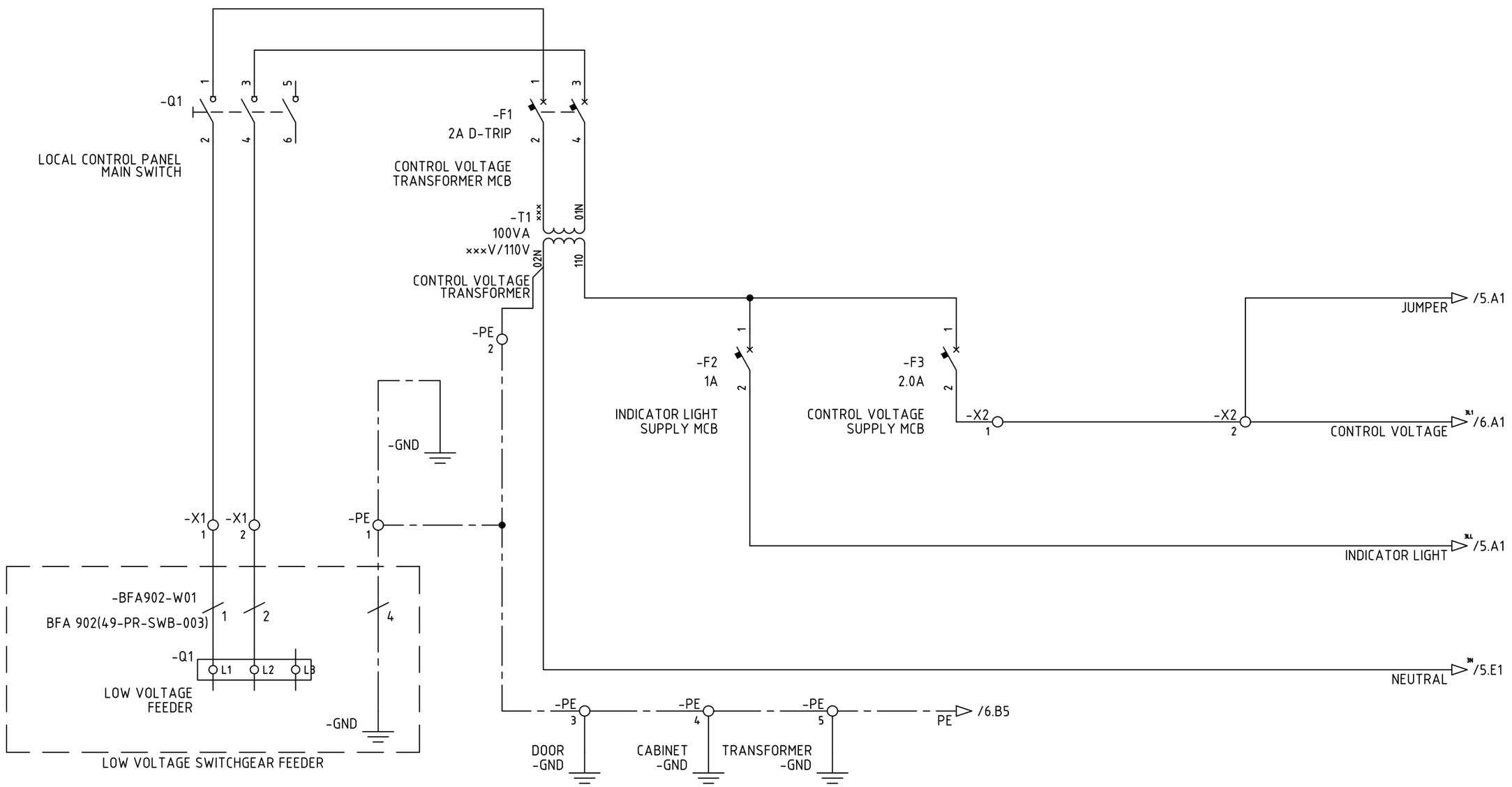


Rev.	Date	Created by	Appr. by	Revision text
a	27.7.2021	RSW004	KSU004	Location Box Updated

Project name: Mansourah - Massarah Gold Mine			OVERVIEW DIAGRAM	Created by: MBAX46	Create date: 8 Jun 2021
Project id: P/20002	Product id: 6X20V32TS	WÄRTSILÄ	OILY WATER TRANSFER UNIT LOCAL CONTROL PANEL SINGLE LINE	Reviewed by: RSW004	Review date: 8 Jun 2021
Based on document id: DAAB198406	Customer document id: -			Approved by: KSU004	Approval date: 8 Jun 2021

Language	Object designation	Sheet size	Page	Document Id:	Rev.
EN	BJU911	A3	3 / 10	DECA0006028	C

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SUPPLY					
Un	400	V	2-phase	60	Hz
In	1	A		10	A

Rev.	Date	Created by	Appr. by	Revision text
a	27.7.2021	RSW004	KSU004	Location Box Updated

Project name	Mansourah - Massarah Gold Mine		
Project id	P/20002	Product id	6X20V32TS
Based on document id	DAAB198406	Customer document id	-

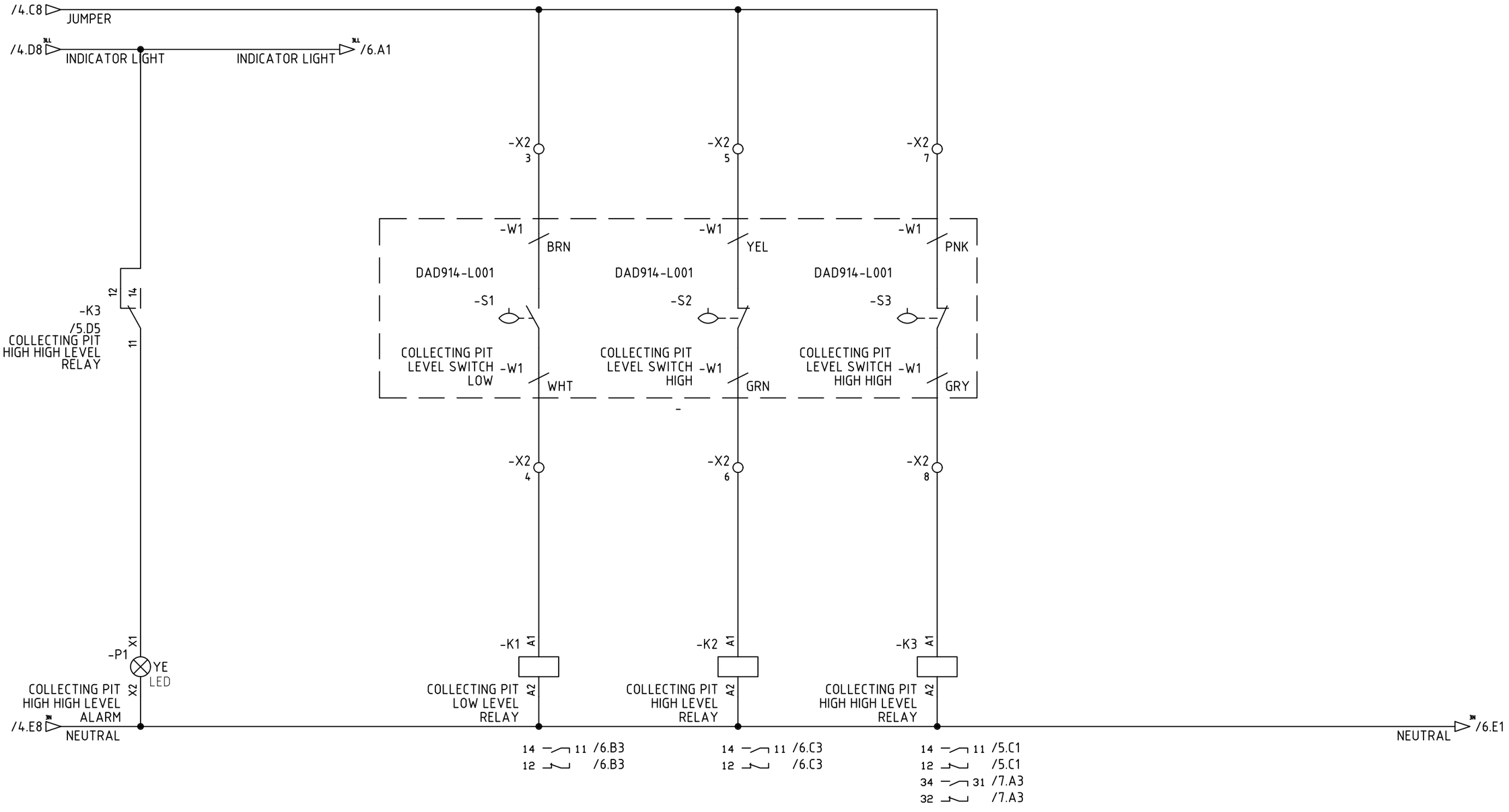
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CIRCUIT DIAGRAM

OILY WATER TRANSFER UNIT
LOCAL CONTROL PANEL
LOW VOLTAGE FEEDER

Created by	MBAX46	Create date	8 Jun 2021
Reviewed by	RSW004	Review date	8 Jun 2021
Approved by	KSU004	Approval date	8 Jun 2021
Language	EN	Object designation	BJU911
Sheet size	A3	Page	4 / 10
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a	27.7.2021	RSW004	KSU004	Location Box Updated

Project name	Mansourah - Massarah Gold Mine		
Project id	P/20002	Product id	6X20V32TS
Based on document id	DAAB198406	Customer document id	-

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CIRCUIT DIAGRAM

OILY WATER TRANSFER UNIT
LOCAL CONTROL PANEL
COMMON INDICATOR/CONTROL

Created by	MBAX46	Create date	8 Jun 2021
Reviewed by	RSW004	Review date	8 Jun 2021
Approved by	KSU004	Approval date	8 Jun 2021
Language	EN	Object designation	BJU911
Sheet size	A3	Page	5 / 10
Document Id:	DECA00006028	Rev.	C

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/4.D8 3L1
CONTROL VOLTAGE

/5.A2 3L1
INDICATOR LIGHT

-K4
/6.D3
TRANSFER PUMP
CONTROL
RELAY 1
12 14 11

-K1
/5.D3
COLLECTING PIT
LOW LEVEL
RELAY
12 14 11

-S1 0-1-S
TRANSFER PUMP
CONTROL
0-1-S
21 22

-S1
/6.B4
TRANSFER PUMP
CONTROL
0-1-S
13 14

-S1					
0	1	S	T1	T2	REF
X			21	22	B4
			X1	X2	6E1
		X	13	14	6C4

-K2
/5.D4
COLLECTING PIT
HIGH LEVEL
RELAY
12 14 11

-K4
/6.D3
TRANSFER PUMP
CONTROL
RELAY 1
22 24 21

-K5
/6.D4
TRANSFER PUMP
CONTROL
RELAY2
22 24 21

/4.E6
PE

-PE 6
-X2 9
-W01 3
-W01 1
-GND
DAD914-CV001
-Y1 X1 X2
TRANSFER PUMP
CONTROL VALVE
-W01 2
-X2 10

-S1 X1
/6.B4
WH
LED
TRANSFER PUMP X2

/5.E8 3N
0-1-S
NEUTRAL

-K4 A1 A2
TRANSFER PUMP
CONTROL
RELAY 1

-K5 A1 A2
TRANSFER PUMP
CONTROL
RELAY2

14 — 11 /6.C1
12 — /6.C1
24 — 21 /6.C3
22 — /6.C3

24 — 21 /6.A6
22 — /6.A6
34 — 31 /7.A4
32 — /7.A4

Rev.	Date	Created by	Appr. by	Revision text
a	27.7.2021	RSW004	KSU004	Location Box Updated

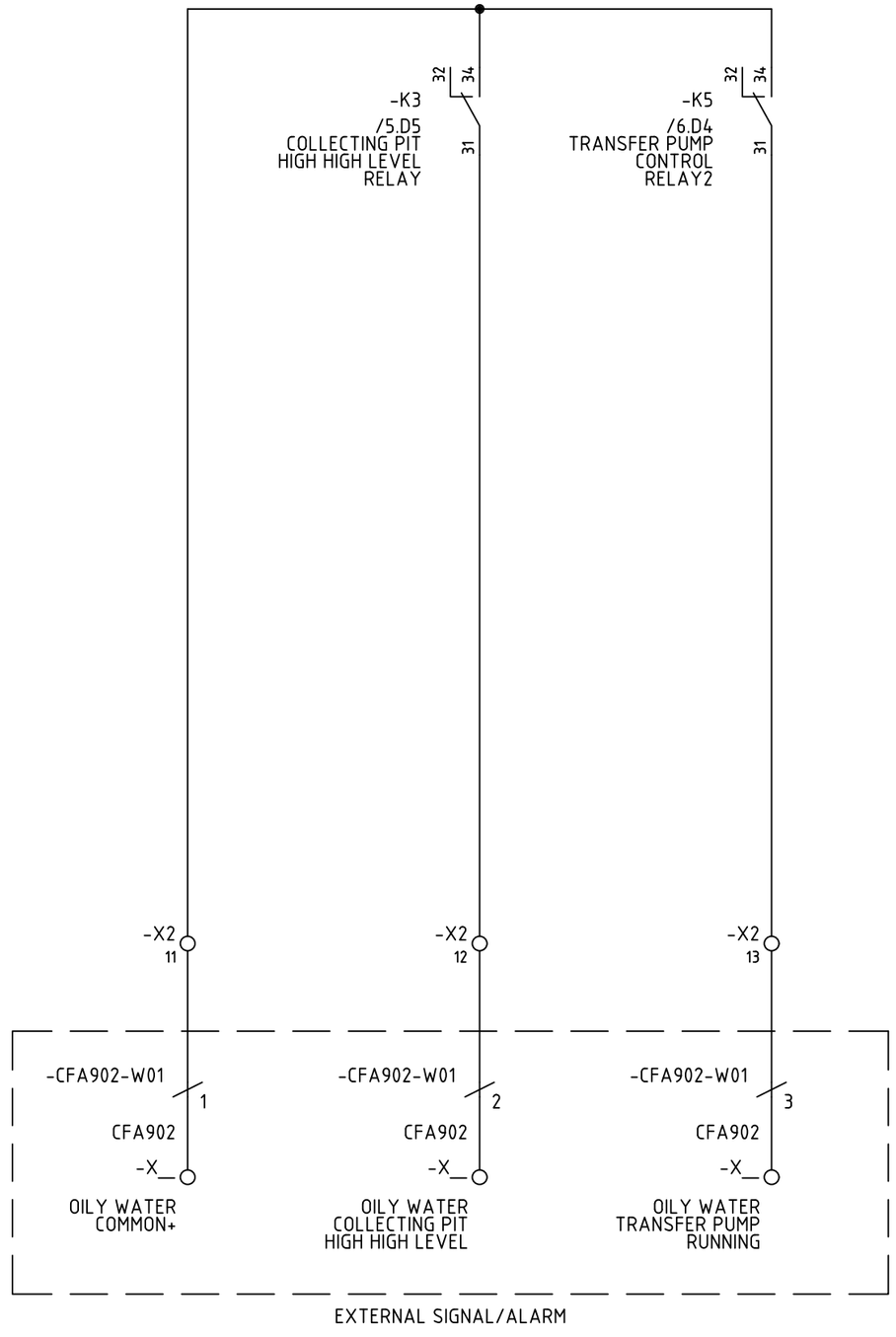
Project name: Mansourah - Massarah Gold Mine
Project id: P/20002
Product id: 6X20V32TS
Based on document id: DAAB198406
Customer document id: -

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CIRCUIT DIAGRAM
OILY WATER TRANSFER UNIT
LOCAL CONTROL PANEL
PUMP CONTROL

Created by: MBAX46	Create date: 8 Jun 2021	Language: EN	Object designation: BJU911
Reviewed by: RSW004	Review date: 8 Jun 2021	Sheet size: A3	Page: 6 / 10
Approved by: KSU004	Approval date: 8 Jun 2021	Document Id: DECA00006028	Rev.: C

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Rev.	Date	Created by	Appr. by	Revision text
a	27.7.2021	RSW004	KSU004	Location Box Updated

Project name	Mansourah - Massarah Gold Mine			
Project id	P/20002	Product id	6X20V32TS	
Based on document id	DAAB198406	Customer document id	-	

WÄRTSILÄ

CIRCUIT DIAGRAM

OILY WATER TRANSFER UNIT
LOCAL CONTROL PANEL
EXTERNAL ALARM

Created by	MBAX46	Create date	8 Jun 2021
Reviewed by	RSW004	Review date	8 Jun 2021
Approved by	KSU004	Approval date	8 Jun 2021
Language	EN	Object designation	BJU911
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Document Id:	DECA00006028	Rev.	C

FACTORY INSTALLED CABLES TO BE SUPPLIED BY HELKAMA IF NOT OTHERWISE STATED

CABLE SPECIFICATION	CABLE TYPE	CABLE INSTALLATION	CABLE NAME	WIRE	FROM COMPONENT	PIN1	TO COMPONENT	PIN2	SH1	REF1	CABLE REMARK
1kV YY	3G1,5rm		-W01	1	-X2	9	=DAD914-CV001-Y1	X1	6	B6	INSTALLED AT FSITE
			-W01	2	-X2	10	=DAD914-CV001-Y1	X2	6	C6	INSTALLED AT FSITE
			-W01	3	-PE	6	=DAD914-CV001-GND		6	B6	INSTALLED AT FSITE
1kV YCY	3x__+__	BFA902	-BFA902-W01	1	=BFA902-Q1	L1	-X1	1	4	E2	INSTALLED AT SITE
		BFA902	-BFA902-W01	2	=BFA902-Q1	L2	-X1	2	4	E2	INSTALLED AT SITE
		BFA902	-BFA902-W01	4	=BFA902-GND		-PE	1	4	E3	INSTALLED AT SITE
1kV YY	4x1,5rm	CFA902	-CFA902-W01	1	=CFA902-X__		-X2	11	7	E2	INSTALLED AT SITE
		CFA902	-CFA902-W01	2	=CFA902-X__		-X2	12	7	E3	INSTALLED AT SITE
		CFA902	-CFA902-W01	3	=CFA902-X__		-X2	13	7	E4	INSTALLED AT SITE
		DAD914-L001	-W1	BRN	=DAD914-L001-S1		-X2	3	5	C3	INSTALLED AT SITE (INTEGRATED SILICON CABLE)
		DAD914-L001	-W1	GRN	=DAD914-L001-S2		-X2	6	5	C4	INSTALLED AT SITE (INTEGRATED SILICON CABLE)
		DAD914-L001	-W1	GRY	=DAD914-L001-S3		-X2	8	5	C5	INSTALLED AT SITE (INTEGRATED SILICON CABLE)
		DAD914-L001	-W1	PNK	=DAD914-L001-S3		-X2	7	5	C5	INSTALLED AT SITE (INTEGRATED SILICON CABLE)
		DAD914-L001	-W1	WHT	=DAD914-L001-S1		-X2	4	5	C3	INSTALLED AT SITE (INTEGRATED SILICON CABLE)
		DAD914-L001	-W1	YEL	=DAD914-L001-S2		-X2	5	5	C4	INSTALLED AT SITE (INTEGRATED SILICON CABLE)

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Rev.	Date	Created by	Appr. by	Revision text
a	27.7.2021	RSW004	KSU004	Location Box Updated

Project name: Mansourah - Massarah Gold Mine
Project id: P/20002
Product id: 6X20V32TS
Based on document id: DAAB198406
Customer document id: -

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CONNECTION TABLE
OILY WATER TRANSFER UNIT
LOCAL CONTROL PANEL
CABLE LIST

Created by	Create date	Language	Object designation
MBAX46	8 Jun 2021	EN	BJU911
Reviewed by	**Review date**	**Sheet size**	**Page**
RSW004	8 Jun 2021	A3	8 / 10
Approved by	**Approval date**	**Document Id:**	**Rev.**
KSU004	8 Jun 2021	DECA0006028	C

SH1	REF1	CMP1	PIN1	WLAY1	CMP2	PIN2	SH2	REF2
4	C4	-T1	110	AC CONTROL	-F2	1	4	C5
4	C4	-T1	02N	AC CONTROL	-K1	A2	5	E3
4	C5	-F2	1	AC CONTROL	-F3	1	4	C6
4	C5	-F2	2	AC CONTROL	-K3	12	5	C1
4	C6	-F3	2	AC CONTROL	-X2	1	4	D6
4	D6	-X2	1	AC CONTROL	-X2	2	4	D7
4	D7	-X2	2	AC CONTROL	-K1	14	6	B3
5	C1	-K3	11	AC CONTROL	-P1	X1	5	D1
5	C1	-K3	12	AC CONTROL	-K4	14	6	C1
5	E1	-P1	X2	AC CONTROL	-S1	X2	6	E1
5	C3	-X2	4	AC CONTROL	-K1	A1	5	D3
5	E3	-K1	A2	AC CONTROL	-K2	A2	5	E4
5	C4	-X2	6	AC CONTROL	-K2	A1	5	D4
5	E4	-K2	A2	AC CONTROL	-K3	A2	5	E5
5	C5	-X2	8	AC CONTROL	-K3	A1	5	D5
5	E5	-K3	A2	AC CONTROL	-K4	A2	6	E3
6	C1	-K4	11	AC CONTROL	-S1	X1	6	D1
6	B3	-K1	11	AC CONTROL	-S1	21	6	C3
6	B3	-K1	14	AC CONTROL	-K5	24	6	A6
6	C3	-K2	12	AC CONTROL	-K4	24	6	C3
6	C3	-K4	24	AC CONTROL	-S1	22	6	C3
6	D3	-K2	11	AC CONTROL	-K4	21	6	D3
6	D3	-K4	21	AC CONTROL	-K4	A1	6	D3
6	D3	-K4	A1	AC CONTROL	-K5	A1	6	D4
6	E3	-K4	A2	AC CONTROL	-K5	A2	6	E4
6	D4	-K5	A1	AC CONTROL	-S1	14	6	C4
6	E4	-K5	A2	AC CONTROL	-X2	10	6	C6
6	A6	-K5	21	AC CONTROL	-X2	9	6	B6
6	A6	-K5	24	AC CONTROL	-S1	13	6	C4
6	C6	-X2	10	AC CONTROL	-P1	X2	5	E1
7	D2	-X2	11	AC CONTROL	-K3	34	7	A3
7	A3	-K3	34	AC CONTROL	-K5	34	7	A4
7	A3	-K3	31	AC CONTROL	-X2	12	7	D3
7	A4	-K5	31	AC CONTROL	-X2	13	7	D4
4	D7	-X2	2	AC CONTROL JMP	-X2	5	5	B4
5	B3	-X2	3	AC CONTROL JMP	-X2	2	4	D7
5	B4	-X2	5	AC CONTROL JMP	-X2	7	5	B5
4	B2	-Q1	2	AC POWER	-X1	1	4	D2
4	B2	-Q1	1	AC POWER	-F1	1	4	B4
4	B2	-Q1	4	AC POWER	-X1	2	4	D2
4	B2	-Q1	3	AC POWER	-F1	3	4	B4
4	B4	-F1	2	AC POWER	-T1	xxx	4	C4
4	B4	-F1	4	AC POWER	-T1	01N	4	C4
5	C3	-S1		EXTERNAL CONTROL (INTERLOCK)	-X2	3	5	B3
5	C3	-S1		EXTERNAL CONTROL (INTERLOCK)	-X2	4	5	C3
5	C4	-S2		EXTERNAL CONTROL (INTERLOCK)	-X2	5	5	B4
5	C4	-S2		EXTERNAL CONTROL (INTERLOCK)	-X2	6	5	C4
5	C5	-S3		EXTERNAL CONTROL (INTERLOCK)	-X2	7	5	B5
5	C5	-S3		EXTERNAL CONTROL (INTERLOCK)	-X2	8	5	C5
6	B6	-X2	9	EXTERNAL CONTROL (INTERLOCK)	-Y1	X1	6	C6
6	C6	-X2	10	EXTERNAL CONTROL (INTERLOCK)	-Y1	X2	6	C6
7	E2	-X_		EXTERNAL CONTROL (INTERLOCK)	-X2	11	7	D2
7	E3	-X_		EXTERNAL CONTROL (INTERLOCK)	-X2	12	7	D3
7	E4	-X_		EXTERNAL CONTROL (INTERLOCK)	-X2	13	7	D4
4	E2	-Q1	L1	EXTERNAL POWER	-X1	1	4	D2
4	E2	-Q1	L2	EXTERNAL POWER	-X1	2	4	D2
4	E3	-GND		EXTERNAL PROTECTIVE	-PE	1	4	D3
6	B6	-PE	6	EXTERNAL PROTECTIVE	-GND		6	B6
4	D3	-PE	1	PROTECTIVE	-GND		4	D3
4	C4	-T1	02N	PROTECTIVE	-PE	2	4	C3

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Rev.	Date	Created by	Appr. by	Revision text
a	27.7.2021	RSW004	KSU004	Location Box Updated

Project name: Mansourah - Massarah Gold Mine
Project id: P/20002
Product id: 6X20V32TS
Based on document id: DAAB198406
Customer document id: -

WÄRTSILÄ

CONNECTION TABLE
OILY WATER TRANSFER UNIT
LOCAL CONTROL PANEL
INTERNAL CONNECTION LIST

Created by: MBAX46	Create date: 8 Jun 2021	Language: EN	Object designation: BJU911
Reviewed by: RSW004	Review date: 8 Jun 2021	Sheet size: A3	Page: 9 / 10
Approved by: KSU004	Approval date: 8 Jun 2021	Document Id: DECA00006028	Rev.: C

COMPONENT NAME	QTY	MANUFACTURER	PART CODE	RATING	DESC
-E01	1	RITTAL	1038000		ENCLOSURE, 380x600x210
	4	RITTAL	2508020		WALL MOUNTING BRACKET
-F1	1	SCHNEIDER ELECTRIC	A9F05202	2A D-TRIP	MINIATURE CIRCUIT BREAKER
-F2	1	SCHNEIDER ELECTRIC	A9F04101	1A	MINIATURE CIRCUIT BREAKER
-F3	1	SCHNEIDER ELECTRIC	A9F04102	2.0A	MINIATURE CIRCUIT BREAKER
-K1	1	CARLO GAVAZZI	RCP11003115120VAC		UNIVERSAL PLUG-IN RELAY, 3 CO
	1	CARLO GAVAZZI	ZPD12A		SOCKET
-K2	1	CARLO GAVAZZI	RCP11003115120VAC		UNIVERSAL PLUG-IN RELAY, 3 CO
	1	CARLO GAVAZZI	ZPD12A		SOCKET
-K3	1	CARLO GAVAZZI	RCP11003115120VAC		UNIVERSAL PLUG-IN RELAY, 3 CO
	1	CARLO GAVAZZI	ZPD12A		SOCKET
-K4	1	CARLO GAVAZZI	RCP11003115120VAC		UNIVERSAL PLUG-IN RELAY, 3 CO
	1	CARLO GAVAZZI	ZPD12A		SOCKET
-K5	1	CARLO GAVAZZI	RCP11003115120VAC		UNIVERSAL PLUG-IN RELAY, 3 CO
	1	CARLO GAVAZZI	ZPD12A		SOCKET
-P1	1	SCHNEIDER ELECTRIC	XB7EV05GP		MONOLITHIC INDICATOR LIGHT, YELLOW, INTEGRAL LED
-PE	2	PHOENIX CONTACT	0404428	AB/SS	END PIECE (FOR PE TERMINAL BLOCK)
-PE	15	PHOENIX CONTACT	0421029		PE TERMINAL BLOCK
-Q1	1	ABB	OT16FT3+OHYS2PJ		MAIN BREAKER WITH IP65 HANDLE
-S1	1	SCHNEIDER ELECTRIC	ZB5AK1813+ZB5AW0G15		CONTROL SWITCH, 3 POSITION
-T1	1	TRAFOMIC	0423192	100VA	CONTROL VOLTAGE TRANSFORMER
-U1	1	AGRO	60010031	M20	CABLE GLAND, M20
-U2	1	AGRO	60010031	M20	CABLE GLAND, M20
-U3	1	AGRO	60010031	M20	CABLE GLAND, M20
-U4	1	AGRO	60010031	M20	CABLE GLAND, M20
-U5	1	AGRO	60010031	M20	CABLE GLAND, M20
-U6	1	AGRO	60010031	M20	CABLE GLAND, M20
-X1	2	WEIDMULLER	1608830000	ZAP/TW5	END PIECE (FOR TERMINAL BLOCK)
	3	WEIDMULLER	1608620000	ZDU 6	TERMINAL BLOCK, FEED-THROUGH, 6MM2
-X2	2	WEIDMULLER	1632090000	ZAP/TW4	END PIECE (FOR TERMINAL BLOCK)
	20	WEIDMULLER	1632050000	ZDU 4	TERMINAL BLOCK, FEED-THROUGH, 4MM2

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Rev.	Date	Created by	Appr. by	Revision text
b	07.04.2022	MBA123	RSW004	Part list updated
a	27.7.2021	RSW004	KSU004	Location Box Updated

Project name: Mansourah - Massarah Gold Mine
Project id: P/20002
Product id: 6X20V32TS
Based on document id: DAAB198406
Customer document id: -

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PARTS LIST
OILY WATER TRANSFER UNIT
LOCAL CONTROL PANEL
PARTS LIST

Created by: MBAX46	Create date: 8 Jun 2021	Language: EN	Object designation: BJU911
Reviewed by: RSW004	Review date: 8 Jun 2021	Sheet size: A3	Page: 10 / 10
Approved by: KSU004	Approval date: 8 Jun 2021	Document Id: DECA00006028	Rev.: C

PANEL CODE	DOCUMENT NAME	DOCUMENT DESCRIPTION	SHEET N:o	DOCUMENT N:o	REVISION
BJU901	LIST OF DOCUMENTS	DRAWING LIST	1	DBAF492188	c
BJU901	DIMENSIONAL DRAWING	LAYOUT 1:10	2	DBAF492188	c
BJU901	OVERVIEW DIAGRAM	SINGLE LINE	3	DBAF492188	c
BJU901	CIRCUIT DIAGRAM	LOW VOLTAGE FEEDER	4	DBAF492188	c
BJU901	CIRCUIT DIAGRAM	COMMON INDICATOR/CONTROL	5	DBAF492188	c
BJU901	CIRCUIT DIAGRAM	PUMP CONTROL	6	DBAF492188	c
BJU901	CIRCUIT DIAGRAM	EXTERNAL ALARM	7	DBAF492188	d
BJU901	CONNECTION TABLE	CABLE LIST	8	DBAF492188	d
BJU901	CONNECTION TABLE	INTERNAL CONNECTION LIST	9	DBAF492188	c
BJU901	PARTS LIST	PARTS LIST	10	DBAF492188	c

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Rev.	Date	Created by	Appr. by	Revision text
c	10.7.2020	RSW004	KSU004	As manufactured update
b	29.6.2020	KKO070	KSU004	As per customer inputs
a	3.6.2020	KKO070	KSU004	As per customer inputs

Project name: Mansourah - Massarah Gold Mine
Project id: P/20002
Product id: 6X20V32TS
Based on document id: DAAB198406 d
Customer document id:

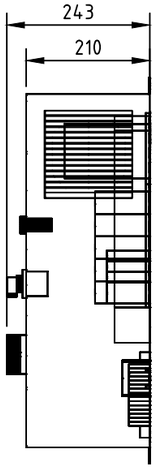
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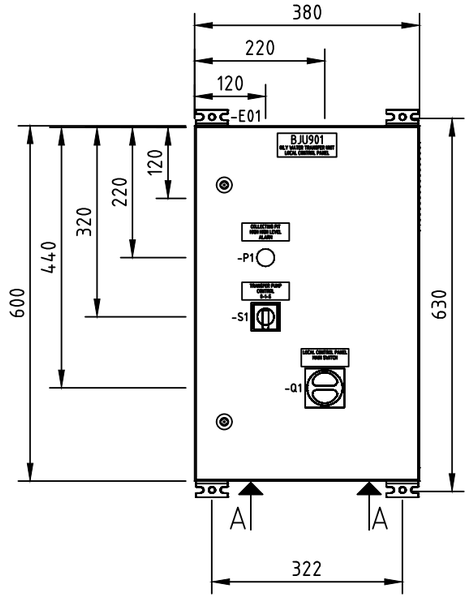
LIST OF DOCUMENTS
OILY WATER TRANSFER UNIT
LOCAL CONTROL PANEL
DRAWING LIST
CONFIDENTIAL

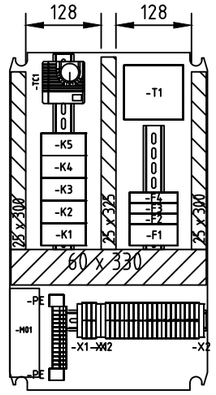
Created by	Create date	Language	Object designation
RSW004	20.4.2020	EN	BJU901
Reviewed by	**Review date**	**Sheet size**	**Page**
KSU004	20.4.2020	A3	1 / 10
Approved by	**Approval date**	**Document Id:**	**Rev.**
KSU004	20.4.2020	DBAF492188	D

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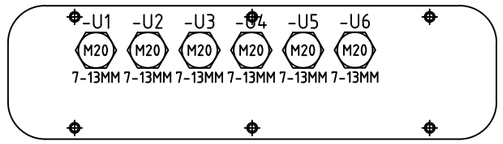
LOCATION - ON OILY WATER TRANSFER UNIT MODULE, NEAR ENGINE HALL & EEB







SECTION A-A SCALE 1:4



PANEL INFORMATION					
WIDTH	380	mm	PROTECTION CLASS		IP54
HEIGHT	600	mm	COLOR		RAL 7035
DEPTH	210	mm	Ik	6	kA/1s

-E01	BJU901	OILY WATER TRANSFER UNIT	LOCAL CONTROL PANEL
-P1	COLLECTING PIT	HIGH HIGH LEVEL	ALARM
-Q1	LOCAL CONTROL PANEL	MAIN SWITCH	
-S1	TRANSFER PUMP	CONTROL	0-1-S

Rev.	Date	Created by	Appr. by	Revision text
c	10.7.2020	RSW004	KSU004	As manufactured update
b	29.6.2020	KKO070	KSU004	As per customer inputs
a	3.6.2020	KKO070	KSU004	As per customer inputs

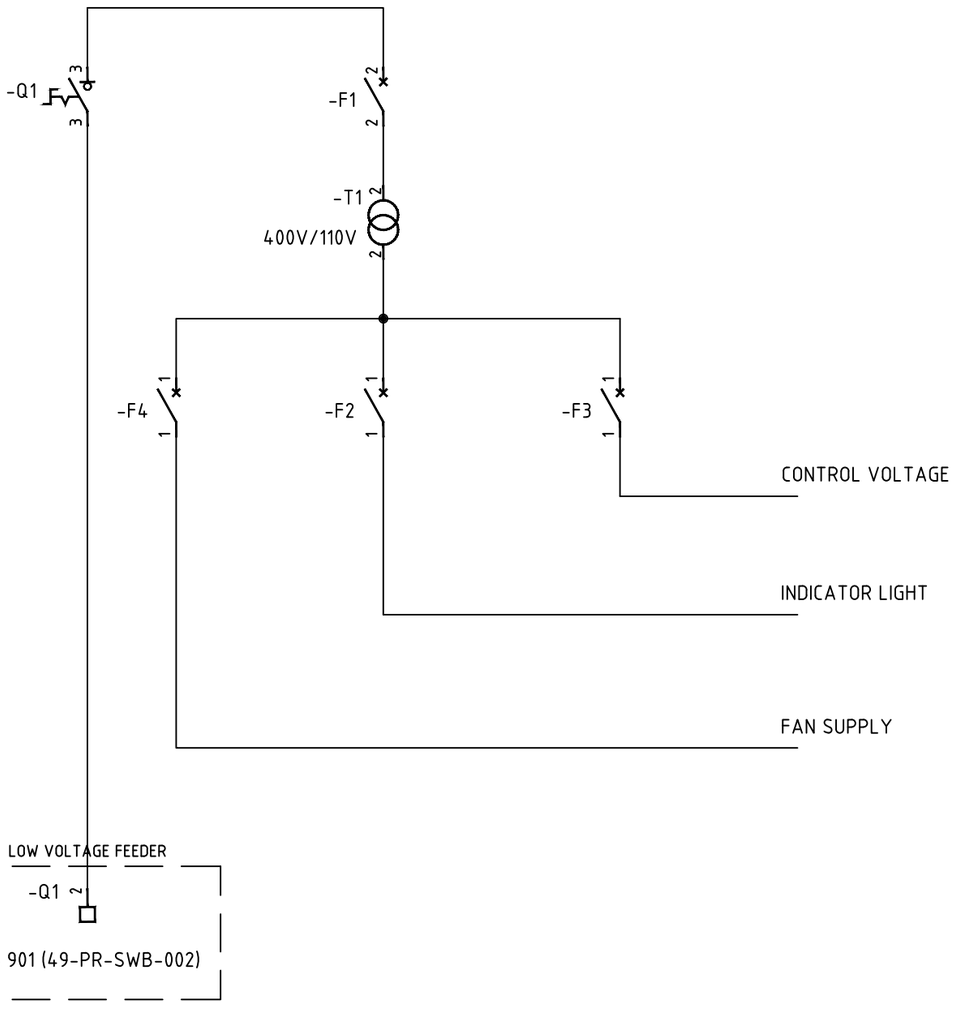
Project name: Mansourah - Massarah Gold Mine
Project id: P/20002
Product id: 6X20V32TS
Based on document id: DAAB198406 d
Customer document id:

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OILY WATER TRANSFER UNIT
LOCAL CONTROL PANEL
CONFIDENTIAL

Created by: RSW004	Create date: 20.4.2020	Language: EN	Object designation: BJU901
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Approved by: KSU004	Approval date: 20.4.2020	Document Id: DBAF492188	Rev.: D

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Rev.	Date	Created by	Appr. by	Revision text
c	10.7.2020	RSW004	KSU004	As manufactured update
b	29.6.2020	KKO070	KSU004	As per customer inputs
a	3.6.2020	KKO070	KSU004	As per customer inputs

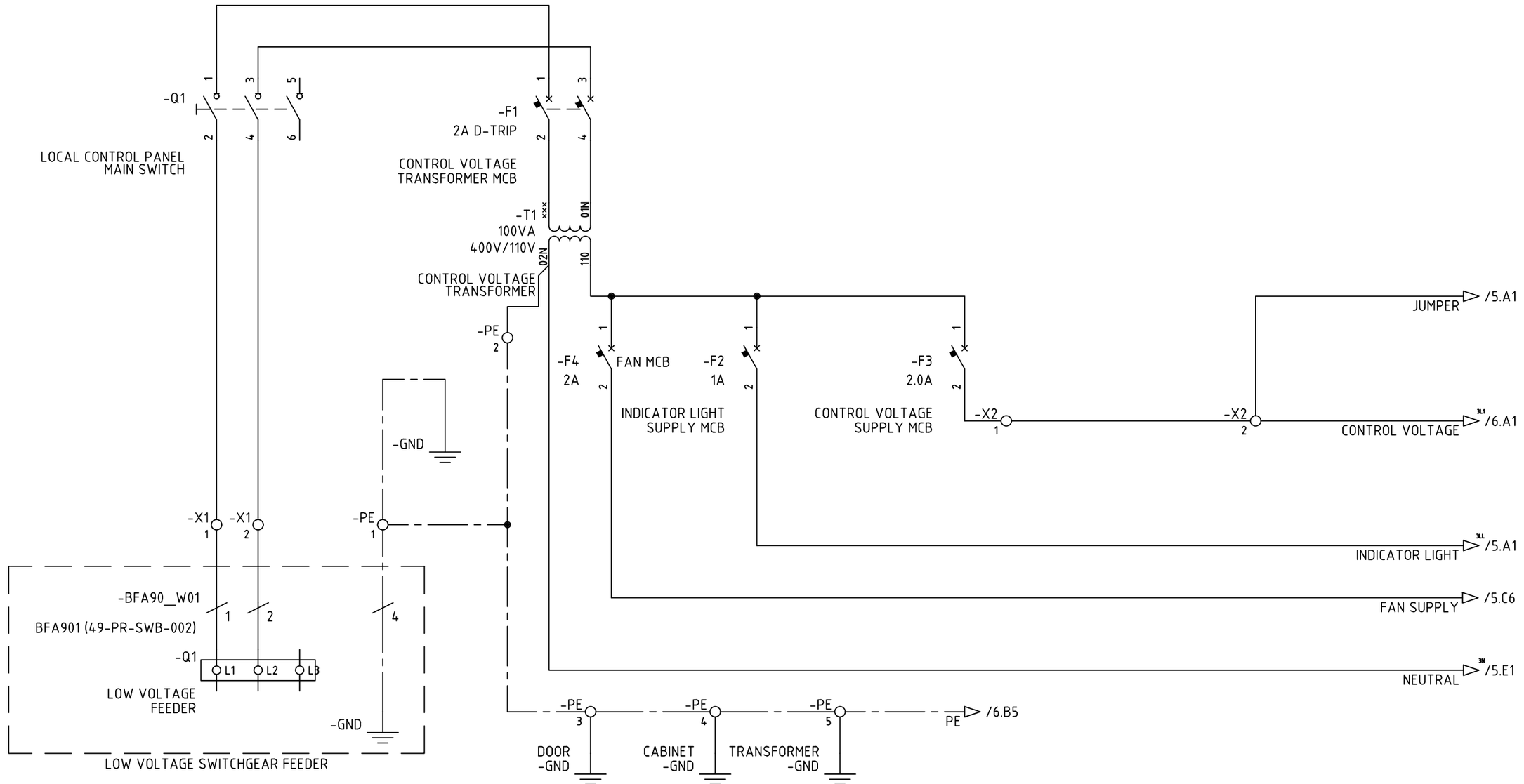
Project name: Mansourah - Massarah Gold Mine
Project id: P/20002
Product id: 6X20V32TS
Based on document id: DAAB198406 d
Customer document id:

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OVERVIEW DIAGRAM
OILY WATER TRANSFER UNIT
LOCAL CONTROL PANEL
SINGLE LINE
CONFIDENTIAL

Created by: RSW004	Create date: 20.4.2020	Language: EN	Object designation: BJU901
Reviewed by: KSU004	Review date: 20.4.2020	Sheet size: A3	Page: 3 / 10
Approved by: KSU004	Approval date: 20.4.2020	Document Id: DBAF492188	Rev.: D

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SUPPLY					
Un	400V	V	2-phase	60	Hz
In	1	A		10	A

Rev.	Date	Created by	Appr. by	Revision text
c	10.7.2020	RSW004	KSU004	As manufactured update
b	29.6.2020	KKO070	KSU004	As per customer inputs
a	3.6.2020	KKO070	KSU004	As per customer inputs

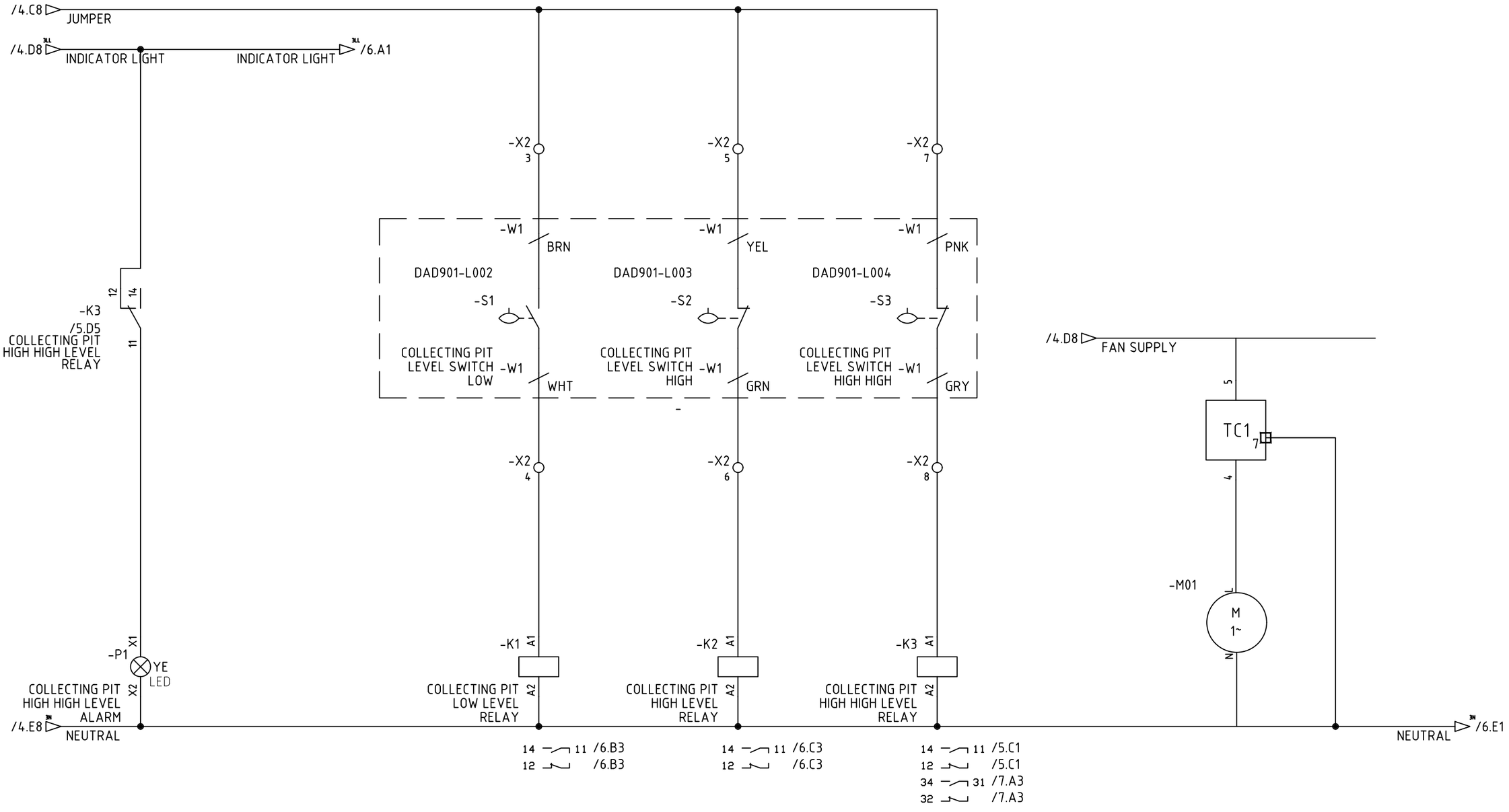
Project name: Mansourah - Massarah Gold Mine
Project id: P/20002
Product id: 6X20V32TS
Based on document id: DAAB198406 d
Customer document id:

WÄRTSILÄ

CIRCUIT DIAGRAM
OILY WATER TRANSFER UNIT
LOCAL CONTROL PANEL
LOW VOLTAGE FEEDER
CONFIDENTIAL

Created by	Create date	Language	Object designation
RSW004	20.4.2020	EN	BJU901
Reviewed by	Review date	Sheet size	Page
KSU004	20.4.2020	A3	4 / 10
Approved by	Approval date	Document Id:	Rev.
KSU004	20.4.2020	DBAF492188	D

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Rev.	Date	Created by	Appr. by	Revision text
c	10.7.2020	RSW004	KSU004	As manufactured update
b	29.6.2020	KKO070	KSU004	As per customer inputs
a	3.6.2020	KKO070	KSU004	As per customer inputs

Project name	Mansourah - Massarah Gold Mine
Project id	P/20002
Product id	6X20V32TS
Based on document id	DAAB198406 d
Customer document id	

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CIRCUIT DIAGRAM

OILY WATER TRANSFER UNIT
LOCAL CONTROL PANEL
COMMON INDICATOR/CONTROL

CONFIDENTIAL

Created by	Create date	Language	Object designation
RSW004	20.4.2020	EN	BJU901
Reviewed by	Review date	Sheet size	Page
KSU004	20.4.2020	A3	5 / 10
Approved by	Approval date	Document Id:	Rev.
KSU004	20.4.2020	DBAF492188	D

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/4.D8 3L1 CONTROL VOLTAGE

/5.A2 3L1 INDICATOR LIGHT

-K4 /6.D3 TRANSFER PUMP CONTROL RELAY 1 (12, 14, 11)

-K1 /5.D3 COLLECTING PIT LOW LEVEL RELAY (12, 14, 11)

-S1 0-1-S (21, 22) TRANSFER PUMP CONTROL 0-1-S

-S1 (13, 14) /6.B4 TRANSFER PUMP CONTROL 0-1-S

-S1					
0	1	S	T1	T2	REF
X			21	22	B4
			X1	X2	6E1
		X	13	14	6C4

-K2 /5.D4 COLLECTING PIT HIGH LEVEL RELAY (12, 14, 11)

-K4 /6.D3 TRANSFER PUMP CONTROL RELAY 1 (22, 24, 21)

-K5 /6.D4 TRANSFER PUMP CONTROL RELAY2 (22, 24, 21)

/4.E6 PE

-PE 6 -X2 9

-W01 3 -W01 1

-GND DAD901-CV001

-Y1 X1 X2 TRANSFER PUMP CONTROL VALVE

-W01 2

-X2 10

-S1 X1 /6.B4 WH LED TRANSFER PUMP X2

/5.E8 3N 0-1-S NEUTRAL

-K4 A1 A2 TRANSFER PUMP CONTROL RELAY 1

-K5 A1 A2 TRANSFER PUMP CONTROL RELAY2

14 — 11 /6.C1
12 — /6.C1
24 — 21 /6.C3
22 — /6.C3

24 — 21 /6.A6
22 — /6.A6
34 — 31 /7.A4
32 — /7.A4

Rev.	Date	Created by	Appr. by	Revision text
c	10.7.2020	RSW004	KSU004	As manufactured update
b	29.6.2020	KKO070	KSU004	As per customer inputs
a	3.6.2020	KKO070	KSU004	As per customer inputs

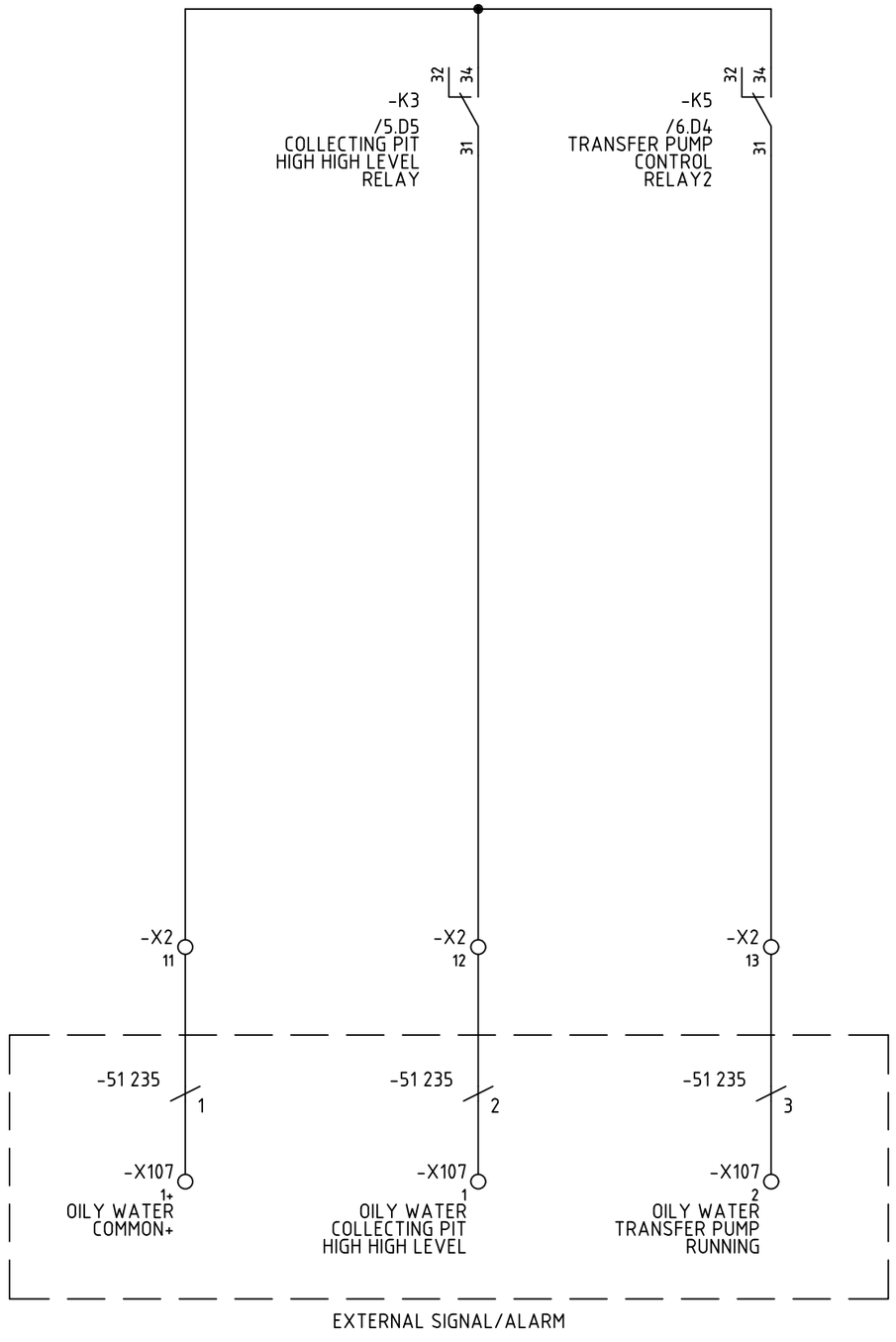
Project name: Mansourah - Massarah Gold Mine
Project id: P/20002 | Product id: 6X20V32TS
Based on document id: DAAB198406 d | Customer document id:

WÄRTSILÄ

CIRCUIT DIAGRAM
OILY WATER TRANSFER UNIT
LOCAL CONTROL PANEL
PUMP CONTROL
CONFIDENTIAL

Created by	Create date	Language	Object designation
RSW004	20.4.2020	EN	BJU901
Reviewed by	Review date	Sheet size	Page
KSU004	20.4.2020	A3	6 / 10
Approved by	Approval date	Document Id:	Rev.
KSU004	20.4.2020	DBAF492188	D

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Rev.	Date	Created by	Appr. by	Revision text
d	16.Feb.2021	MHA186	KSU004	Terminal update
c	10.7.2020	RSW004	KSU004	As manufactured update
b	29.6.2020	KKO070	KSU004	As per customer inputs
a	3.6.2020	KKO070	KSU004	As per customer inputs

Project name: Mansourah - Massarah Gold Mine
Project id: P/20002
Product id: 6X20V32TS
Based on document id: DAAB198406 d
Customer document id:

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CIRCUIT DIAGRAM
OILY WATER TRANSFER UNIT
LOCAL CONTROL PANEL
EXTERNAL ALARM

Created by	Create date	Language	Object designation
RSW004	20.4.2020	EN	BJU901
Reviewed by	**Review date**	**Sheet size**	**Page**
KSU004	20.4.2020	A3	7 / 10
Approved by	**Approval date**	**Document Id:**	**Rev.**
KSU004	20.4.2020	DBAF492188	E

FACTORY INSTALLED CABLES TO BE SUPPLIED BY HELKAMA IF NOT OTHERWISE STATED

CABLE SPECIFICATION	CABLE TYPE	CABLE INSTALLATION	CABLE NAME	WIRE	FROM COMPONENT	PIN1	TO COMPONENT	PIN2	SH1	REF1	CABLE REMARK
1kV YY	3G1,5rm		-W01	1	-X2	9	=DAD901-CV001-Y1	X1	6	B6	INSTALLED AT FACTORY
			-W01	2	-X2	10	=DAD901-CV001-Y1	X2	6	C6	INSTALLED AT FACTORY
			-W01	3	-PE	6	=DAD901-CV001-GND		6	B6	INSTALLED AT FACTORY
1kV YCY	3x__+__	BFA901 (49-PR-SWB-002)	-BFA90__W01	1	=BFA901 (49-PR-SWB-002)-Q1	L1	-X1	1	4	E2	INSTALLED AT SITE
		BFA901 (49-PR-SWB-002)	-BFA90__W01	2	=BFA901 (49-PR-SWB-002)-Q1	L2	-X1	2	4	E2	INSTALLED AT SITE
		BFA901 (49-PR-SWB-002)	-BFA90__W01	4	BFA901 (49-PR-SWB-002)		-PE	1	4	E3	INSTALLED AT SITE
1kV YY	4x1,5rm	CFA901	-51 235	1	=CFA901-X107	1+	-X2	11	7	E2	INSTALLED AT SITE
		CFA901	-51 235	2	=CFA901-X107	1	-X2	12	7	E3	INSTALLED AT SITE
		CFA901	-51 235	3	=CFA901-X107	2	-X2	13	7	E4	INSTALLED AT SITE
		DAD901-L001	-W1	BRN	=DAD901-L002-S1		-X2	3	5	C3	INSTALLED AT SITE (INTEGRATED SILICON CABLE)
		DAD901-L001	-W1	GRN	=DAD901-L003-S2		-X2	6	5	C4	INSTALLED AT SITE (INTEGRATED SILICON CABLE)
		DAD901-L001	-W1	GRY	=DAD901-L004-S3		-X2	8	5	C5	INSTALLED AT SITE (INTEGRATED SILICON CABLE)
		DAD901-L001	-W1	PNK	=DAD901-L004-S3		-X2	7	5	C5	INSTALLED AT SITE (INTEGRATED SILICON CABLE)
		DAD901-L001	-W1	WHT	=DAD901-L002-S1		-X2	4	5	C3	INSTALLED AT SITE (INTEGRATED SILICON CABLE)
		DAD901-L001	-W1	YEL	=DAD901-L003-S2		-X2	5	5	C4	INSTALLED AT SITE (INTEGRATED SILICON CABLE)

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Rev.	Date	Created by	Appr. by	Revision text
d	16.Feb.2021	MHA186	KSU004	Terminal update
c	10.7.2020	RSW004	KSU004	As manufactured update
b	29.6.2020	KKO070	KSU004	As per customer inputs
a	3.6.2020	KKO070	KSU004	As per customer inputs

Project name: Mansourah - Massarah Gold Mine
Project id: P/20002
Product id: 6X20V32TS
Based on document id: DAAB198406 d
Customer document id:

WÄRTSILÄ

CONNECTION TABLE
OILY WATER TRANSFER UNIT
LOCAL CONTROL PANEL
CABLE LIST

Created by	Create date	Language	Object designation
RSW004	20.4.2020	EN	BJU901
Reviewed by	**Review date**	**Sheet size**	**Page**
KSU004	20.4.2020	A3	8 / 10
Approved by	**Approval date**	**Document Id:**	**Rev.**
KSU004	20.4.2020	DBAF492188	E

SH1	REF1	CMP1	PIN1	WLAY1	CMP2	PIN2	SH2	REF2
4	C4	-T1	110	AC CONTROL	-F2	1	4	C5
4	C4	-T1	02N	AC CONTROL	-K1	A2	5	E3
4	C5	-F2	2	AC CONTROL	-K3	12	5	C1
4	C5	-F2	1	AC CONTROL	-F3	1	4	C6
4	C6	-F3	2	AC CONTROL	-X2	1	4	D6
4	D6	-X2	1	AC CONTROL	-X2	2	4	D7
4	D7	-X2	2	AC CONTROL	-K1	14	6	B3
5	C1	-K3	11	AC CONTROL	-P1	X1	5	D1
5	C1	-K3	12	AC CONTROL	-K4	14	6	C1
5	E1	-P1	X2	AC CONTROL	-S1	X2	6	E1
5	C3	-X2	4	AC CONTROL	-K1	A1	5	D3
5	E3	-K1	A2	AC CONTROL	-K2	A2	5	E4
5	C4	-X2	6	AC CONTROL	-K2	A1	5	D4
5	E4	-K2	A2	AC CONTROL	-K3	A2	5	E5
5	C5	-X2	8	AC CONTROL	-K3	A1	5	D5
5	E5	-K3	A2	AC CONTROL	-K4	A2	6	E3
6	C1	-K4	11	AC CONTROL	-S1	X1	6	D1
6	B3	-K1	11	AC CONTROL	-S1	21	6	C3
6	B3	-K1	14	AC CONTROL	-K5	24	6	A6
6	C3	-K2	12	AC CONTROL	-K4	24	6	C3
6	C3	-K4	24	AC CONTROL	-S1	22	6	C3
6	D3	-K2	11	AC CONTROL	-S1	14	6	C4
6	D3	-K4	21	AC CONTROL	-K4	A1	6	D3
6	D3	-K4	A1	AC CONTROL	-K5	A1	6	D4
6	E3	-K4	A2	AC CONTROL	-K5	A2	6	E4
6	E4	-K5	A2	AC CONTROL	-X2	10	6	C6
6	A6	-K5	21	AC CONTROL	-X2	9	6	B6
6	A6	-K5	24	AC CONTROL	-S1	13	6	C4
6	C6	-X2	10	AC CONTROL	-P1	X2	5	E1
7	D2	-X2	11	AC CONTROL	-K3	34	7	A3
7	A3	-K3	31	AC CONTROL	-X2	12	7	D3
7	A3	-K3	34	AC CONTROL	-K5	34	7	A4
7	A4	-K5	31	AC CONTROL	-X2	13	7	D4
4	D7	-X2	2	AC CONTROL JMP	-X2	5	5	B4
5	B3	-X2	3	AC CONTROL JMP	-X2	2	4	D7
5	B4	-X2	5	AC CONTROL JMP	-X2	7	5	B5
4	B2	-Q1	4	AC POWER	-X1	2	4	D2
4	B2	-Q1	3	AC POWER	-F1	3	4	B4
4	B2	-Q1	2	AC POWER	-X1	1	4	D2
4	B2	-Q1	1	AC POWER	-F1	1	4	B4
4	B4	-F1	4	AC POWER	-T1	01N	4	C4
4	B4	-F1	2	AC POWER	-T1	xxx	4	C4
5	C3	-S1		EXTERNAL CONTROL (INTERLOCK)	-X2	4	5	C3
5	C3	-S1		EXTERNAL CONTROL (INTERLOCK)	-X2	3	5	B3
5	C4	-S2		EXTERNAL CONTROL (INTERLOCK)	-X2	6	5	C4
5	C4	-S2		EXTERNAL CONTROL (INTERLOCK)	-X2	5	5	B4
5	C5	-S3		EXTERNAL CONTROL (INTERLOCK)	-X2	8	5	C5
5	C5	-S3		EXTERNAL CONTROL (INTERLOCK)	-X2	7	5	B5
6	B6	-X2	9	EXTERNAL CONTROL (INTERLOCK)	-Y1	X1	6	C6
6	C6	-X2	10	EXTERNAL CONTROL (INTERLOCK)	-Y1	X2	6	C6
7	E2	-X__		EXTERNAL CONTROL (INTERLOCK)	-X2	11	7	D2
7	E3	-X__		EXTERNAL CONTROL (INTERLOCK)	-X2	12	7	D3
7	E4	-X__		EXTERNAL CONTROL (INTERLOCK)	-X2	13	7	D4
4	E2	-Q1	L2	EXTERNAL POWER	-X1	2	4	D2
4	E2	-Q1	L1	EXTERNAL POWER	-X1	1	4	D2
4	E3	-GND		EXTERNAL PROTECTIVE	-PE	1	4	D3
6	B6	-PE	6	EXTERNAL PROTECTIVE	-GND		6	B6
4	D3	-PE	1	PROTECTIVE	-GND		4	D3
4	C4	-T1	02N	PROTECTIVE	-PE	2	4	C3

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b	29.6.2020	KKO070	KSU004	As per customer inputs
a	3.6.2020	KKO070	KSU004	As per customer inputs

Project name: Mansourah - Massarah Gold Mine
Project id: P/20002
Product id: 6X20V32TS
Based on document id: DAAB198406 d
Customer document id:

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CONNECTION TABLE
OILY WATER TRANSFER UNIT
LOCAL CONTROL PANEL
INTERNAL CONNECTION LIST
CONFIDENTIAL

Created by: RSW004	Create date: 20.4.2020	Language: EN	Object designation: BJU901
Reviewed by: KSU004	Review date: 20.4.2020	Sheet size: A3	Page: 9 / 10
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COMPONENT NAME	QTY	MANUFACTURER	PART CODE	RATING	DESC
-E01	1	RITTAL	1038000		
-E01	4	RITTAL	2508020		WALL MOUNTING BRACKET
-E01	1	RITTAL			SECURITY LOCK
-F1	1	Schneider Electric	A9F05202	2A D-TRIP	MINIATURE CIRCUIT BREAKER
-F2	1	Schneider Electric	A9F04101	1A	MINIATURE CIRCUIT BREAKER
-F3	1	Schneider Electric	A9F04102	2.0A	MINIATURE CIRCUIT BREAKER
-K1	1	Carlo Gavazzi	RCP11003115120VAC		UNIVERSAL PLUG-IN RELAY, 3 CO
-K1	1	Carlo Gavazzi	ZPD12A		SOCKET
-K2	1	Carlo Gavazzi	RCP11003115120VAC		UNIVERSAL PLUG-IN RELAY, 3 CO
-K2	1	Carlo Gavazzi	ZPD12A		SOCKET
-K3	1	Carlo Gavazzi	RCP11003115120VAC		UNIVERSAL PLUG-IN RELAY, 3 CO
-K3	1	Carlo Gavazzi	ZPD12A		SOCKET
-K4	1	Carlo Gavazzi	RCP11003115120VAC		UNIVERSAL PLUG-IN RELAY, 3 CO
-K4	1	Carlo Gavazzi	ZPD12A		SOCKET
-K5	1	Carlo Gavazzi	RCP11003115120VAC		UNIVERSAL PLUG-IN RELAY, 3 CO
-K5	1	Carlo Gavazzi	ZPD12A		SOCKET
-P1	1	Schneider Electric	XB7EV05GP		MONOLITHIC INDICATOR LIGHT, YELLOW, INTEGRAL LED
-PE	1	Phoenix Contact	NLS-CU 3/10		BUSBAR (FOR PE TERMINAL BLOCK)
-PE	2	Phoenix Contact	AB/SS		END PIECE (FOR PE TERMINAL BLOCK)
-PE	15	Phoenix Contact	AKG 4 GNYE		PE TERMINAL BLOCK
-Q1	1	BACO	0172001		
-S1	1	Schneider Electric	ZB5AK1813 + ZB5AW0G15		CONTROL SWITCH, 3 POSITION
-T1	1	Trafomic	0423193	200VA	CONTROL VOLTAGE TRANSFORMER
-U1	1	Agro	1555.20.13	M20	CABLE GLAND, M20
-U2	1	Agro	1555.20.13	M20	CABLE GLAND, M20
-U3	1	Agro	1555.20.13	M20	CABLE GLAND, M20
-U4	1	Agro	1555.20.13	M20	CABLE GLAND, M20
-U5	1	Agro	1555.20.13	M20	CABLE GLAND, M20
-U6	1	Agro	1555.20.13	M20	CABLE GLAND, M20
-X1	2	Phoenix Contact	CLIPFIX 35		END PIECE (FOR TERMINAL BLOCK)
-X1	3	Weidmuller	ZDU 6		
-X2	2	Phoenix Contact	CLIPFIX 35		END PIECE (FOR TERMINAL BLOCK)
-X2	20	Weidmuller	ZDU 4		
-F4	1	Schneider Electric	A9F04102		MINIATURE CIRCUIT BREAKER
-FILTER	1	Kryos	GSF15		FILTER
-M01	1	Kryos	GSV1500203		FAN
-TC1	1	RITTAL	3100000		THERMOSTAT

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a	3.6.2020	KKO070	KSU004	As per customer inputs

Project name: Mansourah - Massarah Gold Mine
Project id: P/20002
Product id: 6X20V32TS
Based on document id: DAAB198406 d
Customer document id:

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PARTS LIST
OILY WATER TRANSFER UNIT
LOCAL CONTROL PANEL
PARTS LIST
CONFIDENTIAL

Created by	Create date	Language	Object designation
RSW004	20.4.2020	EN	BJU901
Reviewed by	**Review date**	**Sheet size**	**Page**
KSU004	20.4.2020	A3	10 / 10
Approved by	**Approval date**	**Document Id:**	**Rev.**
KSU004	20.4.2020	DBAF492188	D

PANEL CODE	DOCUMENT NAME	DOCUMENT DESCRIPTION	SHEET N:o	DOCUMENT N:o	REVISION
BJU902	LIST OF DOCUMENTS	DRAWING LIST	1	DBAF492190	c
BJU902	DIMENSIONAL DRAWING	LAYOUT 1:10	2	DBAF492190	c
BJU902	OVERVIEW DIAGRAM	SINGLE LINE	3	DBAF492190	c
BJU902	CIRCUIT DIAGRAM	LOW VOLTAGE FEEDER	4	DBAF492190	c
BJU902	CIRCUIT DIAGRAM	COMMON INDICATOR/CONTROL	5	DBAF492190	c
BJU902	CIRCUIT DIAGRAM	PUMP CONTROL	6	DBAF492190	c
BJU902	CIRCUIT DIAGRAM	EXTERNAL ALARM	7	DBAF492190	d
BJU902	CONNECTION TABLE	CABLE LIST	8	DBAF492190	d
BJU902	CONNECTION TABLE	INTERNAL CONNECTION LIST	9	DBAF492190	c
BJU902	PARTS LIST	PARTS LIST	10	DBAF492190	c

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Project name: Mansourah - Massarah Gold Mine
Project id: P/20002
Product id: 6X20V32TS
Based on document id: DAAB198406 d
Customer document id:

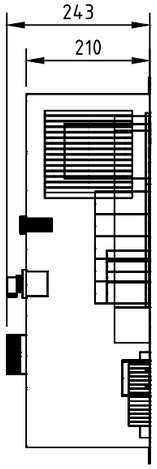
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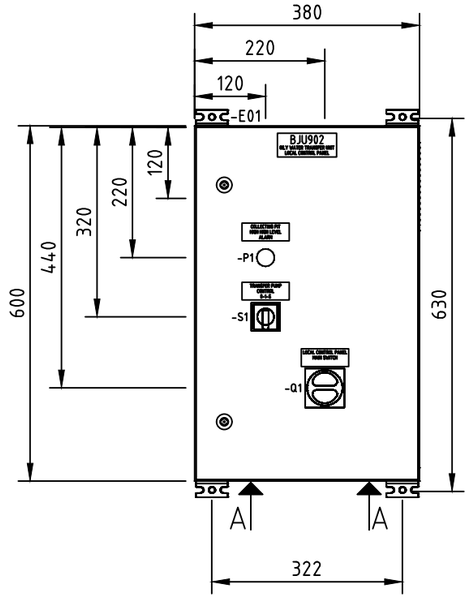
LIST OF DOCUMENTS
OILY WATER TRANSFER UNIT
LOCAL CONTROL PANEL
DRAWING LIST
CONFIDENTIAL

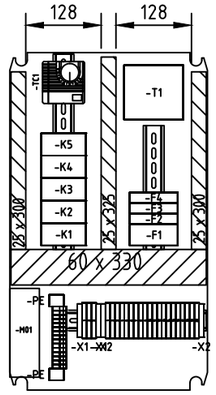
Created by	Create date	Language	Object designation
RSW004	5.5.2020	EN	BJU902
Reviewed by	**Review date**	**Sheet size**	**Page**
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KSU004	5.5.2020	DBAF492190	D

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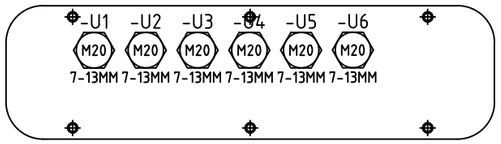
LOCATION - ON OILY WATER TRANSFER UNIT MODULE, NEAR ENGINE HALL & WORKSHOP BUILDING







SECTION A-A SCALE 1:4



PANEL INFORMATION					
WIDTH	380	mm	PROTECTION CLASS		IP54
HEIGHT	600	mm	COLOR		RAL 7035
DEPTH	210	mm	Ik	6	kA/1s

-E01	BJU902	OILY WATER TRANSFER UNIT	LOCAL CONTROL PANEL
-P1	COLLECTING PIT	HIGH HIGH LEVEL	ALARM
-Q1	LOCAL CONTROL PANEL	MAIN SWITCH	
-S1	TRANSFER PUMP	CONTROL	0-1-S

Rev.	Date	Created by	Appr. by	Revision text
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a	3.6.2020	KKO070	KSU004	As per customer inputs

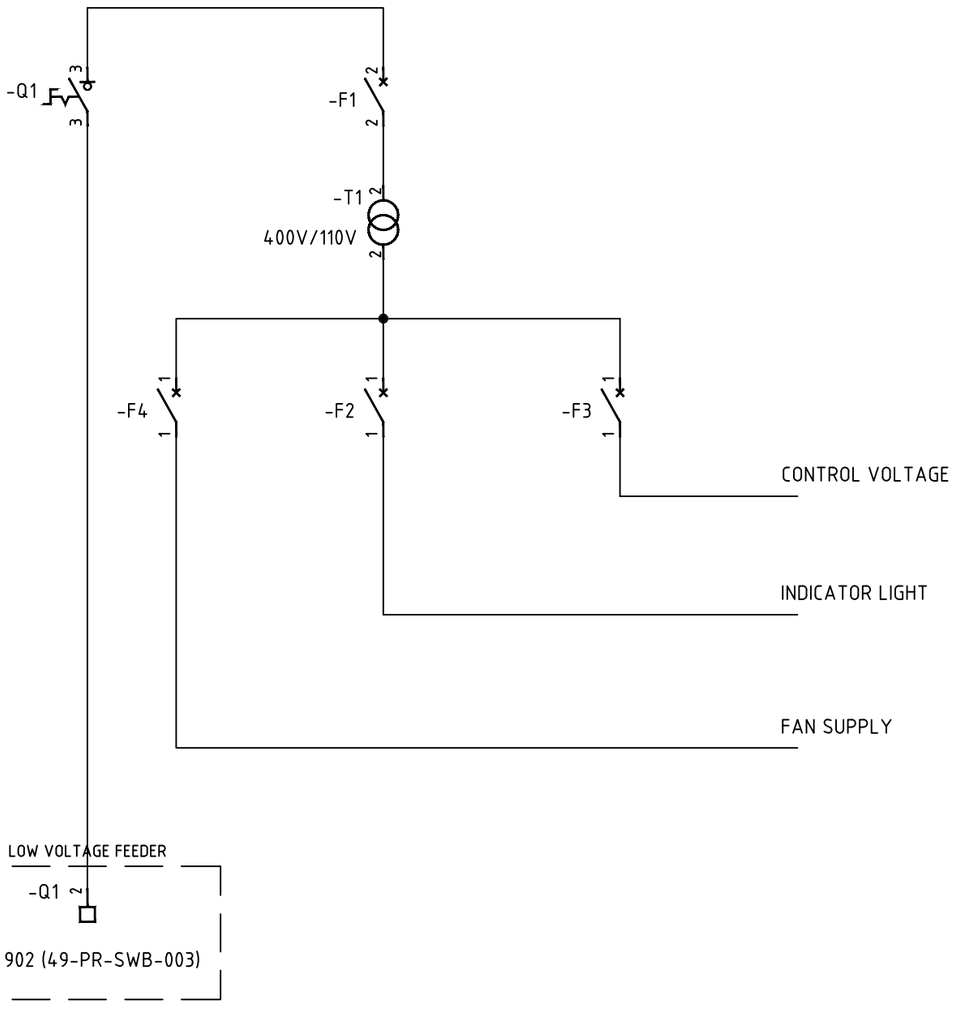
Project name: Mansourah - Massarah Gold Mine
Project id: P/20002
Product id: 6X20V32TS
Based on document id: DAAB198406 d
Customer document id:

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OILY WATER TRANSFER UNIT
LOCAL CONTROL PANEL
CONFIDENTIAL

Created by: RSW004	Create date: 5.5.2020	Language: EN	Object designation: BJU902
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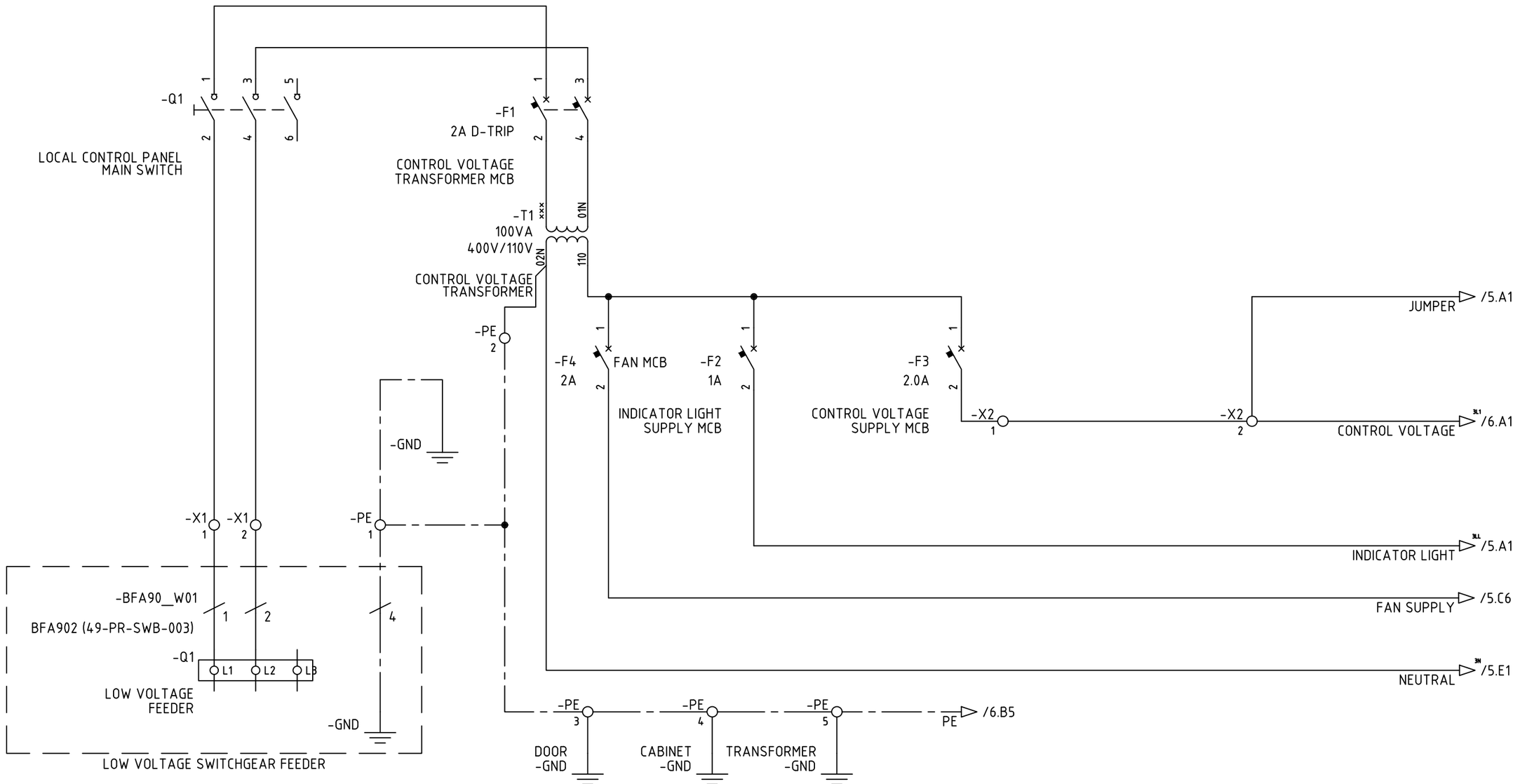
Project name: Mansourah - Massarah Gold Mine
Project id: P/20002
Product id: 6X20V32TS
Based on document id: DAAB198406 d
Customer document id:

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OVERVIEW DIAGRAM
OILY WATER TRANSFER UNIT
LOCAL CONTROL PANEL
SINGLE LINE
CONFIDENTIAL

Created by	Create date	Language	Object designation
RSW004	5.5.2020	EN	BJU902
Reviewed by	**Review date**	**Sheet size**	**Page**
KSU004	5.5.2020	A3	3 / 10
Approved by	**Approval date**	**Document Id:**	**Rev.**
KSU004	5.5.2020	DBAF492190	D

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SUPPLY					
Un	400V	V	2-phase	60	Hz
In	1	A		10	A

Rev.	Date	Created by	Appr. by	Revision text
c	21.7.2020	KKO070	KSU004	As manufactured update
b	29.6.2020	KKO070	KSU004	As per customer inputs
a	3.6.2020	KKO070	KSU004	As per customer inputs

Project name	Mansourah - Massarah Gold Mine
Project id	P/20002
Product id	6X20V32TS
Based on document id	DAAB198406 d
Customer document id	

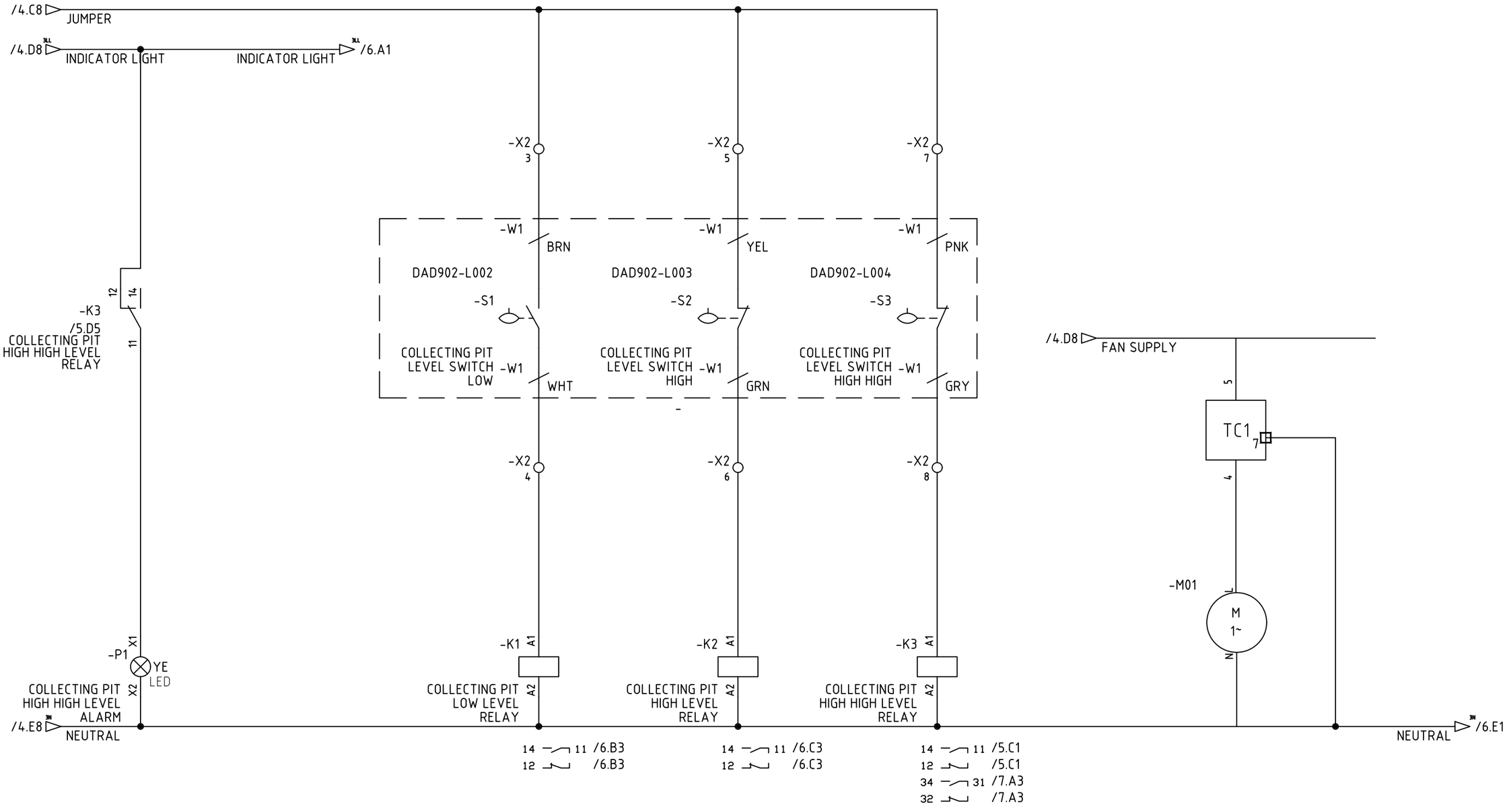
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CIRCUIT DIAGRAM

OILY WATER TRANSFER UNIT
LOCAL CONTROL PANEL
LOW VOLTAGE FEEDER
CONFIDENTIAL

Created by RSW004	Create date 5.5.2020	Language EN	Object designation BJU902
Reviewed by KSU004	Review date 5.5.2020	Sheet size A3	Page 4 / 10
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c	21.7.2020	KKO070	KSU004	As manufactured update
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a	3.6.2020	KKO070	KSU004	As per customer inputs

Project name: Mansourah - Massarah Gold Mine
Project id: P/20002
Product id: 6X20V32TS
Based on document id: DAAB198406 d
Customer document id:

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CIRCUIT DIAGRAM
OILY WATER TRANSFER UNIT
LOCAL CONTROL PANEL
COMMON INDICATOR/CONTROL
CONFIDENTIAL

Created by	Create date	Language	Object designation
RSW004	5.5.2020	EN	BJU902
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KSU004	5.5.2020	A3	5 / 10
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/4.D8 3L1 CONTROL VOLTAGE

/5.A2 3L1 INDICATOR LIGHT

-K4 /6.D3 TRANSFER PUMP CONTROL RELAY 1 (12, 14, 11)

-K1 /5.D3 COLLECTING PIT LOW LEVEL RELAY (12, 14, 11)

-S1 0-1-S (21, 22) TRANSFER PUMP CONTROL 0-1-S

-S1 (13, 14) /6.B4 TRANSFER PUMP CONTROL 0-1-S

-S1					
0	1	S	T1	T2	REF
X			21	22	B4
			X1	X2	6E1
		X	13	14	6C4

-K2 /5.D4 COLLECTING PIT HIGH LEVEL RELAY (12, 14, 11)

-K4 /6.D3 TRANSFER PUMP CONTROL RELAY 1 (22, 24, 21)

-K5 /6.D4 TRANSFER PUMP CONTROL RELAY2 (22, 24, 21)

/4.E6 PE

-PE 6 -GND DAD902-CV001

-X2 9 -W01 (1) -W01 (3)

-Y1 X1 X2 TRANSFER PUMP CONTROL VALVE

-W01 (2) -X2 10

-S1 X1 /6.B4 WH LED X2 TRANSFER PUMP

/5.E8 0-1-S NEUTRAL

-K4 A1 A2 TRANSFER PUMP CONTROL RELAY 1

-K5 A1 A2 TRANSFER PUMP CONTROL RELAY2

14 — 11 /6.C1
12 — /6.C1
24 — 21 /6.C3
22 — /6.C3

24 — 21 /6.A6
22 — /6.A6
34 — 31 /7.A4
32 — /7.A4

Rev.	Date	Created by	Appr. by	Revision text
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b	29.6.2020	KKO070	KSU004	As per customer inputs
a	3.6.2020	KKO070	KSU004	As per customer inputs

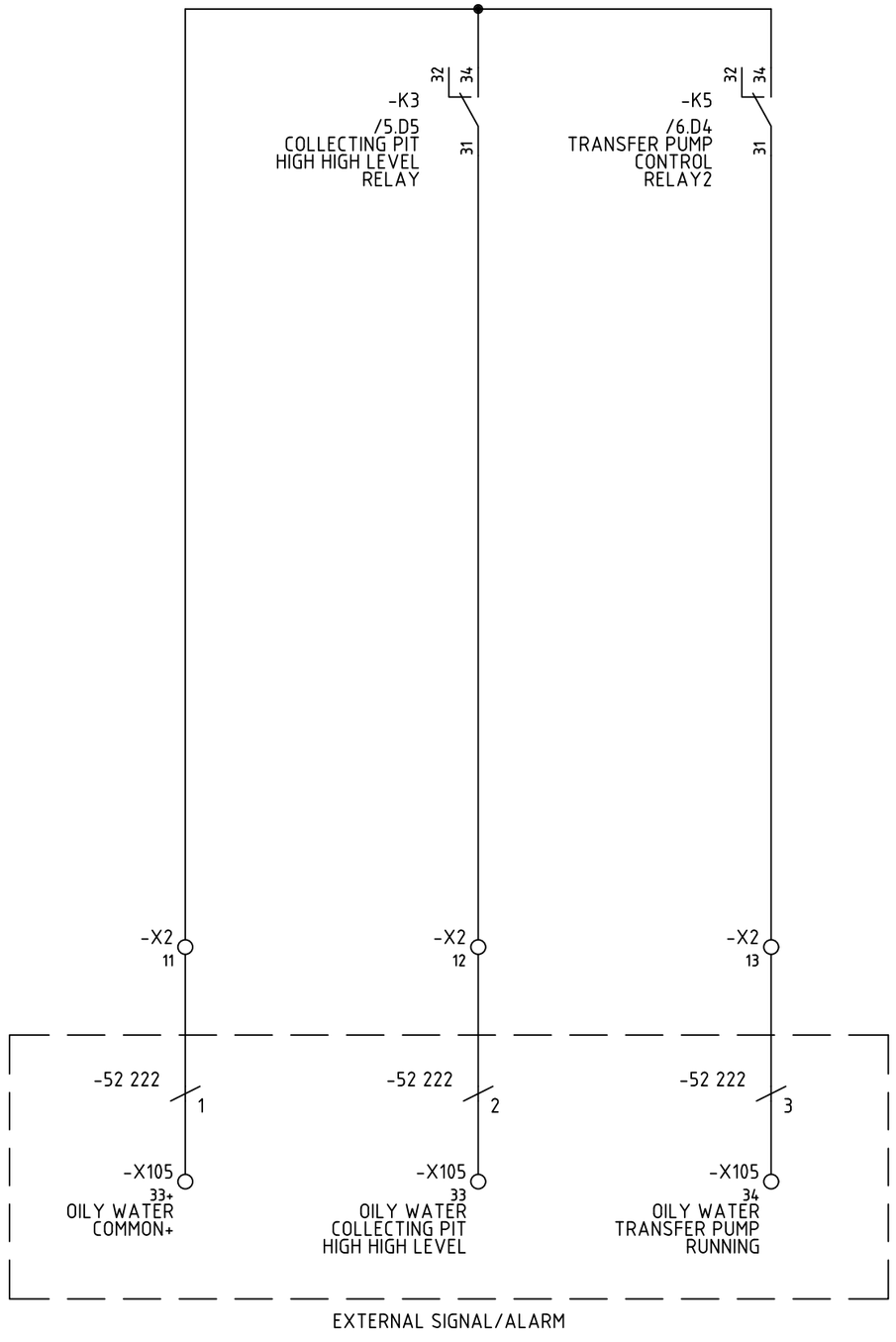
Project name: Mansourah - Massarah Gold Mine
Project id: P/20002 | Product id: 6X20V32TS
Based on document id: DAAB198406 d | Customer document id:

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CIRCUIT DIAGRAM
OILY WATER TRANSFER UNIT
LOCAL CONTROL PANEL
PUMP CONTROL
CONFIDENTIAL

Created by	Create date	Language	Object designation
RSW004	5.5.2020	EN	BJU902
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KSU004	5.5.2020	A3	6 / 10
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KSU004	5.5.2020	DBAF492190	D

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Rev.	Date	Created by	Appr. by	Revision text
d	16.Feb	MHA186	KSU004	Terminal update
c	21.7.2020	KKO070	KSU004	As manufactured update
b	29.6.2020	KKO070	KSU004	As per customer inputs
a	3.6.2020	KKO070	KSU004	As per customer inputs

Project name: Mansourah - Massarah Gold Mine
Project id: P/20002
Product id: 6X20V32TS
Based on document id: DAAB198406 d
Customer document id:

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CIRCUIT DIAGRAM
OILY WATER TRANSFER UNIT
LOCAL CONTROL PANEL
EXTERNAL ALARM

Created by	Create date	Language	Object designation
RSW004	5.5.2020	EN	BJU902
Reviewed by	Review date	Sheet size	Page
KSU004	5.5.2020	A3	7 / 10
Approved by	Approval date	Document Id:	Rev.
KSU004	5.5.2020	DBAF492190	E

FACTORY INSTALLED CABLES TO BE SUPPLIED BY HELKAMA IF NOT OTHERWISE STATED

CABLE SPECIFICATION	CABLE TYPE	CABLE INSTALLATION	CABLE NAME	WIRE	FROM COMPONENT	PIN1	TO COMPONENT	PIN2	SH1	REF1	CABLE REMARK
1kV YY	3G1,5rm		-W01	1	-X2	9	=DAD902-CV001-Y1	X1	6	B6	INSTALLED AT FACTORY
			-W01	2	-X2	10	=DAD902-CV001-Y1	X2	6	C6	INSTALLED AT FACTORY
			-W01	3	-PE	6	=DAD902-CV001-GND		6	B6	INSTALLED AT FACTORY
1kV YCY	3x__+__	BFA902 (49-PR-SWB-003)	-BFA90__W01	1	=BFA902 (49-PR-SWB-003)-Q1	L1	-X1	1	4	E2	INSTALLED AT SITE
		BFA902 (49-PR-SWB-003)	-BFA90__W01	2	=BFA902 (49-PR-SWB-003)-Q1	L2	-X1	2	4	E2	INSTALLED AT SITE
		BFA902 (49-PR-SWB-003)	-BFA90__W01	4	=BFA902 (49-PR-SWB-003)-GND		-PE	1	4	E3	INSTALLED AT SITE
1kV YY	4x1,5rm	CFA902	-52 222	1	=CFA902-X105	33+	-X2	11	7	E2	INSTALLED AT SITE
		CFA902	-52 222	2	=CFA902-X105	33	-X2	12	7	E3	INSTALLED AT SITE
		CFA902	-52 222	3	=CFA902-X105	34	-X2	13	7	E4	INSTALLED AT SITE
		DAD902-L001	-W1	BRN	=DAD902-L002-S1		-X2	3	5	C3	INSTALLED AT SITE (INTEGRATED SILICON CABLE)
		DAD902-L001	-W1	GRN	=DAD902-L003-S2		-X2	6	5	C4	INSTALLED AT SITE (INTEGRATED SILICON CABLE)
		DAD902-L001	-W1	GRY	=DAD902-L004-S3		-X2	8	5	C5	INSTALLED AT SITE (INTEGRATED SILICON CABLE)
		DAD902-L001	-W1	PNK	=DAD902-L004-S3		-X2	7	5	C5	INSTALLED AT SITE (INTEGRATED SILICON CABLE)
		DAD902-L001	-W1	WHT	=DAD902-L002-S1		-X2	4	5	C3	INSTALLED AT SITE (INTEGRATED SILICON CABLE)
		DAD902-L001	-W1	YEL	=DAD902-L003-S2		-X2	5	5	C4	INSTALLED AT SITE (INTEGRATED SILICON CABLE)

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Project name: Mansourah - Massarah Gold Mine
Project id: P/20002
Product id: 6X20V32TS
Based on document id: DAAB198406 d
Customer document id:

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CONNECTION TABLE
OILY WATER TRANSFER UNIT
LOCAL CONTROL PANEL
CABLE LIST

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SH1	REF1	CMP1	PIN1	WLAY1	CMP2	PIN2	SH2	REF2
4	C4	-T1	110	AC CONTROL	-F2	1	4	C5
4	C4	-T1	02N	AC CONTROL	-K4	A2	6	E3
4	C4	-F4	2	AC CONTROL	-TC1	5	5	C7
4	C4	-F4	1	AC CONTROL	-F3	1	4	C6
4	C5	-F2	2	AC CONTROL	-K3	12	5	C1
4	C5	-F2	1	AC CONTROL	-F4	1	4	C4
4	C6	-F3	2	AC CONTROL	-X2	1	4	D6
4	D6	-X2	1	AC CONTROL	-X2	2	4	D7
4	D7	-X2	2	AC CONTROL	-K1	14	6	B3
5	C1	-K3	11	AC CONTROL	-P1	X1	5	D1
5	C1	-K3	12	AC CONTROL	-K4	14	6	C1
5	E1	-P1	X2	AC CONTROL	-S1	X2	6	E1
5	C3	-X2	4	AC CONTROL	-K1	A1	5	D3
5	E3	-K1	A2	AC CONTROL	-K2	A2	5	E4
5	C4	-X2	6	AC CONTROL	-K2	A1	5	D4
5	E4	-K2	A2	AC CONTROL	-K3	A2	5	E5
5	C5	-X2	8	AC CONTROL	-K3	A1	5	D5
5	E5	-K3	A2	AC CONTROL	-P1	X2	5	E1
5	D7	-TC1	4	AC CONTROL	-M01	L	5	D7
6	C1	-K4	11	AC CONTROL	-S1	X1	6	D1
6	E1	-S1	X2	AC CONTROL	-M01	N	5	D7
6	B3	-K1	11	AC CONTROL	-S1	21	6	C3
6	B3	-K1	14	AC CONTROL	-K5	24	6	A6
6	C3	-K2	12	AC CONTROL	-K4	24	6	C3
6	C3	-K4	24	AC CONTROL	-S1	22	6	C3
6	D3	-K2	11	AC CONTROL	-S1	14	6	C4
6	D3	-K4	21	AC CONTROL	-K4	A1	6	D3
6	D3	-K4	A1	AC CONTROL	-K5	A1	6	D4
6	E3	-K4	A2	AC CONTROL	-K5	A2	6	E4
6	E4	-K5	A2	AC CONTROL	-X2	10	6	C6
6	A6	-K5	21	AC CONTROL	-X2	9	6	B6
6	A6	-K5	24	AC CONTROL	-S1	13	6	C4
6	C6	-X2	10	AC CONTROL	-K1	A2	5	E3
7	D2	-X2	11	AC CONTROL	-K3	34	7	A3
7	A3	-K3	31	AC CONTROL	-X2	12	7	D3
7	A3	-K3	34	AC CONTROL	-K5	34	7	A4
7	A4	-K5	31	AC CONTROL	-X2	13	7	D4
4	D7	-X2	2	AC CONTROL JMP	-X2	5	5	B4
5	B3	-X2	3	AC CONTROL JMP	-X2	2	4	D7
5	B4	-X2	5	AC CONTROL JMP	-X2	7	5	B5
4	B2	-Q1	4	AC POWER	-X1	2	4	D2
4	B2	-Q1	3	AC POWER	-F1	3	4	B4
4	B2	-Q1	2	AC POWER	-X1	1	4	D2
4	B2	-Q1	1	AC POWER	-F1	1	4	B4
4	B4	-F1	4	AC POWER	-T1	01N	4	C4
4	B4	-F1	2	AC POWER	-T1	×××	4	C4
5	C3	-S1		EXTERNAL CONTROL (INTERLOCK)	-X2	4	5	C3
5	C3	-S1		EXTERNAL CONTROL (INTERLOCK)	-X2	3	5	B3
5	C4	-S2		EXTERNAL CONTROL (INTERLOCK)	-X2	6	5	C4
5	C4	-S2		EXTERNAL CONTROL (INTERLOCK)	-X2	5	5	B4
5	C5	-S3		EXTERNAL CONTROL (INTERLOCK)	-X2	8	5	C5
5	C5	-S3		EXTERNAL CONTROL (INTERLOCK)	-X2	7	5	B5
6	B6	-X2	9	EXTERNAL CONTROL (INTERLOCK)	-Y1	X1	6	C6
6	C6	-X2	10	EXTERNAL CONTROL (INTERLOCK)	-Y1	X2	6	C6
7	E2	-X_		EXTERNAL CONTROL (INTERLOCK)	-X2	11	7	D2
7	E3	-X_		EXTERNAL CONTROL (INTERLOCK)	-X2	12	7	D3
7	E4	-X_		EXTERNAL CONTROL (INTERLOCK)	-X2	13	7	D4
4	E2	-Q1	L2	EXTERNAL POWER	-X1	2	4	D2
4	E2	-Q1	L1	EXTERNAL POWER	-X1	1	4	D2
4	E3	-GND		EXTERNAL PROTECTIVE	-PE	1	4	D3
6	B6	-PE	6	EXTERNAL PROTECTIVE	-GND		6	B6
4	D3	-PE	1	PROTECTIVE	-GND		4	D3
4	C4	-T1	02N	PROTECTIVE	-PE	2	4	C3

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Rev.	Date	Created by	Appr. by	Revision text
c	21.7.2020	KKO070	KSU004	As manufactured update
b	29.6.2020	KKO070	KSU004	As per customer inputs
a	3.6.2020	KKO070	KSU004	As per customer inputs

Project name: Mansourah - Massarah Gold Mine
Project id: P/20002
Product id: 6X20V32TS
Based on document id: DAAB198406 d
Customer document id:

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CONNECTION TABLE
OILY WATER TRANSFER UNIT
LOCAL CONTROL PANEL
INTERNAL CONNECTION LIST
CONFIDENTIAL

Created by: RSW004	Create date: 5.5.2020	Language: EN	Object designation: BJU902
Reviewed by: KSU004	Review date: 5.5.2020	Sheet size: A3	Page: 9 / 10
Approved by: KSU004	Approval date: 5.5.2020	Document Id: DBAF492190	Rev.: D

COMPONENT NAME	QTY	MANUFACTURER	PART CODE	RATING	DESC
-E01	1	RITTAL	1038000		
-E01	4	RITTAL	2508020		WALL MOUNTING BRACKET
-E01	1	RITTAL			SECURITY LOCK
-F1	1	Schneider Electric	A9F05202	2A D-TRIP	MINIATURE CIRCUIT BREAKER
-F2	1	Schneider Electric	A9F04101	1A	MINIATURE CIRCUIT BREAKER
-F3	1	Schneider Electric	A9F04102	2.0A	MINIATURE CIRCUIT BREAKER
-K1	1	Carlo Gavazzi	RCP11003115120VAC		UNIVERSAL PLUG-IN RELAY, 3 CO
-K1	1	Carlo Gavazzi	ZPD12A		SOCKET
-K2	1	Carlo Gavazzi	RCP11003115120VAC		UNIVERSAL PLUG-IN RELAY, 3 CO
-K2	1	Carlo Gavazzi	ZPD12A		SOCKET
-K3	1	Carlo Gavazzi	RCP11003115120VAC		UNIVERSAL PLUG-IN RELAY, 3 CO
-K3	1	Carlo Gavazzi	ZPD12A		SOCKET
-K4	1	Carlo Gavazzi	RCP11003115120VAC		UNIVERSAL PLUG-IN RELAY, 3 CO
-K4	1	Carlo Gavazzi	ZPD12A		SOCKET
-K5	1	Carlo Gavazzi	RCP11003115120VAC		UNIVERSAL PLUG-IN RELAY, 3 CO
-K5	1	Carlo Gavazzi	ZPD12A		SOCKET
-P1	1	Schneider Electric	XB7EV05GP		MONOLITHIC INDICATOR LIGHT, YELLOW, INTEGRAL LED
-PE	1	Phoenix Contact	NLS-CU 3/10		BUSBAR (FOR PE TERMINAL BLOCK)
-PE	2	Phoenix Contact	AB/SS		END PIECE (FOR PE TERMINAL BLOCK)
-PE	15	Phoenix Contact	AKG 4 GNYE		PE TERMINAL BLOCK
-Q1	1	BACO	0172001		
-S1	1	Schneider Electric	ZB5AK1813 + ZB5AW0G15		CONTROL SWITCH, 3 POSITION
-T1	1	Trafomic	0423193	200VA	CONTROL VOLTAGE TRANSFORMER
-U1	1	Agro	1555.20.13	M20	CABLE GLAND, M20
-U2	1	Agro	1555.20.13	M20	CABLE GLAND, M20
-U3	1	Agro	1555.20.13	M20	CABLE GLAND, M20
-U4	1	Agro	1555.20.13	M20	CABLE GLAND, M20
-U5	1	Agro	1555.20.13	M20	CABLE GLAND, M20
-U6	1	Agro	1555.20.13	M20	CABLE GLAND, M20
-X1	2	Phoenix Contact	CLIPFIX 35		END PIECE (FOR TERMINAL BLOCK)
-X1	3	Weidmuller	ZDU 6		
-X2	2	Phoenix Contact	CLIPFIX 35		END PIECE (FOR TERMINAL BLOCK)
-X2	20	Weidmuller	ZDU 4		
-F4	1	Schneider Electric	A9F04102		MINIATURE CIRCUIT BREAKER
-FILTER	1	Kryos	GSF15		FILTER
-M01	1	Kryos	GSV1500203		FAN
-TC1	1	RITTAL	3100000		THERMOSTAT

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b	29.6.2020	KKO070	KSU004	As per customer inputs
a	3.6.2020	KKO070	KSU004	As per customer inputs

Project name: Mansourah - Massarah Gold Mine
Project id: P/20002
Product id: 6X20V32TS
Based on document id: DAAB198406 d
Customer document id:

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PARTS LIST
OILY WATER TRANSFER UNIT
LOCAL CONTROL PANEL
PARTS LIST
CONFIDENTIAL

Created by	Create date	Language	Object designation
RSW004	5.5.2020	EN	BJU902
Reviewed by	**Review date**	**Sheet size**	**Page**
KSU004	5.5.2020	A3	10 / 10
Approved by	**Approval date**	**Document Id:**	**Rev.**
KSU004	5.5.2020	DBAF492190	D

PANEL CODE	DOCUMENT NAME	DOCUMENT DESCRIPTION	SHEET N:o	DOCUMENT N:o	REVISION
BJU903	LIST OF DOCUMENTS	DRAWING LIST	1	DBAF492191	c
BJU903	DIMENSIONAL DRAWING	LAYOUT 1:10	2	DBAF492191	c
BJU903	OVERVIEW DIAGRAM	SINGLE LINE	3	DBAF492191	c
BJU903	CIRCUIT DIAGRAM	LOW VOLTAGE FEEDER	4	DBAF492191	c
BJU903	CIRCUIT DIAGRAM	COMMON INDICATOR/CONTROL	5	DBAF492191	c
BJU903	CIRCUIT DIAGRAM	PUMP CONTROL	6	DBAF492191	c
BJU903	CIRCUIT DIAGRAM	EXTERNAL ALARM	7	DBAF492191	d
BJU903	CONNECTION TABLE	CABLE LIST	8	DBAF492191	d
BJU903	CONNECTION TABLE	INTERNAL CONNECTION LIST	9	DBAF492191	c
BJU903	PARTS LIST	PARTS LIST	10	DBAF492191	c

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Project name: Mansourah - Massarah Gold Mine
Project id: P/20002
Product id: 6X20V32TS
Based on document id: DAAB198406 d
Customer document id:

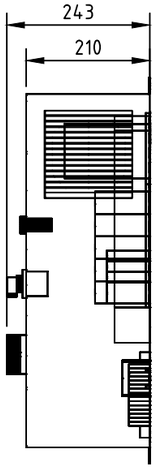
WÄRTSILÄ

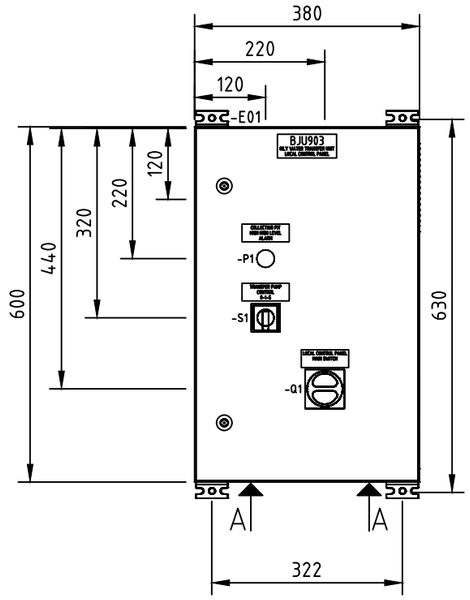
LIST OF DOCUMENTS
OILY WATER TRANSFER UNIT
LOCAL CONTROL PANEL
DRAWING LIST
CONFIDENTIAL

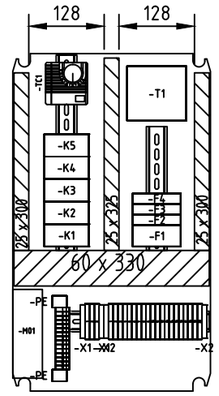
Created by	Create date	Language	Object designation
RSW004	5.5.2020	EN	BJU903
Reviewed by	**Review date**	**Sheet size**	**Page**
KSU004	5.5.2020	A3	1 / 10
Approved by	**Approval date**	**Document Id:**	**Rev.**
KSU004	5.5.2020	DBAF492191	D

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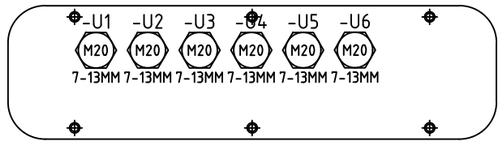
LOCATION - ON OILY WATER TRANSFER UNIT MODULE, NEAR PUMP SHELTER







SECTION A-A SCALE 1:4



PANEL INFORMATION					
WIDTH	380	mm	PROTECTION CLASS		IP54
HEIGHT	600	mm	COLOR		RAL 7035
DEPTH	210	mm	Ik	6	kA/1s

-E01	BJU903	OILY WATER TRANSFER UNIT	LOCAL CONTROL PANEL
-P1	COLLECTING PIT	HIGH HIGH LEVEL	ALARM
-Q1	LOCAL CONTROL PANEL	MAIN SWITCH	
-S1	TRANSFER PUMP	CONTROL	0-1-S

Rev.	Date	Created by	Appr. by	Revision text
c	21.7.2020	KKO070	KSU004	As manufactured update
b	29.6.2020	KKO070	KSU004	As per customer inputs
a	3.6.2020	KKO070	KSU004	As per customer inputs

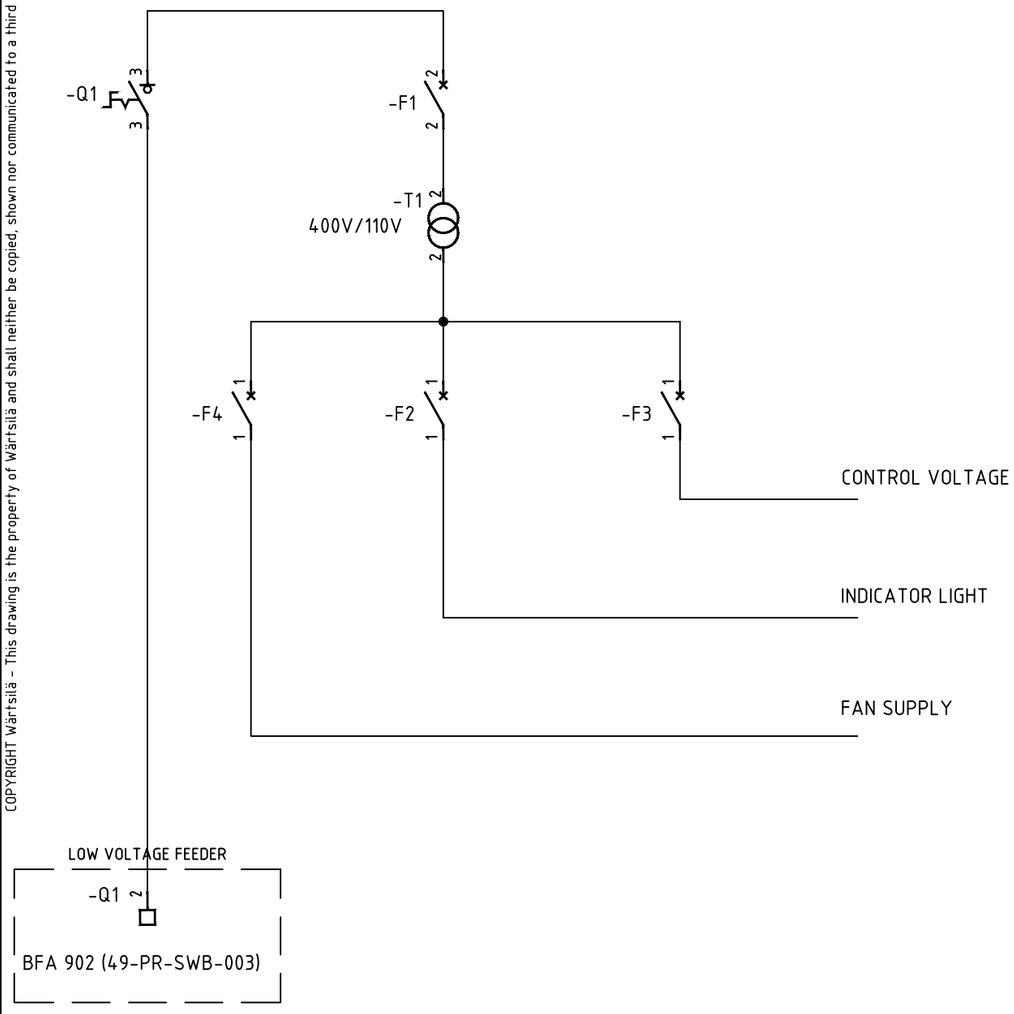
Project name: Mansourah - Massarah Gold Mine
Project id: P/20002
Product id: 6X20V32TS
Based on document id: DAAB198406 d
Customer document id:

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OILY WATER TRANSFER UNIT
LOCAL CONTROL PANEL
CONFIDENTIAL

Created by: RSW004	Create date: 5.5.2020	Language: EN	Object designation: BJU903
Reviewed by: KSU004	Review date: 5.5.2020	Sheet size: A3	Page: 2 / 10
Approved by: KSU004	Approval date: 5.5.2020	Document Id: DBAF492191	Rev.: D

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Rev.	Date	Created by	Appr. by	Revision text
c	21.7.2020	KKO070	KSU004	As manufactured update
b	29.6.2020	KKO070	KSU004	As per customer inputs
a	3.6.2020	KKO070	KSU004	As per customer inputs

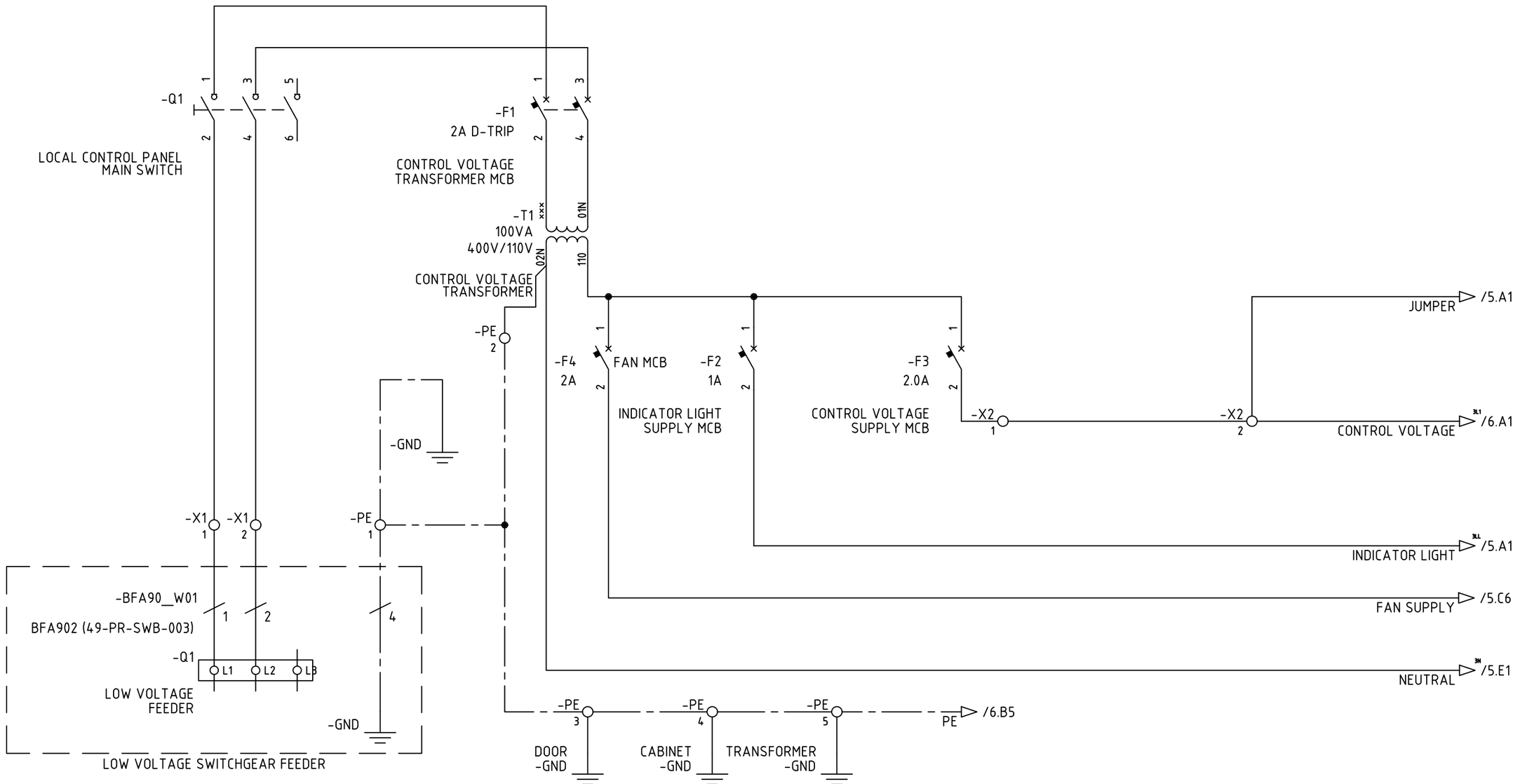
Project name: Mansourah - Massarah Gold Mine
Project id: P/20002
Product id: 6X20V32TS
Based on document id: DAAB198406 d
Customer document id:

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OVERVIEW DIAGRAM
OILY WATER TRANSFER UNIT
LOCAL CONTROL PANEL
SINGLE LINE
CONFIDENTIAL

Created by	Create date	Language	Object designation
RSW004	5.5.2020	EN	BJU903
Reviewed by	**Review date**	**Sheet size**	**Page**
KSU004	5.5.2020	A3	3 / 10
Approved by	**Approval date**	**Document Id:**	**Rev.**
KSU004	5.5.2020	DBAF492191	D

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SUPPLY					
Un	400V	V	2-phase	60	Hz
In	1	A		10	A

Rev.	Date	Created by	Appr. by	Revision text
b	29.6.2020	KKO070	KSU004	As per customer inputs
a	3.6.2020	KKO070	KSU004	As per customer inputs

Project name	Mansourah - Massarah Gold Mine		
Project id	P/20002	Product id	6X20V32TS
Based on document id	DAAB198406 d	Customer document id	

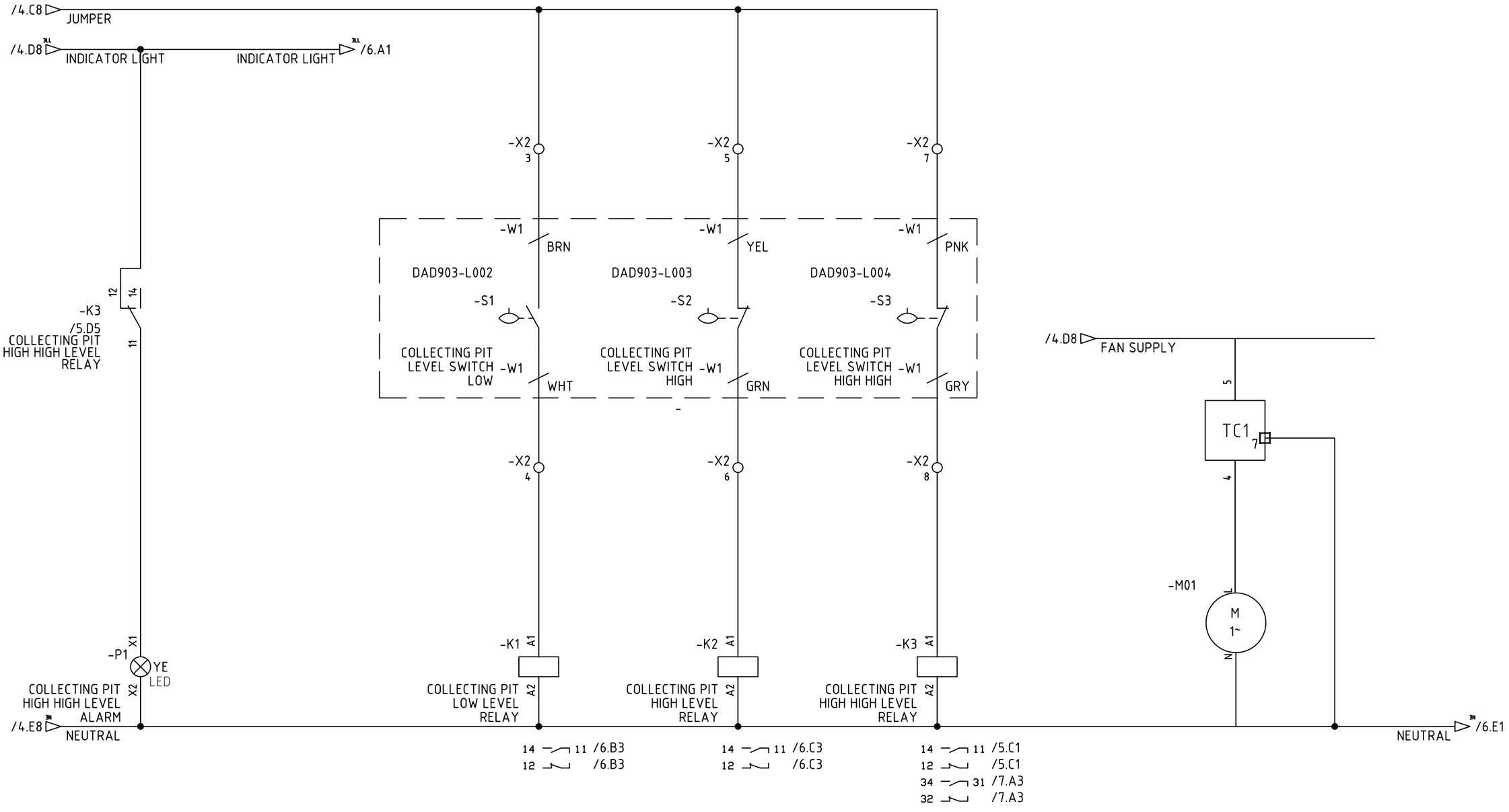
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CIRCUIT DIAGRAM

OILY WATER TRANSFER UNIT
LOCAL CONTROL PANEL
LOW VOLTAGE FEEDER
CONFIDENTIAL

Created by	RSW004	Create date	5.5.2020
Reviewed by	KSU004	Review date	5.5.2020
Approved by	KSU004	Approval date	5.5.2020
Language	EN	Object designation	BJU903
Sheet size	A3	Page	4 / 10
Document Id:	DBAF492191	Rev.	C

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Rev.	Date	Created by	Appr. by	Revision text
c	21.7.2020	KKO070	KSU004	As manufactured update
b	29.6.2020	KKO070	KSU004	As per customer inputs
a	3.6.2020	KKO070	KSU004	As per customer inputs

Project name	Mansourah - Massarah Gold Mine
Project id	P/20002
Product id	6X20V32TS
Based on document id	DAAB198406 d
Customer document id	

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CIRCUIT DIAGRAM

OILY WATER TRANSFER UNIT
LOCAL CONTROL PANEL
COMMON INDICATOR/CONTROL
CONFIDENTIAL

Created by: RSW004	Create date: 5.5.2020	Language: EN	Object designation: BJU903
Reviewed by: KSU004	Review date: 5.5.2020	Sheet size: A3	Page: 5 / 10
Approved by: KSU004	Approval date: 5.5.2020	Document Id: DBAF492191	Rev.: D

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/4.D8 3L1 CONTROL VOLTAGE

/5.A2 3L1 INDICATOR LIGHT

-K4 /6.D3 TRANSFER PUMP CONTROL RELAY 1 (12, 14, 11)

-K1 /5.D3 COLLECTING PIT LOW LEVEL RELAY (12, 14, 11)

-S1 0-1-S (21, 22) TRANSFER PUMP CONTROL 0-1-S

-S1 (13, 14) /6.B4 TRANSFER PUMP CONTROL 0-1-S

-S1					
0	1	S	T1	T2	REF
X			21	22	B4
			X1	X2	6E1
		X	13	14	6C4

-K2 /5.D4 COLLECTING PIT HIGH LEVEL RELAY (12, 14, 11)

-K4 /6.D3 TRANSFER PUMP CONTROL RELAY 1 (22, 24, 21)

-K5 /6.D4 TRANSFER PUMP CONTROL RELAY2 (22, 24, 21)

/4.E6 PE

-PE 6

-X2 9

-W01 3

-W01 1

-GND

DAD903-CV001

-Y1 X1 X2 TRANSFER PUMP CONTROL VALVE

-W01 2

-X2 10

-S1 X1 /6.B4 WH LED TRANSFER PUMP X2

/5.E8 3N 0-1-S NEUTRAL

-K4 A1 A2 TRANSFER PUMP CONTROL RELAY 1

-K5 A1 A2 TRANSFER PUMP CONTROL RELAY2

14 — 11 /6.C1
12 — /6.C1
24 — 21 /6.C3
22 — /6.C3

24 — 21 /6.A6
22 — /6.A6
34 — 31 /7.A4
32 — /7.A4

Rev.	Date	Created by	Appr. by	Revision text
c	21.7.2020	KKO070	KSU004	As manufactured update
b	29.6.2020	KKO070	KSU004	As per customer inputs
a	3.6.2020	KKO070	KSU004	As per customer inputs

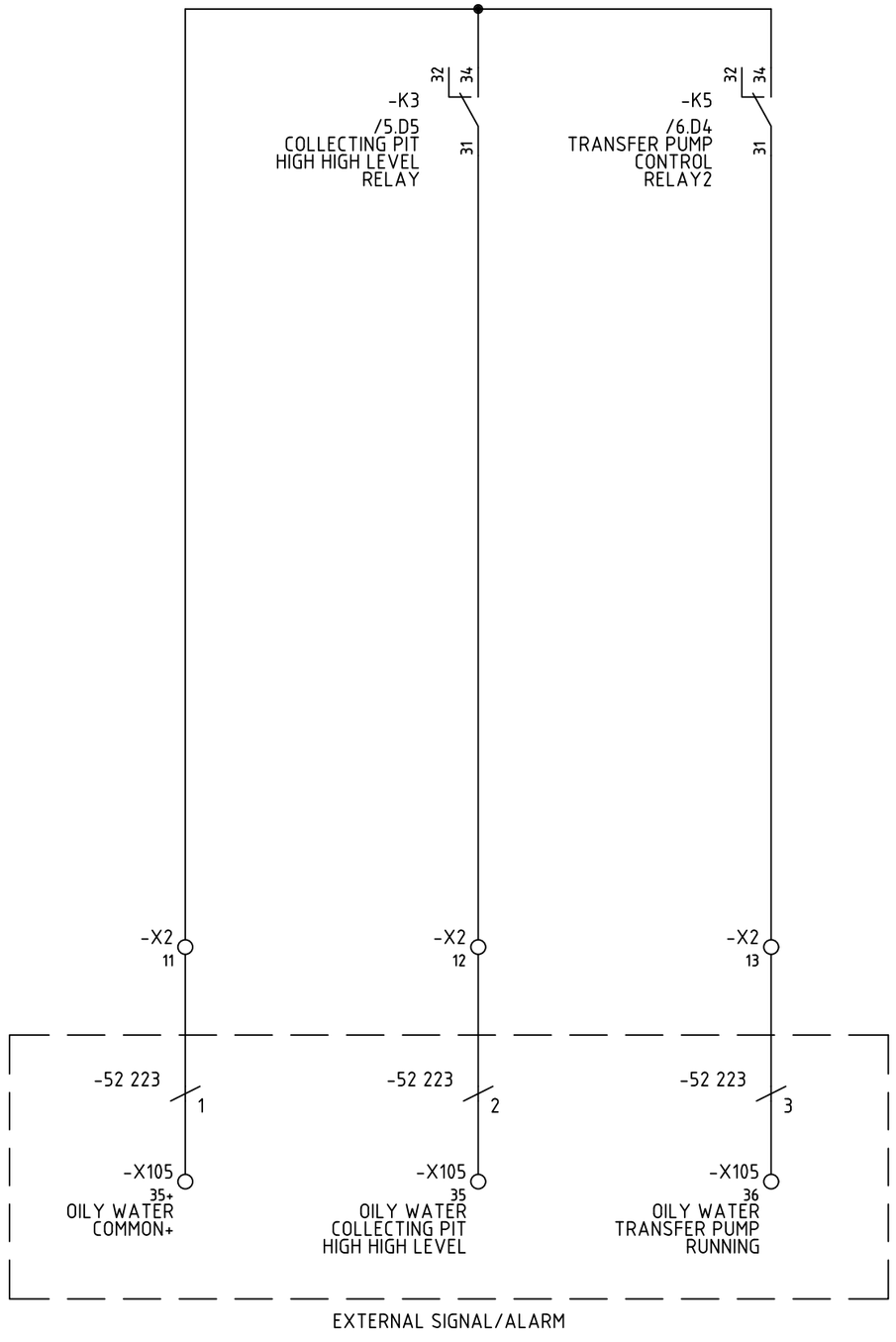
Project name: Mansourah - Massarah Gold Mine
Project id: P/20002
Product id: 6X20V32TS
Based on document id: DAAB198406 d
Customer document id:

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CIRCUIT DIAGRAM
OILY WATER TRANSFER UNIT
LOCAL CONTROL PANEL
PUMP CONTROL
CONFIDENTIAL

Created by: RSW004	Create date: 5.5.2020	Language: EN	Object designation: BJU903
Reviewed by: KSU004	Review date: 5.5.2020	Sheet size: A3	Page: 6 / 10
Approved by: KSU004	Approval date: 5.5.2020	Document Id: DBAF492191	Rev.: D

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Rev.	Date	Created by	Appr. by	Revision text
d	16.Feb.2021	MHA186	KSU004	Terminal Update
c	21.7.2020	KKO070	KSU004	As manufactured update
b	29.6.2020	KKO070	KSU004	As per customer inputs
a	3.6.2020	KKO070	KSU004	As per customer inputs

Project name: Mansourah - Massarah Gold Mine
Project id: P/20002
Product id: 6X20V32TS
Based on document id: DAAB198406 d
Customer document id:

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CIRCUIT DIAGRAM
OILY WATER TRANSFER UNIT
LOCAL CONTROL PANEL
EXTERNAL ALARM

Created by	Create date	Language	Object designation
RSW004	5.5.2020	EN	BJU903
Reviewed by	**Review date**	**Sheet size**	**Page**
KSU004	5.5.2020	A3	7 / 10
Approved by	**Approval date**	**Document Id:**	**Rev.**
KSU004	5.5.2020	DBAF492191	E

FACTORY INSTALLED CABLES TO BE SUPPLIED BY HELKAMA IF NOT OTHERWISE STATED

CABLE SPECIFICATION	CABLE TYPE	CABLE INSTALLATION	CABLE NAME	WIRE	FROM COMPONENT	PIN1	TO COMPONENT	PIN2	SH1	REF1	CABLE REMARK
1kV YY	3G1,5rm		-W01	1	-X2	9	=DAD903-CV001-Y1	X1	6	B6	INSTALLED AT FACTORY
			-W01	2	-X2	10	=DAD903-CV001-Y1	X2	6	C6	INSTALLED AT FACTORY
			-W01	3	-PE	6	=DAD903-CV001-GND		6	B6	INSTALLED AT FACTORY
1kV YCY	3x__+__	BFA902 (49-PR-SWB-003)	-BFA90__W01	1	=BFA902 (49-PR-SWB-003)-Q1	L1	-X1	1	4	E2	INSTALLED AT SITE
		BFA902 (49-PR-SWB-003)	-BFA90__W01	2	=BFA902 (49-PR-SWB-003)-Q1	L2	-X1	2	4	E2	INSTALLED AT SITE
		BFA902 (49-PR-SWB-003)	-BFA90__W01	4	=BFA902 (49-PR-SWB-003)-GND		-PE	1	4	E3	INSTALLED AT SITE
1kV YY	4x1,5rm	CFA902	-52 223	1	=CFA902-X105	35+	-X2	11	7	E2	INSTALLED AT SITE
		CFA902	-52 223	2	=CFA902-X105	35	-X2	12	7	E3	INSTALLED AT SITE
		CFA902	-52 223	3	=CFA902-X105	36	-X2	13	7	E4	INSTALLED AT SITE
		DAD903-L001	-W1	BRN	=DAD903-L002-S1		-X2	3	5	C3	INSTALLED AT SITE (INTEGRATED SILICON CABLE)
		DAD903-L001	-W1	GRN	=DAD903-L003-S2		-X2	6	5	C4	INSTALLED AT SITE (INTEGRATED SILICON CABLE)
		DAD903-L001	-W1	GRY	=DAD903-L004-S3		-X2	8	5	C5	INSTALLED AT SITE (INTEGRATED SILICON CABLE)
		DAD903-L001	-W1	PNK	=DAD903-L004-S3		-X2	7	5	C5	INSTALLED AT SITE (INTEGRATED SILICON CABLE)
		DAD903-L001	-W1	WHT	=DAD903-L002-S1		-X2	4	5	C3	INSTALLED AT SITE (INTEGRATED SILICON CABLE)
		DAD903-L001	-W1	YEL	=DAD903-L003-S2		-X2	5	5	C4	INSTALLED AT SITE (INTEGRATED SILICON CABLE)

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Rev.	Date	Created by	Appr. by	Revision text
d	16.Feb.2021	MHA186	KSU004	Terminal Update
c	21.7.2020	KKO070	KSU004	As manufactured update
b	29.6.2020	KKO070	KSU004	As per customer inputs
a	3.6.2020	KKO070	KSU004	As per customer inputs

Project name: Mansourah - Massarah Gold Mine
Project id: P/20002
Product id: 6X20V32TS
Based on document id: DAAB198406 d
Customer document id:

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CONNECTION TABLE
OILY WATER TRANSFER UNIT
LOCAL CONTROL PANEL
CABLE LIST

Created by: RSW004	Create date: 5.5.2020	Language: EN	Object designation: BJU903
Reviewed by: KSU004	Review date: 5.5.2020	Sheet size: A3	Page: 8 / 10
Approved by: KSU004	Approval date: 5.5.2020	Document Id: DBAF492191	Rev.: E

SH1	REF1	CMP1	PIN1	WLAY1	CMP2	PIN2	SH2	REF2
4	C4	-T1	110	AC CONTROL	-F2	1	4	C5
4	C4	-T1	02N	AC CONTROL	-K4	A2	6	E3
4	C4	-F4	2	AC CONTROL	-TC1	5	5	C7
4	C4	-F4	1	AC CONTROL	-F3	1	4	C6
4	C5	-F2	2	AC CONTROL	-K3	12	5	C1
4	C5	-F2	1	AC CONTROL	-F4	1	4	C4
4	C6	-F3	2	AC CONTROL	-X2	1	4	D6
4	D6	-X2	1	AC CONTROL	-X2	2	4	D7
4	D7	-X2	2	AC CONTROL	-K1	14	6	B3
5	C1	-K3	11	AC CONTROL	-P1	X1	5	D1
5	C1	-K3	12	AC CONTROL	-K4	14	6	C1
5	E1	-P1	X2	AC CONTROL	-S1	X2	6	E1
5	C3	-X2	4	AC CONTROL	-K1	A1	5	D3
5	E3	-K1	A2	AC CONTROL	-K2	A2	5	E4
5	C4	-X2	6	AC CONTROL	-K2	A1	5	D4
5	E4	-K2	A2	AC CONTROL	-K3	A2	5	E5
5	C5	-X2	8	AC CONTROL	-K3	A1	5	D5
5	E5	-K3	A2	AC CONTROL	-P1	X2	5	E1
5	D7	-TC1	4	AC CONTROL	-M01	L	5	D7
6	C1	-K4	11	AC CONTROL	-S1	X1	6	D1
6	E1	-S1	X2	AC CONTROL	-M01	N	5	D7
6	B3	-K1	11	AC CONTROL	-S1	21	6	C3
6	B3	-K1	14	AC CONTROL	-K5	24	6	A6
6	C3	-K2	12	AC CONTROL	-K4	24	6	C3
6	C3	-K4	24	AC CONTROL	-S1	22	6	C3
6	D3	-K2	11	AC CONTROL	-S1	14	6	C4
6	D3	-K4	21	AC CONTROL	-K4	A1	6	D3
6	D3	-K4	A1	AC CONTROL	-K5	A1	6	D4
6	E3	-K4	A2	AC CONTROL	-K5	A2	6	E4
6	E4	-K5	A2	AC CONTROL	-X2	10	6	C6
6	A6	-K5	21	AC CONTROL	-X2	9	6	B6
6	A6	-K5	24	AC CONTROL	-S1	13	6	C4
6	C6	-X2	10	AC CONTROL	-K1	A2	5	E3
7	D2	-X2	11	AC CONTROL	-K3	34	7	A3
7	A3	-K3	31	AC CONTROL	-X2	12	7	D3
7	A3	-K3	34	AC CONTROL	-K5	34	7	A4
7	A4	-K5	31	AC CONTROL	-X2	13	7	D4
4	D7	-X2	2	AC CONTROL JMP	-X2	5	5	B4
5	B3	-X2	3	AC CONTROL JMP	-X2	2	4	D7
5	B4	-X2	5	AC CONTROL JMP	-X2	7	5	B5
4	B2	-Q1	4	AC POWER	-X1	2	4	D2
4	B2	-Q1	3	AC POWER	-F1	3	4	B4
4	B2	-Q1	2	AC POWER	-X1	1	4	D2
4	B2	-Q1	1	AC POWER	-F1	1	4	B4
4	B4	-F1	4	AC POWER	-T1	01N	4	C4
4	B4	-F1	2	AC POWER	-T1	×××	4	C4
5	C3	-S1		EXTERNAL CONTROL (INTERLOCK)	-X2	4	5	C3
5	C3	-S1		EXTERNAL CONTROL (INTERLOCK)	-X2	3	5	B3
5	C4	-S2		EXTERNAL CONTROL (INTERLOCK)	-X2	6	5	C4
5	C4	-S2		EXTERNAL CONTROL (INTERLOCK)	-X2	5	5	B4
5	C5	-S3		EXTERNAL CONTROL (INTERLOCK)	-X2	8	5	C5
5	C5	-S3		EXTERNAL CONTROL (INTERLOCK)	-X2	7	5	B5
6	B6	-X2	9	EXTERNAL CONTROL (INTERLOCK)	-Y1	X1	6	C6
6	C6	-X2	10	EXTERNAL CONTROL (INTERLOCK)	-Y1	X2	6	C6
7	E2	-X_		EXTERNAL CONTROL (INTERLOCK)	-X2	11	7	D2
7	E3	-X_		EXTERNAL CONTROL (INTERLOCK)	-X2	12	7	D3
7	E4	-X_		EXTERNAL CONTROL (INTERLOCK)	-X2	13	7	D4
4	E2	-Q1	L2	EXTERNAL POWER	-X1	2	4	D2
4	E2	-Q1	L1	EXTERNAL POWER	-X1	1	4	D2
4	E3	-GND		EXTERNAL PROTECTIVE	-PE	1	4	D3
6	B6	-PE	6	EXTERNAL PROTECTIVE	-GND		6	B6
4	D3	-PE	1	PROTECTIVE	-GND		4	D3
4	C4	-T1	02N	PROTECTIVE	-PE	2	4	C3

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Rev.	Date	Created by	Appr. by	Revision text
c	21.7.2020	KKO070	KSU004	As manufactured update
b	29.6.2020	KKO070	KSU004	As per customer inputs
a	3.6.2020	KKO070	KSU004	As per customer inputs

Project name: Mansourah - Massarah Gold Mine
Project id: P/20002
Product id: 6X20V32TS
Based on document id: DAAB198406 d
Customer document id:

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CONNECTION TABLE
OILY WATER TRANSFER UNIT
LOCAL CONTROL PANEL
INTERNAL CONNECTION LIST
CONFIDENTIAL

Created by	Create date	Language	Object designation
RSW004	5.5.2020	EN	BJU903
Reviewed by	**Review date**	**Sheet size**	**Page**
KSU004	5.5.2020	A3	9 / 10
Approved by	**Approval date**	**Document Id:**	**Rev.**
KSU004	5.5.2020	DBAF492191	D

COMPONENT NAME	QTY	MANUFACTURER	PART CODE	RATING	DESC
-E01	1	RITTAL	1038000		
-E01	4	RITTAL	2508020		WALL MOUNTING BRACKET
-E01	1	RITTAL			SECURITY LOCK
-F1	1	Schneider Electric	A9F05202	2A D-TRIP	MINIATURE CIRCUIT BREAKER
-F2	1	Schneider Electric	A9F04101	1A	MINIATURE CIRCUIT BREAKER
-F3	1	Schneider Electric	A9F04102	2.0A	MINIATURE CIRCUIT BREAKER
-K1	1	Carlo Gavazzi	RCP11003115120VAC		UNIVERSAL PLUG-IN RELAY, 3 CO
-K1	1	Carlo Gavazzi	ZPD12A		SOCKET
-K2	1	Carlo Gavazzi	RCP11003115120VAC		UNIVERSAL PLUG-IN RELAY, 3 CO
-K2	1	Carlo Gavazzi	ZPD12A		SOCKET
-K3	1	Carlo Gavazzi	RCP11003115120VAC		UNIVERSAL PLUG-IN RELAY, 3 CO
-K3	1	Carlo Gavazzi	ZPD12A		SOCKET
-K4	1	Carlo Gavazzi	RCP11003115120VAC		UNIVERSAL PLUG-IN RELAY, 3 CO
-K4	1	Carlo Gavazzi	ZPD12A		SOCKET
-K5	1	Carlo Gavazzi	RCP11003115120VAC		UNIVERSAL PLUG-IN RELAY, 3 CO
-K5	1	Carlo Gavazzi	ZPD12A		SOCKET
-P1	1	Schneider Electric	XB7EV05GP		MONOLITHIC INDICATOR LIGHT, YELLOW, INTEGRAL LED
-PE	1	Phoenix Contact	NLS-CU 3/10		BUSBAR (FOR PE TERMINAL BLOCK)
-PE	2	Phoenix Contact	AB/SS		END PIECE (FOR PE TERMINAL BLOCK)
-PE	15	Phoenix Contact	AKG 4 GNYE		PE TERMINAL BLOCK
-Q1	1	BACO	0172001		
-S1	1	Schneider Electric	ZB5AK1813 + ZB5AW0G15		CONTROL SWITCH, 3 POSITION
-T1	1	Trafomic	0423193	200VA	CONTROL VOLTAGE TRANSFORMER
-U1	1	Agro	1555.20.13	M20	CABLE GLAND, M20
-U2	1	Agro	1555.20.13	M20	CABLE GLAND, M20
-U3	1	Agro	1555.20.13	M20	CABLE GLAND, M20
-U4	1	Agro	1555.20.13	M20	CABLE GLAND, M20
-U5	1	Agro	1555.20.13	M20	CABLE GLAND, M20
-U6	1	Agro	1555.20.13	M20	CABLE GLAND, M20
-X1	2	Phoenix Contact	CLIPFIX 35		END PIECE (FOR TERMINAL BLOCK)
-X1	3	Weidmuller	ZDU 6		
-X2	2	Phoenix Contact	CLIPFIX 35		END PIECE (FOR TERMINAL BLOCK)
-X2	20	Weidmuller	ZDU 4		
-F4	1	Schneider Electric	A9F04102		MINIATURE CIRCUIT BREAKER
-FILTER	1	Kryos	GSF15		FILTER
-M01	1	Kryos	GSV1500203		FAN
-TC1	1	RITTAL	3100000		THERMOSTAT

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Project name: Mansourah - Massarah Gold Mine
Project id: P/20002
Product id: 6X20V32TS
Based on document id: DAAB198406 d
Customer document id:

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PARTS LIST
OILY WATER TRANSFER UNIT
LOCAL CONTROL PANEL
PARTS LIST
CONFIDENTIAL

Created by: RSW004	Create date: 5.5.2020	Language: EN	Object designation: BJU903
Reviewed by: KSU004	Review date: 5.5.2020	Sheet size: A3	Page: 10 / 10
Approved by: KSU004	Approval date: 5.5.2020	Document Id: DBAF492191	Rev.: D

PANEL CODE	DOCUMENT NAME	DOCUMENT DESCRIPTION	SHEET N:o	DOCUMENT N:o	REVISION
BJU904	LIST OF DOCUMENTS	DRAWING LIST	1	DBAF492192	c
BJU904	DIMENSIONAL DRAWING	LAYOUT 1:10	2	DBAF492192	c
BJU904	OVERVIEW DIAGRAM	SINGLE LINE	3	DBAF492192	c
BJU904	CIRCUIT DIAGRAM	LOW VOLTAGE FEEDER	4	DBAF492192	c
BJU904	CIRCUIT DIAGRAM	COMMON INDICATOR/CONTROL	5	DBAF492192	c
BJU904	CIRCUIT DIAGRAM	PUMP CONTROL	6	DBAF492192	c
BJU904	CIRCUIT DIAGRAM	EXTERNAL ALARM	7	DBAF492192	d
BJU904	CONNECTION TABLE	CABLE LIST	8	DBAF492192	d
BJU904	CONNECTION TABLE	INTERNAL CONNECTION LIST	9	DBAF492192	c
BJU904	PARTS LIST	PARTS LIST	10	DBAF492192	c

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Rev.	Date	Created by	Appr. by	Revision text
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Project name: Mansourah - Massarah Gold Mine
Project id: P/20002
Product id: 6X20V32TS
Based on document id: DAAB198406 d
Customer document id:

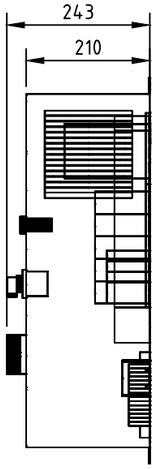
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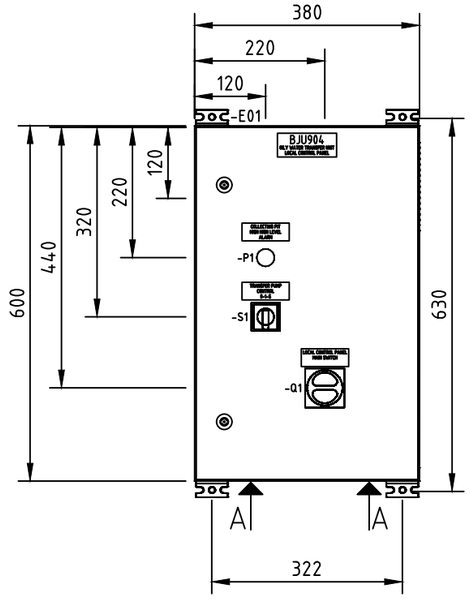
LIST OF DOCUMENTS
OILY WATER TRANSFER UNIT
LOCAL CONTROL PANEL
DRAWING LIST
CONFIDENTIAL

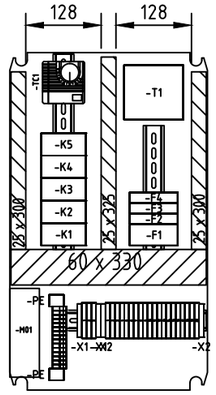
Created by	Create date	Language	Object designation
RSW004	5.5.2020	EN	BJU904
Reviewed by	**Review date**	**Sheet size**	**Page**
KSU004	5.5.2020	A3	1 / 10
Approved by	**Approval date**	**Document Id:**	**Rev.**
KSU004	5.5.2020	DBAF492192	D

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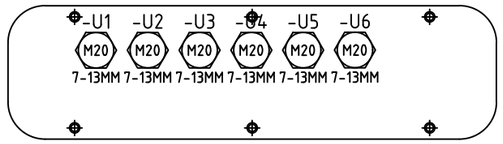
LOCATION - ON OILY WATER TRANSFER UNIT MODULE, NEAR OUTSIDE FUEL TREATMENT HOUSE & LUBE OIL TANK







SECTION A-A SCALE 1:4



PANEL INFORMATION					
WIDTH	380	mm	PROTECTION CLASS		IP54
HEIGHT	600	mm	COLOR		RAL 7035
DEPTH	210	mm	Ik	6	kA/1s

-E01	BJU904	OILY WATER TRANSFER UNIT	LOCAL CONTROL PANEL
-P1	COLLECTING PIT	HIGH HIGH LEVEL	ALARM
-Q1	LOCAL CONTROL PANEL	MAIN SWITCH	
-S1	TRANSFER PUMP	CONTROL	0-1-S

Rev.	Date	Created by	Appr. by	Revision text
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b	29.6.2020	KKO070	KSU004	As per customer inputs
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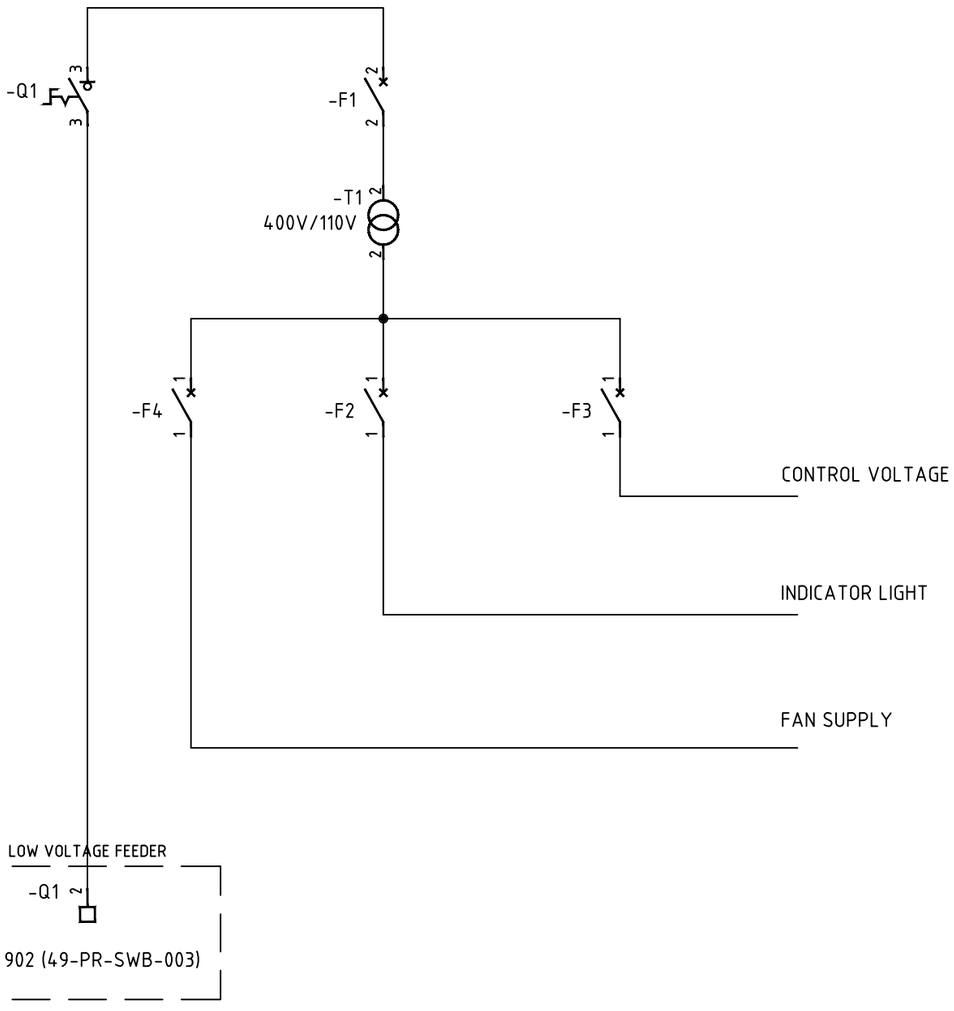
Project name: Mansourah - Massarah Gold Mine
Project id: P/20002
Product id: 6X20V32TS
Based on document id: DAAB198406 d
Customer document id:

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OILY WATER TRANSFER UNIT
LOCAL CONTROL PANEL
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Created by: RSW004	Create date: 5.5.2020	Language: EN	Object designation: BJU904
Reviewed by: KSU004	Review date: 5.5.2020	Sheet size: A3	Page: 2 / 10
Approved by: KSU004	Approval date: 5.5.2020	Document Id: DBAF492192	Rev.: D

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c	21.7.2020	KKO070	KSU004	As manufactured update
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a	3.6.2020	KKO070	KSU004	As per customer inputs

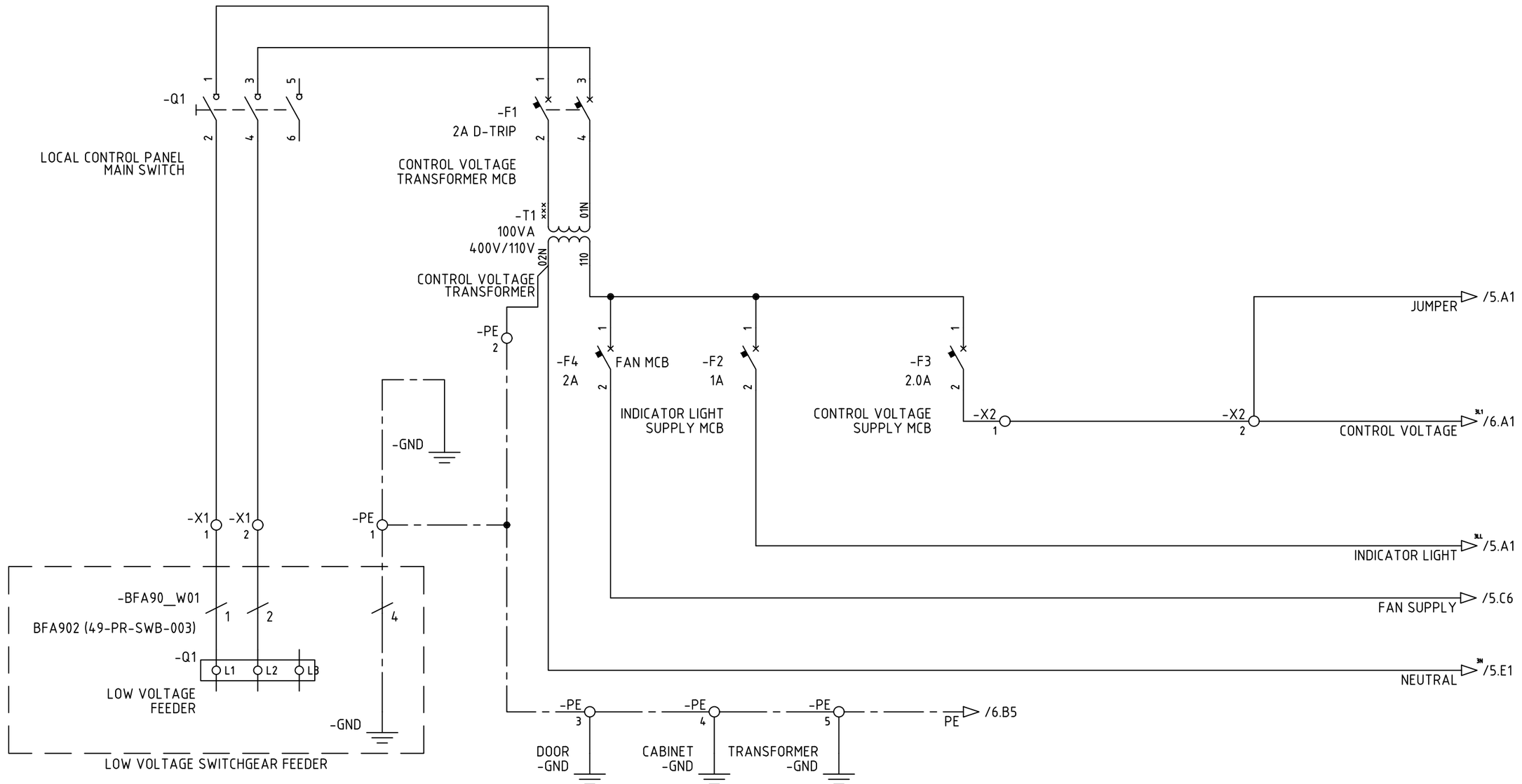
Project name: Mansourah - Massarah Gold Mine
Project id: P/20002
Product id: 6X20V32TS
Based on document id: DAAB198406 d
Customer document id:

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OVERVIEW DIAGRAM
OILY WATER TRANSFER UNIT
LOCAL CONTROL PANEL
SINGLE LINE
CONFIDENTIAL

Created by	Create date	Language	Object designation
RSW004	5.5.2020	EN	BJU904
Reviewed by	**Review date**	**Sheet size**	**Page**
KSU004	5.5.2020	A3	3 / 10
Approved by	**Approval date**	**Document Id:**	**Rev.**
KSU004	5.5.2020	DBAF492192	D

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SUPPLY					
Un	400V	V	2-phase	60	Hz
In	1	A		10	A

Rev.	Date	Created by	Appr. by	Revision text
c	21.7.2020	KKO070	KSU004	As manufactured update
b	29.6.2020	KKO070	KSU004	As per customer inputs
a	3.6.2020	KKO070	KSU004	As per customer inputs

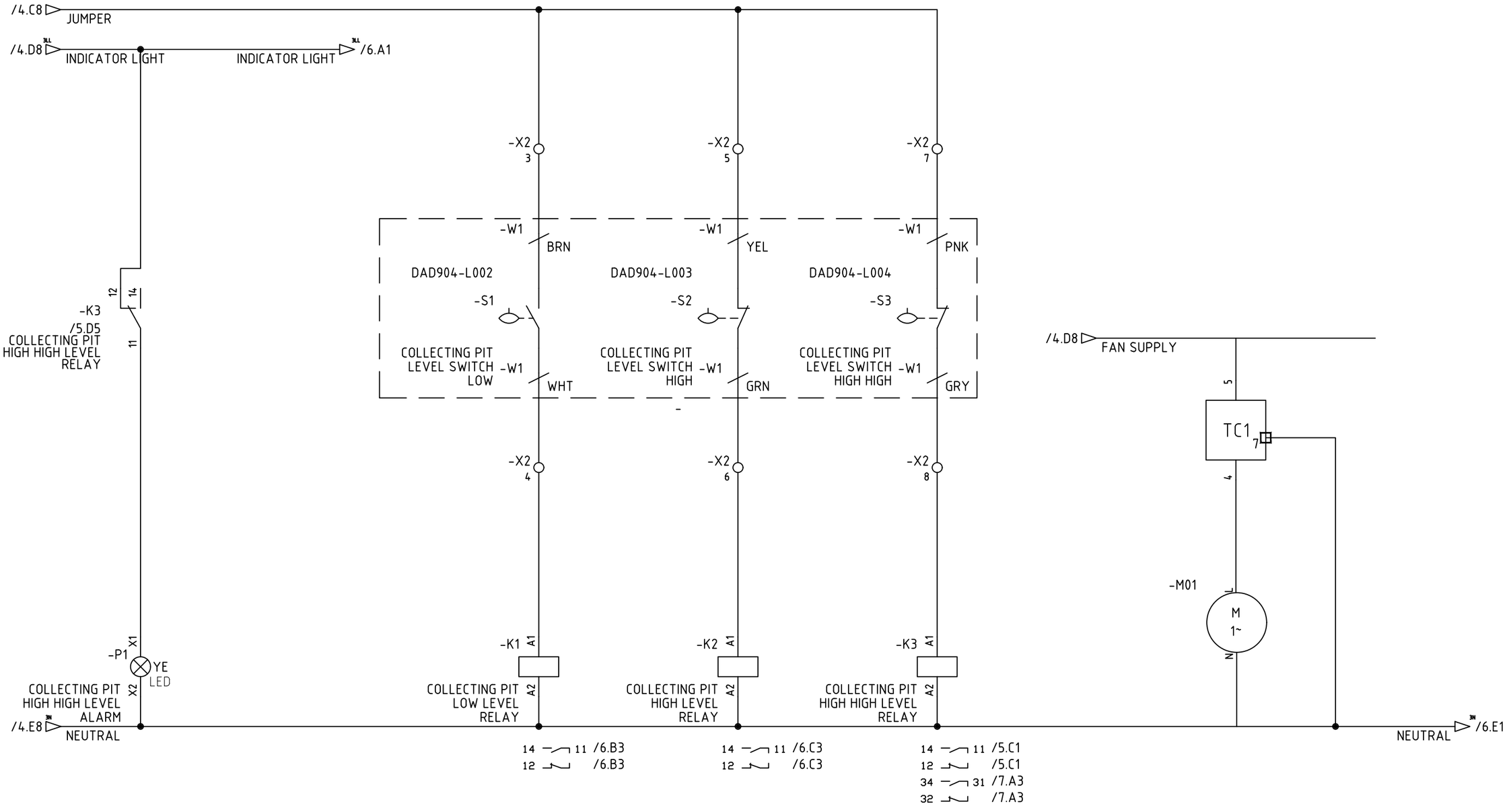
Project name	Mansourah - Massarah Gold Mine
Project id	P/20002
Product id	6X20V32TS
Based on document id	DAAB198406 d
Customer document id	

CIRCUIT DIAGRAM

OILY WATER TRANSFER UNIT
LOCAL CONTROL PANEL
LOW VOLTAGE FEEDER
CONFIDENTIAL

Created by RSW004	Create date 5.5.2020	Language EN	Object designation BJU904
Reviewed by KSU004	Review date 5.5.2020	Sheet size A3	Page 4 / 10
Approved by KSU004	Approval date 5.5.2020	Document Id: DBAF492192	Rev. D

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Project name	Mansourah - Massarah Gold Mine		
Project id	P/20002	Product id	6X20V32TS
Based on document id	DAAB198406	Customer document id	d

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CIRCUIT DIAGRAM			
OILY WATER TRANSFER UNIT LOCAL CONTROL PANEL COMMON INDICATOR/CONTROL			
CONFIDENTIAL			

Created by	Create date	Language	Object designation
RSW004	5.5.2020	EN	BJU904
Reviewed by	Review date	Sheet size	Page
KSU004	5.5.2020	A3	5 / 10
Approved by	Approval date	Document Id:	Rev.
KSU004	5.5.2020	DBAF492192	D

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/4.D8 3L1 CONTROL VOLTAGE

/5.A2 3L1 INDICATOR LIGHT

-K4 /6.D3 TRANSFER PUMP CONTROL RELAY 1 (12, 14, 11)

-K1 /5.D3 COLLECTING PIT LOW LEVEL RELAY (12, 14, 11)

-S1 0-1-S (21, 22) TRANSFER PUMP CONTROL 0-1-S

-S1 (13, 14) /6.B4 TRANSFER PUMP CONTROL 0-1-S

-S1					
0	1	S	T1	T2	REF
X			21	22	B4
			X1	X2	6E1
		X	13	14	6C4

-K2 /5.D4 COLLECTING PIT HIGH LEVEL RELAY (12, 14, 11)

-K4 /6.D3 TRANSFER PUMP CONTROL RELAY 1 (22, 24, 21)

-K5 /6.D4 TRANSFER PUMP CONTROL RELAY2 (22, 24, 21)

/4.E6 PE

-PE 6 -X2 9

-W01 3 -W01 1

-GND

DAD904-CV001

-Y1 X1 X2 TRANSFER PUMP CONTROL VALVE

-W01 2

-X2 10

-S1 X1 /6.B4 WH LED TRANSFER PUMP X2

/5.E8 0-1-S NEUTRAL

-K4 A1 A2 TRANSFER PUMP CONTROL RELAY 1

-K5 A1 A2 TRANSFER PUMP CONTROL RELAY2

14 — 11 /6.C1
12 — /6.C1
24 — 21 /6.C3
22 — /6.C3

24 — 21 /6.A6
22 — /6.A6
34 — 31 /7.A4
32 — /7.A4

Rev.	Date	Created by	Appr. by	Revision text
c	21.7.2020	KKO070	KSU004	As manufactured update
b	29.6.2020	KKO070	KSU004	As per customer inputs
a	3.6.2020	KKO070	KSU004	As per customer inputs

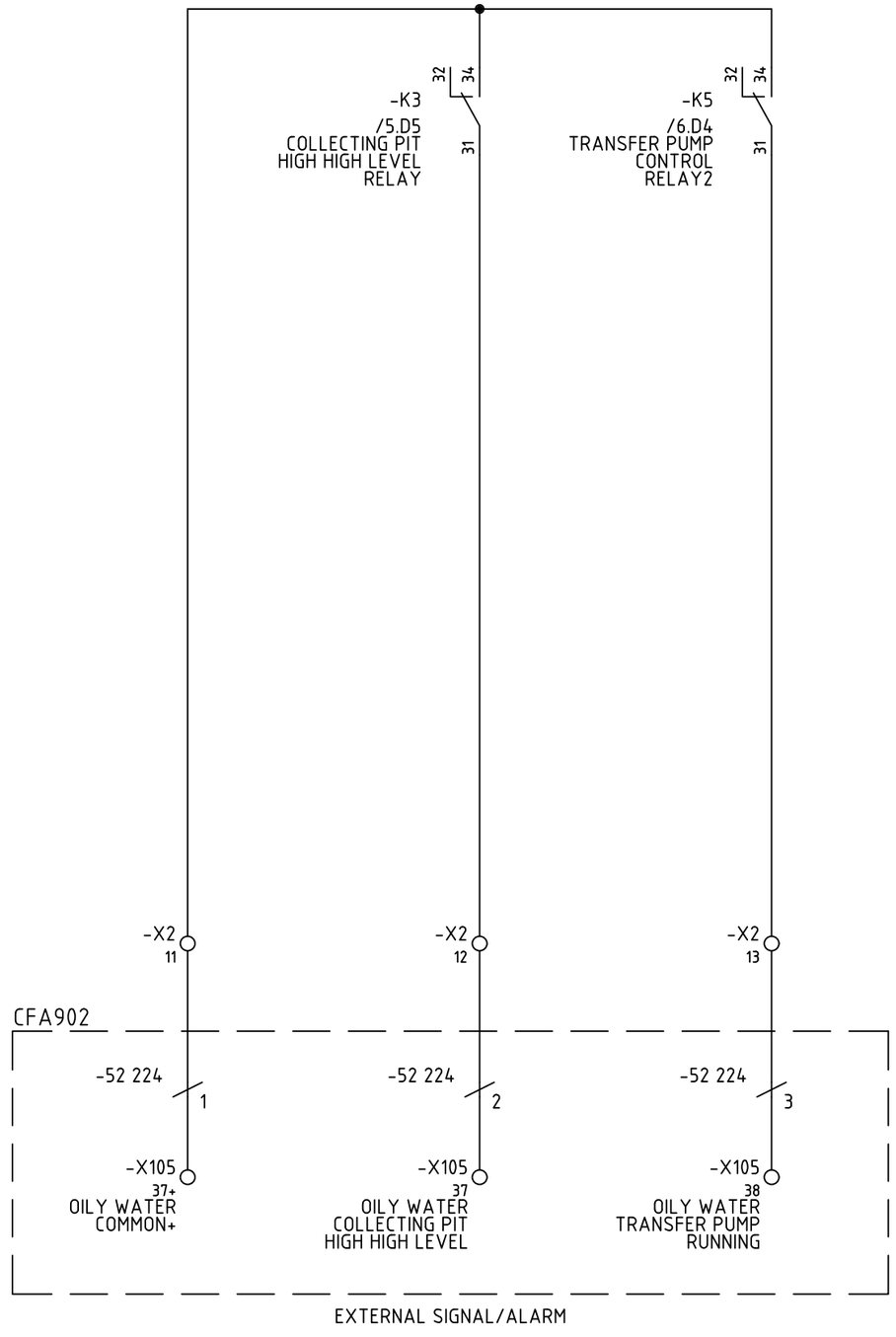
Project name: Mansourah - Massarah Gold Mine
Project id: P/20002
Product id: 6X20V32TS
Based on document id: DAAB198406 d
Customer document id:

WÄRTSILÄ

CIRCUIT DIAGRAM
OILY WATER TRANSFER UNIT
LOCAL CONTROL PANEL
PUMP CONTROL
CONFIDENTIAL

Created by	Create date	Language	Object designation
RSW004	5.5.2020	EN	BJU904
Reviewed by	Review date	Sheet size	Page
KSU004	5.5.2020	A3	6 / 10
Approved by	Approval date	Document Id:	Rev.
KSU004	5.5.2020	DBAF492192	D

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Rev.	Date	Created by	Appr. by	Revision text
d	16.Feb.2021	MHA186	KSU004	Terminal Update
c	21.7.2020	KKO070	KSU004	As manufactured update
b	29.6.2020	KKO070	KSU004	As per customer inputs
a	3.6.2020	KKO070	KSU004	As per customer inputs

Project name: Mansourah - Massarah Gold Mine
Project id: P/20002
Product id: 6X20V32TS
Based on document id: DAAB198406 d
Customer document id:

WÄRTSILÄ

CIRCUIT DIAGRAM
OILY WATER TRANSFER UNIT
LOCAL CONTROL PANEL
EXTERNAL ALARM

Created by	Create date	Language	Object designation
RSW004	5.5.2020	EN	BJU904
Reviewed by	Review date	Sheet size	Page
KSU004	5.5.2020	A3	7 / 10
Approved by	Approval date	Document Id:	Rev.
KSU004	5.5.2020	DBAF492192	E

FACTORY INSTALLED CABLES TO BE SUPPLIED BY HELKAMA IF NOT OTHERWISE STATED

CABLE SPECIFICATION	CABLE TYPE	CABLE INSTALLATION	CABLE NAME	WIRE	FROM COMPONENT	PIN1	TO COMPONENT	PIN2	SH1	REF1	CABLE REMARK
1kV YY	3G1,5rm		-W01	1	-X2	9	=DAD904-CV001-Y1	X1	6	B6	INSTALLED AT FACTORY
			-W01	2	-X2	10	=DAD904-CV001-Y1	X2	6	C6	INSTALLED AT FACTORY
			-W01	3	-PE	6	=DAD904-CV001-GND		6	B6	INSTALLED AT FACTORY
1kV YCY	3x__+__	BFA902 (49-PR-SWB-003)	-BFA90__W01	1	=BFA902 (49-PR-SWB-003)-Q1	L1	-X1	1	4	E2	INSTALLED AT SITE
		BFA902 (49-PR-SWB-003)	-BFA90__W01	2	=BFA902 (49-PR-SWB-003)-Q1	L2	-X1	2	4	E2	INSTALLED AT SITE
		BFA902 (49-PR-SWB-003)	-BFA90__W01	4	=BFA902 (49-PR-SWB-003)-GND		-PE	1	4	E3	INSTALLED AT SITE
1kV YY	4x1,5rm	CFA902	-52 224	1	=CFA902-X105	37+	-X2	11	7	E2	INSTALLED AT SITE
		CFA902	-52 224	2	=CFA902-X105	37	-X2	12	7	E3	INSTALLED AT SITE
		CFA902	-52 224	3	=CFA902-X105	38	-X2	13	7	E4	INSTALLED AT SITE
		DAD904-L001	-W1	BRN	=DAD904-L002-S1		-X2	3	5	C3	INSTALLED AT SITE (INTEGRATED SILICON CABLE)
		DAD904-L001	-W1	GRN	=DAD904-L003-S2		-X2	6	5	C4	INSTALLED AT SITE (INTEGRATED SILICON CABLE)
		DAD904-L001	-W1	GRY	=DAD904-L004-S3		-X2	8	5	C5	INSTALLED AT SITE (INTEGRATED SILICON CABLE)
		DAD904-L001	-W1	PNK	=DAD904-L004-S3		-X2	7	5	C5	INSTALLED AT SITE (INTEGRATED SILICON CABLE)
		DAD904-L001	-W1	WHT	=DAD904-L002-S1		-X2	4	5	C3	INSTALLED AT SITE (INTEGRATED SILICON CABLE)
		DAD904-L001	-W1	YEL	=DAD904-L003-S2		-X2	5	5	C4	INSTALLED AT SITE (INTEGRATED SILICON CABLE)

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Project name: Mansourah - Massarah Gold Mine
Project id: P/20002
Product id: 6X20V32TS
Based on document id: DAAB198406 d
Customer document id:

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CONNECTION TABLE
OILY WATER TRANSFER UNIT
LOCAL CONTROL PANEL
CABLE LIST

Created by: RSW004	Create date: 5.5.2020	Language: EN	Object designation: BJU904
Reviewed by: KSU004	Review date: 5.5.2020	Sheet size: A3	Page: 8 / 10
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SH1	REF1	CMP1	PIN1	WLAY1	CMP2	PIN2	SH2	REF2
4	C4	-T1	110	AC CONTROL	-F2	1	4	C5
4	C4	-T1	02N	AC CONTROL	-K4	A2	6	E3
4	C4	-F4	2	AC CONTROL	-TC1	5	5	C7
4	C4	-F4	1	AC CONTROL	-F3	1	4	C6
4	C5	-F2	2	AC CONTROL	-K3	12	5	C1
4	C5	-F2	1	AC CONTROL	-F4	1	4	C4
4	C6	-F3	2	AC CONTROL	-X2	1	4	D6
4	D6	-X2	1	AC CONTROL	-X2	2	4	D7
4	D7	-X2	2	AC CONTROL	-K1	14	6	B3
5	C1	-K3	11	AC CONTROL	-P1	X1	5	D1
5	C1	-K3	12	AC CONTROL	-K4	14	6	C1
5	E1	-P1	X2	AC CONTROL	-S1	X2	6	E1
5	C3	-X2	4	AC CONTROL	-K1	A1	5	D3
5	E3	-K1	A2	AC CONTROL	-K2	A2	5	E4
5	C4	-X2	6	AC CONTROL	-K2	A1	5	D4
5	E4	-K2	A2	AC CONTROL	-K3	A2	5	E5
5	C5	-X2	8	AC CONTROL	-K3	A1	5	D5
5	E5	-K3	A2	AC CONTROL	-P1	X2	5	E1
5	D7	-TC1	4	AC CONTROL	-M01	L	5	D7
6	C1	-K4	11	AC CONTROL	-S1	X1	6	D1
6	E1	-S1	X2	AC CONTROL	-M01	N	5	D7
6	B3	-K1	11	AC CONTROL	-S1	21	6	C3
6	B3	-K1	14	AC CONTROL	-K5	24	6	A6
6	C3	-K2	12	AC CONTROL	-K4	24	6	C3
6	C3	-K4	24	AC CONTROL	-S1	22	6	C3
6	D3	-K2	11	AC CONTROL	-S1	14	6	C4
6	D3	-K4	21	AC CONTROL	-K4	A1	6	D3
6	D3	-K4	A1	AC CONTROL	-K5	A1	6	D4
6	E3	-K4	A2	AC CONTROL	-K5	A2	6	E4
6	E4	-K5	A2	AC CONTROL	-X2	10	6	C6
6	A6	-K5	21	AC CONTROL	-X2	9	6	B6
6	A6	-K5	24	AC CONTROL	-S1	13	6	C4
6	C6	-X2	10	AC CONTROL	-K1	A2	5	E3
7	D2	-X2	11	AC CONTROL	-K3	34	7	A3
7	A3	-K3	31	AC CONTROL	-X2	12	7	D3
7	A3	-K3	34	AC CONTROL	-K5	34	7	A4
7	A4	-K5	31	AC CONTROL	-X2	13	7	D4
4	D7	-X2	2	AC CONTROL JMP	-X2	5	5	B4
5	B3	-X2	3	AC CONTROL JMP	-X2	2	4	D7
5	B4	-X2	5	AC CONTROL JMP	-X2	7	5	B5
4	B2	-Q1	4	AC POWER	-X1	2	4	D2
4	B2	-Q1	3	AC POWER	-F1	3	4	B4
4	B2	-Q1	2	AC POWER	-X1	1	4	D2
4	B2	-Q1	1	AC POWER	-F1	1	4	B4
4	B4	-F1	4	AC POWER	-T1	01N	4	C4
4	B4	-F1	2	AC POWER	-T1	×××	4	C4
5	C3	-S1		EXTERNAL CONTROL (INTERLOCK)	-X2	4	5	C3
5	C3	-S1		EXTERNAL CONTROL (INTERLOCK)	-X2	3	5	B3
5	C4	-S2		EXTERNAL CONTROL (INTERLOCK)	-X2	6	5	C4
5	C4	-S2		EXTERNAL CONTROL (INTERLOCK)	-X2	5	5	B4
5	C5	-S3		EXTERNAL CONTROL (INTERLOCK)	-X2	8	5	C5
5	C5	-S3		EXTERNAL CONTROL (INTERLOCK)	-X2	7	5	B5
6	B6	-X2	9	EXTERNAL CONTROL (INTERLOCK)	-Y1	X1	6	C6
6	C6	-X2	10	EXTERNAL CONTROL (INTERLOCK)	-Y1	X2	6	C6
7	E2	-X_		EXTERNAL CONTROL (INTERLOCK)	-X2	11	7	D2
7	E3	-X_		EXTERNAL CONTROL (INTERLOCK)	-X2	12	7	D3
7	E4	-X_		EXTERNAL CONTROL (INTERLOCK)	-X2	13	7	D4
4	E2	-Q1	L2	EXTERNAL POWER	-X1	2	4	D2
4	E2	-Q1	L1	EXTERNAL POWER	-X1	1	4	D2
4	E3	-GND		EXTERNAL PROTECTIVE	-PE	1	4	D3
6	B6	-PE	6	EXTERNAL PROTECTIVE	-GND		6	B6
4	D3	-PE	1	PROTECTIVE	-GND		4	D3
4	C4	-T1	02N	PROTECTIVE	-PE	2	4	C3

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Project name: Mansourah - Massarah Gold Mine
Project id: P/20002
Product id: 6X20V32TS
Based on document id: DAAB198406 d
Customer document id:

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CONNECTION TABLE
OILY WATER TRANSFER UNIT
LOCAL CONTROL PANEL
INTERNAL CONNECTION LIST
CONFIDENTIAL

Created by: RSW004	Create date: 5.5.2020	Language: EN	Object designation: BJU904
Reviewed by: KSU004	Review date: 5.5.2020	Sheet size: A3	Page: 9 / 10
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COMPONENT NAME	QTY	MANUFACTURER	PART CODE	RATING	DESC
-E01	1	RITTAL	1038000		
-E01	4	RITTAL	2508020		WALL MOUNTING BRACKET
-E01	1	RITTAL			SECURITY LOCK
-F1	1	Schneider Electric	A9F05202	2A D-TRIP	MINIATURE CIRCUIT BREAKER
-F2	1	Schneider Electric	A9F04101	1A	MINIATURE CIRCUIT BREAKER
-F3	1	Schneider Electric	A9F04102	2.0A	MINIATURE CIRCUIT BREAKER
-K1	1	Carlo Gavazzi	RCP11003115120VAC		UNIVERSAL PLUG-IN RELAY, 3 CO
-K1	1	Carlo Gavazzi	ZPD12A		SOCKET
-K2	1	Carlo Gavazzi	RCP11003115120VAC		UNIVERSAL PLUG-IN RELAY, 3 CO
-K2	1	Carlo Gavazzi	ZPD12A		SOCKET
-K3	1	Carlo Gavazzi	RCP11003115120VAC		UNIVERSAL PLUG-IN RELAY, 3 CO
-K3	1	Carlo Gavazzi	ZPD12A		SOCKET
-K4	1	Carlo Gavazzi	RCP11003115120VAC		UNIVERSAL PLUG-IN RELAY, 3 CO
-K4	1	Carlo Gavazzi	ZPD12A		SOCKET
-K5	1	Carlo Gavazzi	RCP11003115120VAC		UNIVERSAL PLUG-IN RELAY, 3 CO
-K5	1	Carlo Gavazzi	ZPD12A		SOCKET
-P1	1	Schneider Electric	XB7EV05GP		MONOLITHIC INDICATOR LIGHT, YELLOW, INTEGRAL LED
-PE	1	Phoenix Contact	NLS-CU 3/10		BUSBAR (FOR PE TERMINAL BLOCK)
-PE	2	Phoenix Contact	AB/SS		END PIECE (FOR PE TERMINAL BLOCK)
-PE	15	Phoenix Contact	AKG 4 GNYE		PE TERMINAL BLOCK
-Q1	1	BACO	0172001		
-S1	1	Schneider Electric	ZB5AK1813 + ZB5AW0G15		CONTROL SWITCH, 3 POSITION
-T1	1	Trafomic	0423193	200VA	CONTROL VOLTAGE TRANSFORMER
-U1	1	Agro	1555.20.13	M20	CABLE GLAND, M20
-U2	1	Agro	1555.20.13	M20	CABLE GLAND, M20
-U3	1	Agro	1555.20.13	M20	CABLE GLAND, M20
-U4	1	Agro	1555.20.13	M20	CABLE GLAND, M20
-U5	1	Agro	1555.20.13	M20	CABLE GLAND, M20
-U6	1	Agro	1555.20.13	M20	CABLE GLAND, M20
-X1	2	Phoenix Contact	CLIPFIX 35		END PIECE (FOR TERMINAL BLOCK)
-X1	3	Weidmuller	ZDU 6		
-X2	2	Phoenix Contact	CLIPFIX 35		END PIECE (FOR TERMINAL BLOCK)
-X2	20	Weidmuller	ZDU 4		
-F4	1	Schneider Electric	A9F04102		MINIATURE CIRCUIT BREAKER
-FILTER	1	Kryos	GSF15		FILTER
-M01	1	Kryos	GSV1500203		FAN
-TC1	1	RITTAL	3100000		THERMOSTAT

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Project name: Mansourah - Massarah Gold Mine
Project id: P/20002 | Product id: 6X20V32TS
Based on document id: DAAB198406 d | Customer document id:

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PARTS LIST
OILY WATER TRANSFER UNIT
LOCAL CONTROL PANEL
PARTS LIST
CONFIDENTIAL

Created by: RSW004	Create date: 5.5.2020	Language: EN	Object designation: BJU904
Reviewed by: KSU004	Review date: 5.5.2020	Sheet size: A3	Page: 10 / 10
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PANEL CODE	DOCUMENT NAME	DOCUMENT DESCRIPTION	SHEET N:o	DOCUMENT N:o	REVISION
BJU905	LIST OF DOCUMENTS	DRAWING LIST	1	DBAF511317	c
BJU905	DIMENSIONAL DRAWING	LAYOUT 1:10	2	DBAF511317	c
BJU905	OVERVIEW DIAGRAM	SINGLE LINE	3	DBAF511317	c
BJU905	CIRCUIT DIAGRAM	LOW VOLTAGE FEEDER	4	DBAF511317	c
BJU905	CIRCUIT DIAGRAM	COMMON INDICATOR/CONTROL	5	DBAF511317	c
BJU905	CIRCUIT DIAGRAM	PUMP CONTROL	6	DBAF511317	c
BJU905	CIRCUIT DIAGRAM	EXTERNAL ALARM	7	DBAF511317	d
BJU905	CONNECTION TABLE	CABLE LIST	8	DBAF511317	d
BJU905	CONNECTION TABLE	INTERNAL CONNECTION LIST	9	DBAF511317	c
BJU905	PARTS LIST	PARTS LIST	10	DBAF511317	c

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Project name: Mansourah - Massarah Gold Mine
Project id: P/20002
Product id: 6X20V32TS
Based on document id: DAAB198406 d
Customer document id:

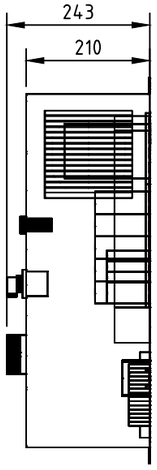
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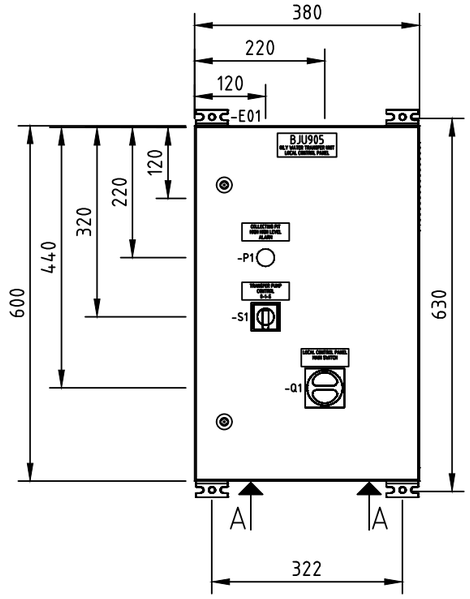
LIST OF DOCUMENTS
OILY WATER TRANSFER UNIT
LOCAL CONTROL PANEL
DRAWING LIST
CONFIDENTIAL

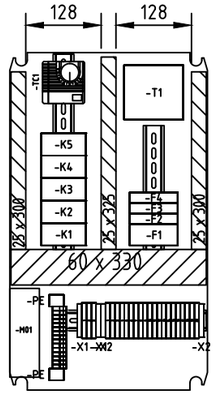
Created by	Create date	Language	Object designation
RSW004	12.5.2020	EN	BJU905
Reviewed by	**Review date**	**Sheet size**	**Page**
KSU004	12.5.2020	A3	1 / 10
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KSU004	12.5.2020	DBAF511317	D

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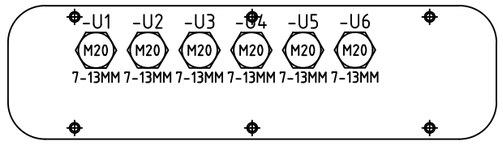
Location - on Oily water transfer unit module, near Unloading station







SECTION A-A SCALE 1:4



PANEL INFORMATION					
WIDTH	380	mm	PROTECTION CLASS		IP54
HEIGHT	600	mm	COLOR		RAL 7035
DEPTH	210	mm	Ik	6	kA/1s

-E01	BJU905	OILY WATER TRANSFER UNIT	LOCAL CONTROL PANEL
-P1	COLLECTING PIT	HIGH HIGH LEVEL	ALARM
-Q1	LOCAL CONTROL PANEL	MAIN SWITCH	
-S1	TRANSFER PUMP	CONTROL	0-1-S

Rev.	Date	Created by	Appr. by	Revision text
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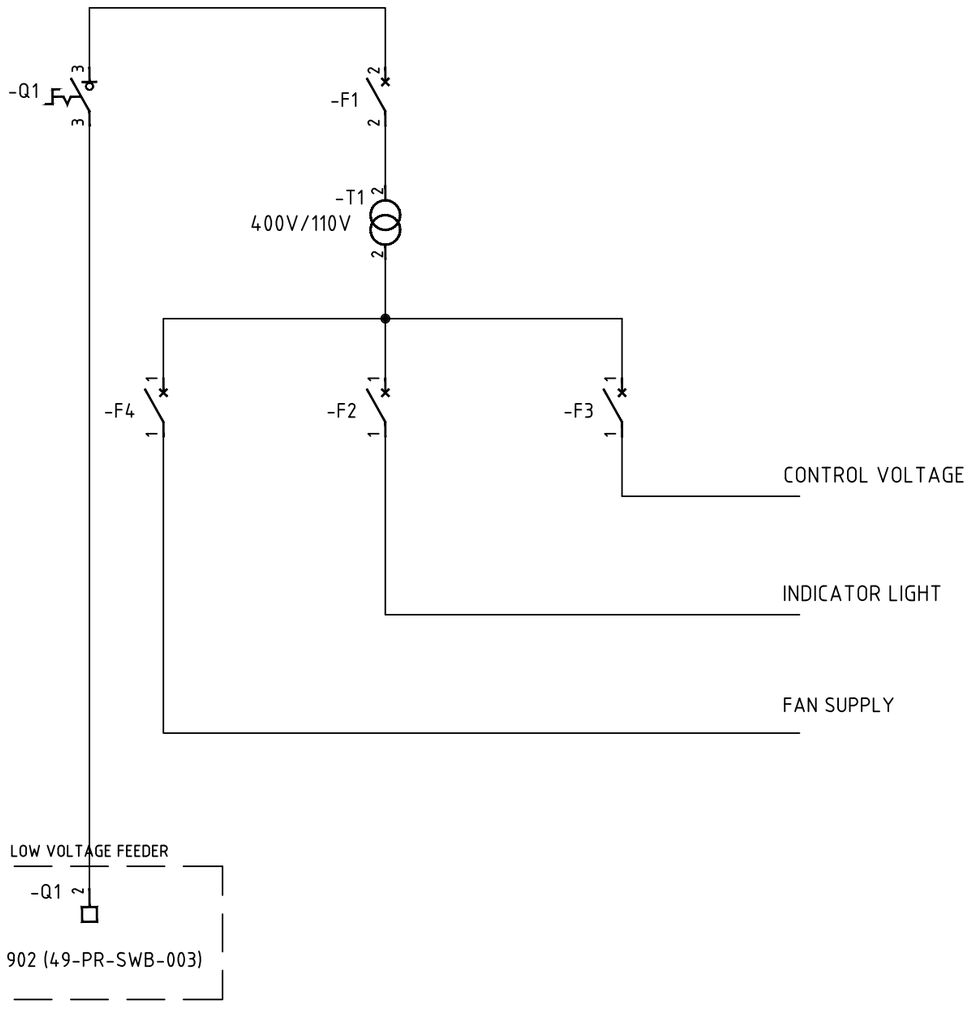
Project name: Mansourah - Massarah Gold Mine
Project id: P/20002
Product id: 6X20V32TS
Based on document id: DAAB198406 d
Customer document id:

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OILY WATER TRANSFER UNIT
LOCAL CONTROL PANEL
CONFIDENTIAL

Created by: RSW004	Create date: 12.5.2020	Language: EN	Object designation: BJU905
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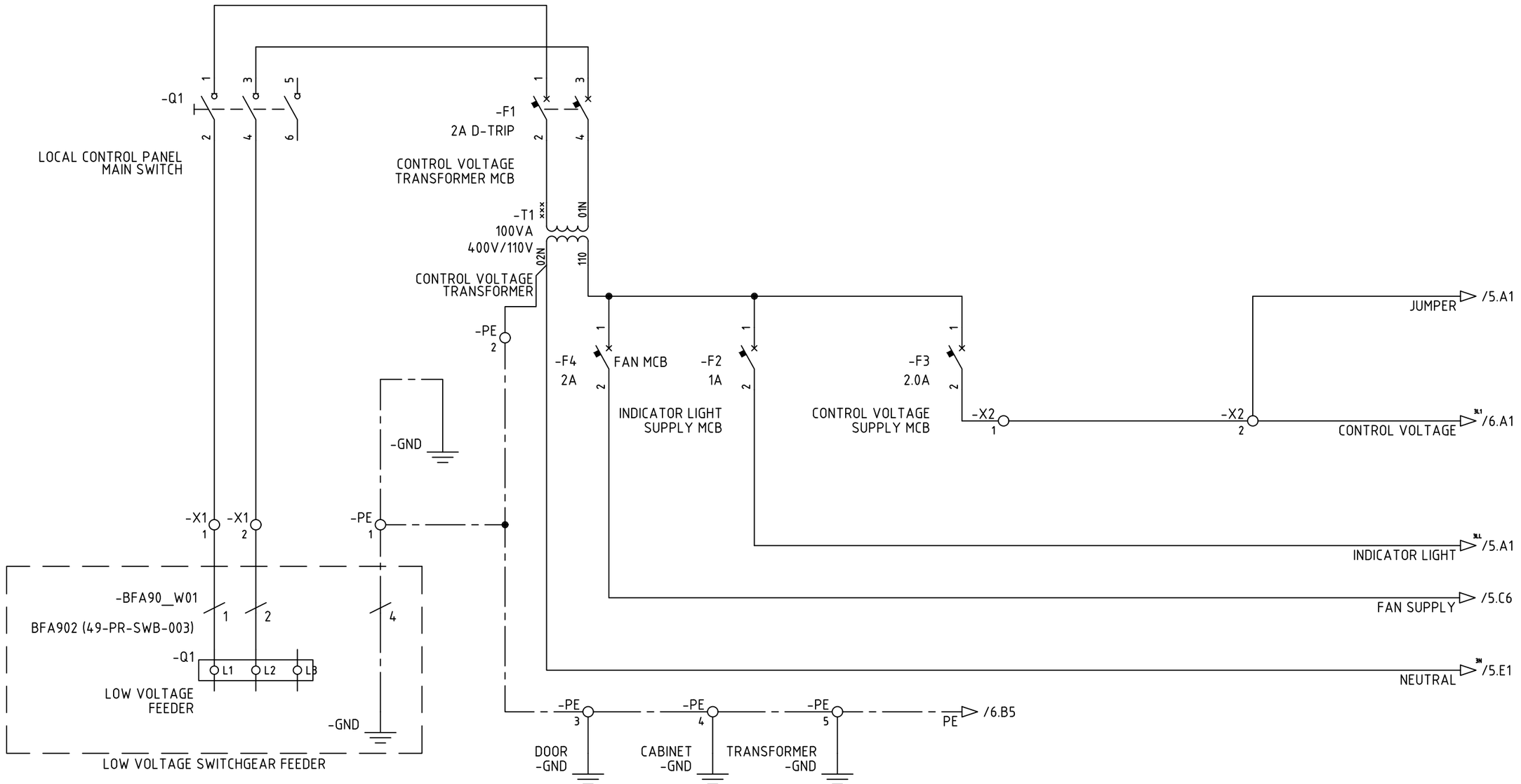
Project name: Mansourah - Massarah Gold Mine
Project id: P/20002
Product id: 6X20V32TS
Based on document id: DAAB198406 d
Customer document id:

WÄRTSILÄ

OVERVIEW DIAGRAM
OILY WATER TRANSFER UNIT
LOCAL CONTROL PANEL
SINGLE LINE
CONFIDENTIAL

Created by	Create date	Language	Object designation
RSW004	12.5.2020	EN	BJU905
Reviewed by	**Review date**	**Sheet size**	**Page**
KSU004	12.5.2020	A3	3 / 10
Approved by	**Approval date**	**Document Id:**	**Rev.**
KSU004	12.5.2020	DBAF511317	D

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SUPPLY					
Un	400V	V	2-phase	60	Hz
In	1	A		10	A

Rev.	Date	Created by	Appr. by	Revision text
c	21.7.2020	KKO070	KSU004	As manufactured update
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Project name	Mansourah - Massarah Gold Mine
Project id	P/20002
Product id	6X20V32TS
Based on document id	DAAB198406 d
Customer document id	

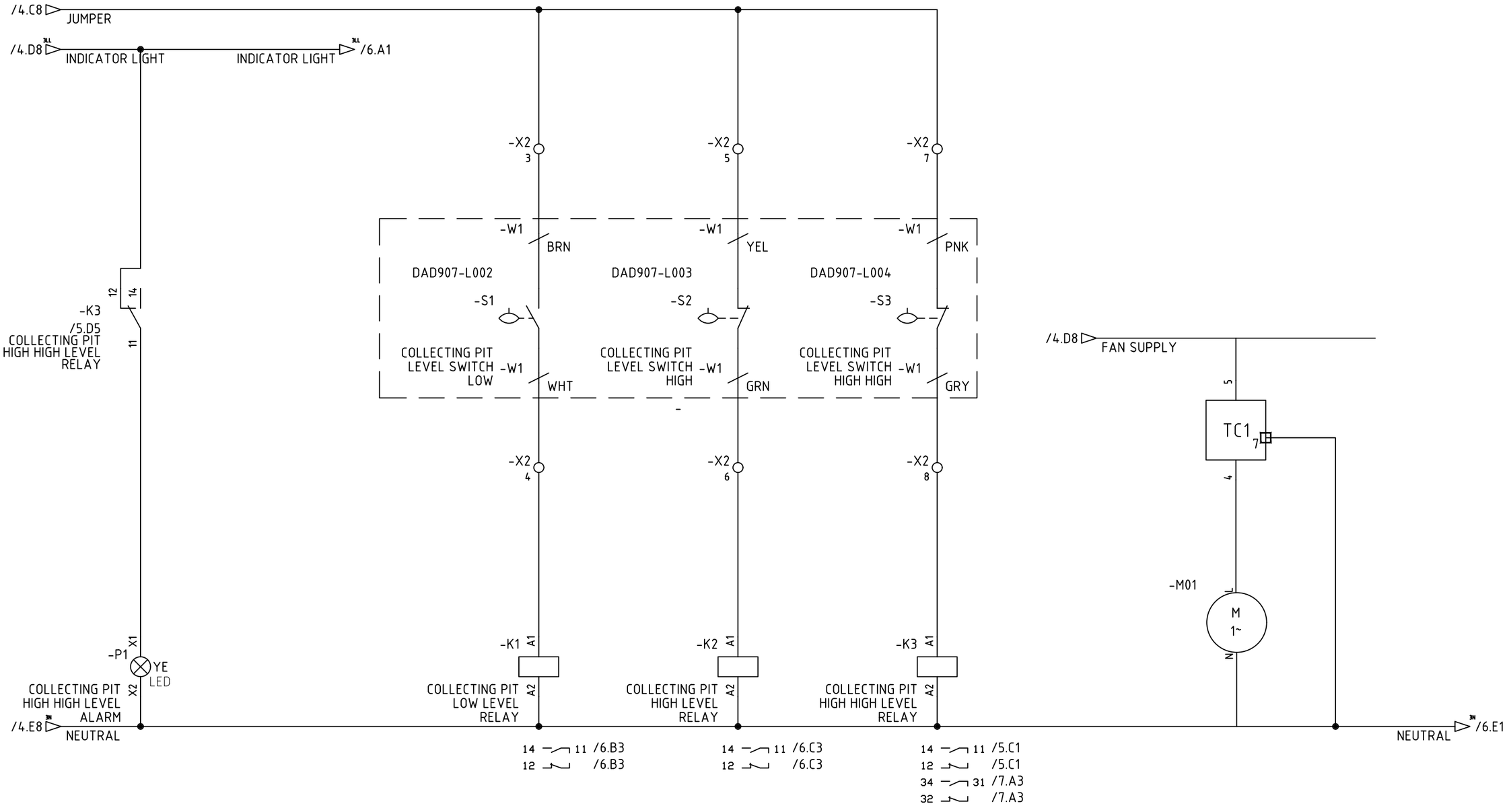
CIRCUIT DIAGRAM

OILY WATER TRANSFER UNIT
LOCAL CONTROL PANEL
LOW VOLTAGE FEEDER

CONFIDENTIAL

Created by RSW004	Create date 12.5.2020	Language EN	Object designation BJU905
Reviewed by KSU004	Review date 12.5.2020	Sheet size A3	Page 4 / 10
Approved by KSU004	Approval date 12.5.2020	Document Id: DBAF511317	Rev. D

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a	5.6.2020	KKO070	KSU004	As per customer inputs

Project name	Mansourah - Massarah Gold Mine		
Project id	P/20002	Product id	6X20V32TS
Based on document id	DAAB198406	Customer document id	d

WÄRTSILÄ

CIRCUIT DIAGRAM

OILY WATER TRANSFER UNIT
LOCAL CONTROL PANEL
COMMON INDICATOR/CONTROL
CONFIDENTIAL

Created by	RSW004	Create date	12.5.2020
Reviewed by	KSU004	Review date	12.5.2020
Approved by	KSU004	Approval date	12.5.2020
Language	EN	Object designation	BJU905
Sheet size	A3	Page	5 / 10
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/4.D8 3L1 CONTROL VOLTAGE

/5.A2 3L1 INDICATOR LIGHT

-K1 /5.D3 COLLECTING PIT LOW LEVEL RELAY (12, 14, 11)

-K4 /6.D3 TRANSFER PUMP CONTROL RELAY 1 (12, 14, 11)

-S1 0-1-S TRANSFER PUMP CONTROL 0-1-S (21, 22)

-S1 /6.B4 TRANSFER PUMP CONTROL 0-1-S (13, 14)

-S1					
0	1	S	T1	T2	REF
X			21	22	B4
			X1	X2	6E1
		X	13	14	6C4

-K2 /5.D4 COLLECTING PIT HIGH LEVEL RELAY (12, 14, 11)

-K4 /6.D3 TRANSFER PUMP CONTROL RELAY 1 (22, 24, 21)

-K5 /6.D4 TRANSFER PUMP CONTROL RELAY2 (22, 24, 21)

/4.E6 PE

-PE 6

-X2 9

-W01 3

-W01 1

-GND

DAD907-CV001

-Y1 X1 X2 TRANSFER PUMP CONTROL VALVE

-W01 2

-X2 10

-S1 X1 /6.B4 WH LED TRANSFER PUMP X2

/5.E8 3N 0-1-S NEUTRAL

-K4 A1 A2 TRANSFER PUMP CONTROL RELAY 1

-K5 A1 A2 TRANSFER PUMP CONTROL RELAY2

14 — 11 /6.C1
12 — /6.C1
24 — 21 /6.C3
22 — /6.C3

24 — 21 /6.A6
22 — /6.A6
34 — 31 /7.A4
32 — /7.A4

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a	5.6.2020	KKO070	KSU004	As per customer inputs

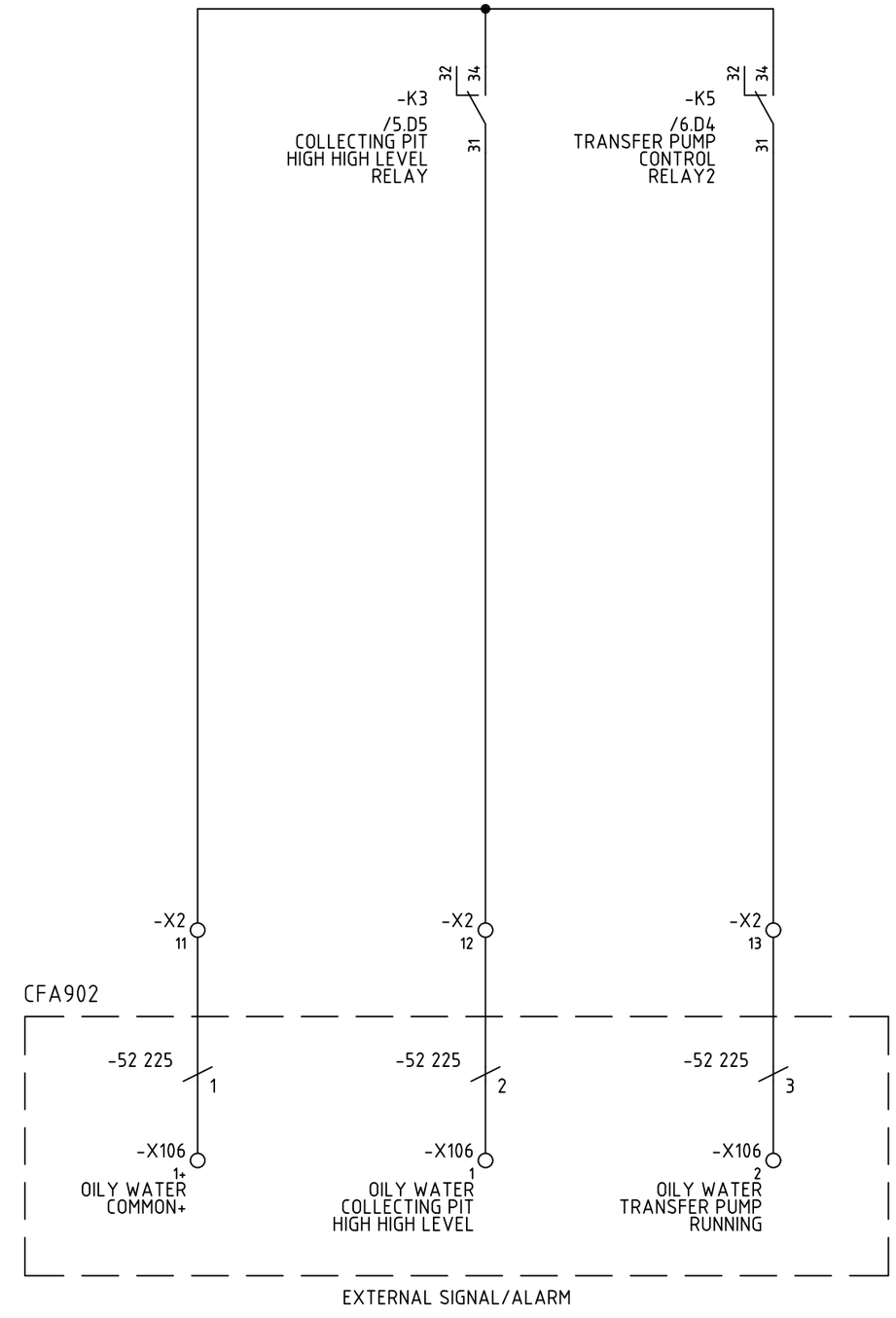
Project name: Mansourah - Massarah Gold Mine
Project id: P/20002
Product id: 6X20V32TS
Based on document id: DAAB198406 d
Customer document id:

WÄRTSILÄ

CIRCUIT DIAGRAM
OILY WATER TRANSFER UNIT
LOCAL CONTROL PANEL
PUMP CONTROL
CONFIDENTIAL

Created by	Create date	Language	Object designation
RSW004	12.5.2020	EN	BJU905
Reviewed by	Review date	Sheet size	Page
KSU004	12.5.2020	A3	6 / 10
Approved by	Approval date	Document Id:	Rev.
KSU004	12.5.2020	DBAF511317	D

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d	16.Feb.2021	MHA186	KSU004	Terminal update
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Project name: Mansourah - Massarah Gold Mine
Project id: P/20002
Product id: 6X20V32TS
Based on document id: DAAB198406 d
Customer document id:

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CIRCUIT DIAGRAM
OILY WATER TRANSFER UNIT
LOCAL CONTROL PANEL
EXTERNAL ALARM

Created by	Create date	Language	Object designation
RSW004	12.5.2020	EN	BJU905
Reviewed by	Review date	Sheet size	Page
KSU004	12.5.2020	A3	7 / 10
Approved by	Approval date	Document Id:	Rev.
KSU004	12.5.2020	DBAF511317	d

FACTORY INSTALLED CABLES TO BE SUPPLIED BY HELKAMA IF NOT OTHERWISE STATED

CABLE SPECIFICATION	CABLE TYPE	CABLE INSTALLATION	CABLE NAME	WIRE	FROM COMPONENT	PIN1	TO COMPONENT	PIN2	SH1	REF1	CABLE REMARK
			-W01	1	-X2	9	=DAD907-CV001-Y1	X1	6	B6	INSTALLED AT FACTORY
			-W01	2	-X2	10	=DAD907-CV001-Y1	X2	6	C6	INSTALLED AT FACTORY
			-W01	3	-PE	6	=DAD907-CV001-GND		6	B6	INSTALLED AT FACTORY
		BFA902 (49-PR-SWB-003)	-BFA90_W01	1	=BFA902 (49-PR-SWB-003)-Q1	L1	-X1	1	4	E2	INSTALLED AT SITE
		BFA902 (49-PR-SWB-003)	-BFA90_W01	2	=BFA902 (49-PR-SWB-003)-Q1	L2	-X1	2	4	E2	INSTALLED AT SITE
		BFA902 (49-PR-SWB-003)	-BFA90_W01	4	=BFA902 (49-PR-SWB-003)-GND		-PE	1	4	E3	INSTALLED AT SITE
		CFA902	-52 225	1	=CFA902-X106	1+	-X2	11	7	E2	INSTALLED AT SITE
		CFA902	-52 225	2	=CFA902-X106	1	-X2	12	7	E3	INSTALLED AT SITE
		CFA902	-52 225	3	=CFA902-X106	2	-X2	13	7	E4	INSTALLED AT SITE
		DAD907-L001	-W1	BRN	=DAD907-L002-S1		-X2	3	5	C3	INSTALLED AT SITE (INTEGRATED SILICON CABLE)
		DAD907-L001	-W1	GRN	=DAD907-L003-S2		-X2	6	5	C4	INSTALLED AT SITE (INTEGRATED SILICON CABLE)
		DAD907-L001	-W1	GRY	=DAD907-L004-S3		-X2	8	5	C5	INSTALLED AT SITE (INTEGRATED SILICON CABLE)
		DAD907-L001	-W1	PNK	=DAD907-L004-S3		-X2	7	5	C5	INSTALLED AT SITE (INTEGRATED SILICON CABLE)
		DAD907-L001	-W1	WHT	=DAD907-L002-S1		-X2	4	5	C3	INSTALLED AT SITE (INTEGRATED SILICON CABLE)
		DAD907-L001	-W1	YEL	=DAD907-L003-S2		-X2	5	5	C4	INSTALLED AT SITE (INTEGRATED SILICON CABLE)

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Project name: Mansourah - Massarah Gold Mine
Project id: P/20002
Product id: 6X20V32TS
Based on document id: DAAB198406 d
Customer document id:

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CONNECTION TABLE
OILY WATER TRANSFER UNIT
LOCAL CONTROL PANEL
CABLE LIST

Created by: RSW004	Create date: 12.5.2020	Language: EN	Object designation: BJU905
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SH1	REF1	CMP1	PIN1	WLAY1	CMP2	PIN2	SH2	REF2
4	C4	-T1	110	AC CONTROL	-F2	1	4	C5
4	C4	-T1	02N	AC CONTROL	-K4	A2	6	E3
4	C4	-F4	2	AC CONTROL	-TC1	5	5	C7
4	C4	-F4	1	AC CONTROL	-F3	1	4	C6
4	C5	-F2	2	AC CONTROL	-K3	12	5	C1
4	C5	-F2	1	AC CONTROL	-F4	1	4	C4
4	C6	-F3	2	AC CONTROL	-X2	1	4	D6
4	D6	-X2	1	AC CONTROL	-X2	2	4	D7
4	D7	-X2	2	AC CONTROL	-K1	14	6	B3
5	C1	-K3	11	AC CONTROL	-P1	X1	5	D1
5	C1	-K3	12	AC CONTROL	-K4	14	6	C1
5	E1	-P1	X2	AC CONTROL	-S1	X2	6	E1
5	C3	-X2	4	AC CONTROL	-K1	A1	5	D3
5	E3	-K1	A2	AC CONTROL	-K2	A2	5	E4
5	C4	-X2	6	AC CONTROL	-K2	A1	5	D4
5	E4	-K2	A2	AC CONTROL	-K3	A2	5	E5
5	C5	-X2	8	AC CONTROL	-K3	A1	5	D5
5	E5	-K3	A2	AC CONTROL	-P1	X2	5	E1
5	D7	-TC1	4	AC CONTROL	-M01	L	5	D7
6	C1	-K4	11	AC CONTROL	-S1	X1	6	D1
6	E1	-S1	X2	AC CONTROL	-M01	N	5	D7
6	B3	-K1	11	AC CONTROL	-S1	21	6	C3
6	B3	-K1	14	AC CONTROL	-K5	24	6	A6
6	C3	-K2	12	AC CONTROL	-K4	24	6	C3
6	C3	-K4	24	AC CONTROL	-S1	22	6	C3
6	D3	-K2	11	AC CONTROL	-S1	14	6	C4
6	D3	-K4	21	AC CONTROL	-K4	A1	6	D3
6	D3	-K4	A1	AC CONTROL	-K5	A1	6	D4
6	E3	-K4	A2	AC CONTROL	-K5	A2	6	E4
6	E4	-K5	A2	AC CONTROL	-X2	10	6	C6
6	A6	-K5	21	AC CONTROL	-X2	9	6	B6
6	A6	-K5	24	AC CONTROL	-S1	13	6	C4
6	C6	-X2	10	AC CONTROL	-K1	A2	5	E3
7	D2	-X2	11	AC CONTROL	-K3	34	7	A3
7	A3	-K3	31	AC CONTROL	-X2	12	7	D3
7	A3	-K3	34	AC CONTROL	-K5	34	7	A4
7	A4	-K5	31	AC CONTROL	-X2	13	7	D4
4	D7	-X2	2	AC CONTROL JMP	-X2	5	5	B4
5	B3	-X2	3	AC CONTROL JMP	-X2	2	4	D7
5	B4	-X2	5	AC CONTROL JMP	-X2	7	5	B5
4	B2	-Q1	4	AC POWER	-X1	2	4	D2
4	B2	-Q1	3	AC POWER	-F1	3	4	B4
4	B2	-Q1	2	AC POWER	-X1	1	4	D2
4	B2	-Q1	1	AC POWER	-F1	1	4	B4
4	B4	-F1	4	AC POWER	-T1	01N	4	C4
4	B4	-F1	2	AC POWER	-T1	×××	4	C4
5	C3	-S1		EXTERNAL CONTROL (INTERLOCK)	-X2	4	5	C3
5	C3	-S1		EXTERNAL CONTROL (INTERLOCK)	-X2	3	5	B3
5	C4	-S2		EXTERNAL CONTROL (INTERLOCK)	-X2	6	5	C4
5	C4	-S2		EXTERNAL CONTROL (INTERLOCK)	-X2	5	5	B4
5	C5	-S3		EXTERNAL CONTROL (INTERLOCK)	-X2	8	5	C5
5	C5	-S3		EXTERNAL CONTROL (INTERLOCK)	-X2	7	5	B5
6	B6	-X2	9	EXTERNAL CONTROL (INTERLOCK)	-Y1	X1	6	C6
6	C6	-X2	10	EXTERNAL CONTROL (INTERLOCK)	-Y1	X2	6	C6
7	E2	-X_		EXTERNAL CONTROL (INTERLOCK)	-X2	11	7	D2
7	E3	-X_		EXTERNAL CONTROL (INTERLOCK)	-X2	12	7	D3
7	E4	-X_		EXTERNAL CONTROL (INTERLOCK)	-X2	13	7	D4
4	E2	-Q1	L2	EXTERNAL POWER	-X1	2	4	D2
4	E2	-Q1	L1	EXTERNAL POWER	-X1	1	4	D2
4	E3	-GND		EXTERNAL PROTECTIVE	-PE	1	4	D3
6	B6	-PE	6	EXTERNAL PROTECTIVE	-GND		6	B6
4	D3	-PE	1	PROTECTIVE	-GND		4	D3
4	C4	-T1	02N	PROTECTIVE	-PE	2	4	C3

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Project name: Mansourah - Massarah Gold Mine
Project id: P/20002
Product id: 6X20V32TS
Based on document id: DAAB198406 d
Customer document id:

WÄRTSILÄ

CONNECTION TABLE
OILY WATER TRANSFER UNIT
LOCAL CONTROL PANEL
INTERNAL CONNECTION LIST
CONFIDENTIAL

Created by: RSW004	Create date: 12.5.2020	Language: EN	Object designation: BJU905
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COMPONENT NAME	QTY	MANUFACTURER	PART CODE	RATING	DESC
-E01	1	RITTAL	1038000		
-E01	4	RITTAL	2508020		WALL MOUNTING BRACKET
-E01	1	RITTAL			SECURITY LOCK
-F1	1	Schneider Electric	A9F05202	2A D-TRIP	MINIATURE CIRCUIT BREAKER
-F2	1	Schneider Electric	A9F04101	1A	MINIATURE CIRCUIT BREAKER
-F3	1	Schneider Electric	A9F04102	2.0A	MINIATURE CIRCUIT BREAKER
-K1	1	Carlo Gavazzi	RCP11003115120VAC		UNIVERSAL PLUG-IN RELAY, 3 CO
-K1	1	Carlo Gavazzi	ZPD12A		SOCKET
-K2	1	Carlo Gavazzi	RCP11003115120VAC		UNIVERSAL PLUG-IN RELAY, 3 CO
-K2	1	Carlo Gavazzi	ZPD12A		SOCKET
-K3	1	Carlo Gavazzi	RCP11003115120VAC		UNIVERSAL PLUG-IN RELAY, 3 CO
-K3	1	Carlo Gavazzi	ZPD12A		SOCKET
-K4	1	Carlo Gavazzi	RCP11003115120VAC		UNIVERSAL PLUG-IN RELAY, 3 CO
-K4	1	Carlo Gavazzi	ZPD12A		SOCKET
-K5	1	Carlo Gavazzi	RCP11003115120VAC		UNIVERSAL PLUG-IN RELAY, 3 CO
-K5	1	Carlo Gavazzi	ZPD12A		SOCKET
-P1	1	Schneider Electric	XB7EV05GP		MONOLITHIC INDICATOR LIGHT, YELLOW, INTEGRAL LED
-PE	1	Phoenix Contact	NLS-CU 3/10		BUSBAR (FOR PE TERMINAL BLOCK)
-PE	2	Phoenix Contact	AB/SS		END PIECE (FOR PE TERMINAL BLOCK)
-PE	15	Phoenix Contact	AKG 4 GNYE		PE TERMINAL BLOCK
-Q1	1	BACO	0172001		
-S1	1	Schneider Electric	ZB5AK1813 + ZB5AW0G15		CONTROL SWITCH, 3 POSITION
-T1	1	Trafomic	0423193	200VA	CONTROL VOLTAGE TRANSFORMER
-U1	1	Agro	1555.20.13	M20	CABLE GLAND, M20
-U2	1	Agro	1555.20.13	M20	CABLE GLAND, M20
-U3	1	Agro	1555.20.13	M20	CABLE GLAND, M20
-U4	1	Agro	1555.20.13	M20	CABLE GLAND, M20
-U5	1	Agro	1555.20.13	M20	CABLE GLAND, M20
-U6	1	Agro	1555.20.13	M20	CABLE GLAND, M20
-X1	2	Phoenix Contact	CLIPFIX 35		END PIECE (FOR TERMINAL BLOCK)
-X1	3	Weidmuller	ZDU 6		
-X2	2	Phoenix Contact	CLIPFIX 35		END PIECE (FOR TERMINAL BLOCK)
-X2	20	Weidmuller	ZDU 4		
-F4	1	Schneider Electric	A9F04102		MINIATURE CIRCUIT BREAKER
-FILTER	1	Kryos	GSF15		FILTER
-M01	1	Kryos	GSV1500203		FAN
-TC1	1	RITTAL	3100000		THERMOSTAT

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Project name: Mansourah - Massarah Gold Mine
Project id: P/20002
Product id: 6X20V32TS
Based on document id: DAAB198406 d
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PARTS LIST
OILY WATER TRANSFER UNIT
LOCAL CONTROL PANEL
PARTS LIST
CONFIDENTIAL

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